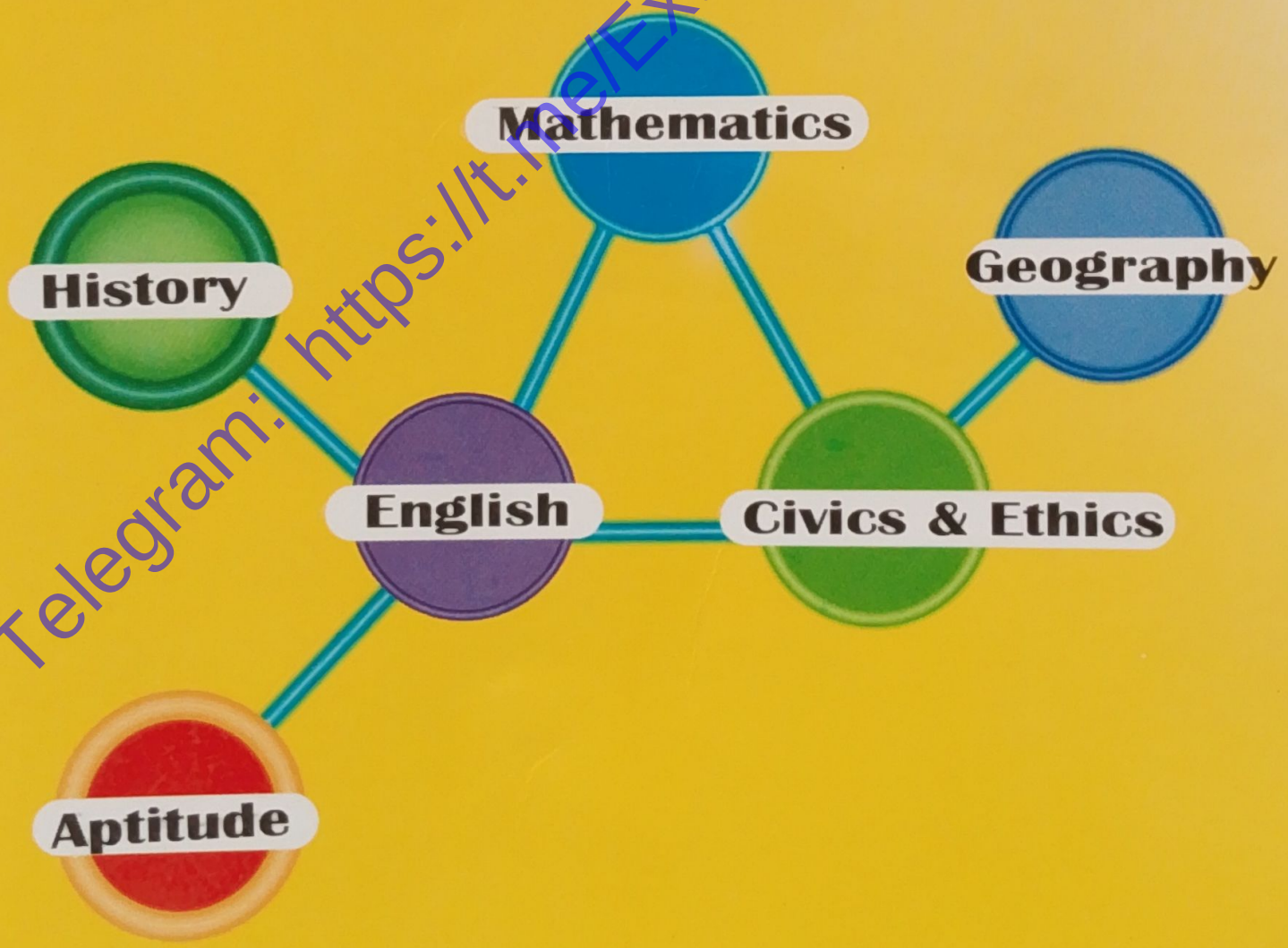
Extreme Series

EXAM BOOK

Social Science

University Entrance Examination



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Extreme Series

2015 E.C

Compiled by:
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Extreme

Social Science University Entrance Exam Book

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Extreme Exam Book Reference

2010 - 2014/15 E. C

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SECTION 1 – MATHEMATICS

1. If a function f is differentiable at a, then what is the value of $\lim_{x\to a} f(x)$?

   A. $f(a)$ B. $f'(a)$ C. $f'(a)-f(a)$ D. 0

   Solution: If f is differentiable at a, then f is continuous at a, the value of $\lim_{x\to a} f(x) = f(a)$ **Answer: A**

2. Which one of the following is a convergent sequence?

   A. $\left\{\frac{1+2^n}{3^n}\right\}$ C. $\left\{\frac{1-3^n}{2^n}\right\}$

   B. $\left\{\frac{1}{n}+\sin(n)\right\}$ D. $\left\{\frac{(-1)^n}{2}\right\}$

   Solution:

   A. $\lim_{n\to\infty}\left(\frac{1+2^n}{3^n}\right)=0$ C. $\lim_{n\to\infty}\left(\frac{1-3^n}{2^n}\right)=-\infty$

   B. $\lim_{n\to\infty}\left(\frac{1}{n}+\sin(n)\right)=\not\exists$ D. $\lim_{n\to\infty}\frac{(-1)^n}{2}=\pm\frac{1}{2}$

   $\therefore \left\{\frac{1+2^n}{3^n}\right\}$ is convergent sequence. **Answer: A**

3. What is the value of $\sum_{n=2}^{20}\left(\frac{1}{n-1}-\frac{1}{n}\right)$?

   A. $\frac{17}{20}$ B. $\frac{23}{20}$ C. $\frac{21}{20}$ D. $\frac{19}{20}$

   Solution: $\sum_{n=2}^{20}\left(\frac{1}{n-1}-\frac{1}{n}\right)$

   $$=\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\ldots+\left(\frac{1}{n-1}-\frac{1}{n}\right)$$

   $\therefore S_n = 1-\frac{1}{n} \Rightarrow S_{20} = 1-\frac{1}{20}=\frac{19}{20}$ **Answer: D**

4. Let A and B be two events. Suppose that the probability that neither event occurs in $3/8$. What is the probability that at least one of the events occurs?

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A. $\frac{1}{8}$ B. $\frac{5}{8}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

Solution: A and B be two events P(at least one of the events A and B occur) = 1 – P(neither events occur)

P(at least one of the events A and B occur) $= 1-\frac{3}{8}=\frac{5}{8}$

Answer: B

5. If the truth value of $(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r]$ is True, then which one of the following must be True?

A. q B. p C. $\neg q$ D. $\neg r$

Solution: The truth value of

$(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv$ True (T)

$p \wedge \neg p \equiv F$ from the definition of negation and conjunction

$q \vee \neg q \equiv T$ from the definition of negation and disjunction

$(q \vee \neg q) \Rightarrow r \equiv$ False because from the given

$(p \vee \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv T$ and from the definition of bi implication $r \equiv F$ whereas $\neg r \equiv T$

Answer: D

6. What is the maximum value of the function $f(x) = x^4 - 2x^2$ on $[-2, 1]$?

A. 8 B. 12 C. 24 D. 40

Solution: Here at, we set $f'(x) = 0$

$f(x) = x^4 - 2x^2 \Rightarrow f'(x) = 4x^3 - 4x = 0$

$\Rightarrow 4x(x^2 - 1) = 4x(x-1)(x+1) = 0$

Hence, the critical numbers are –1, 0, and 1.

We evaluate f at the critical points and the end points of the interval:

$f(-2) = (-2)^4 - 2(-2)^2 = 16 - 8 = 8 \rightarrow$ is the maximum value

$f(-1) = (-1)^4 - 2(-1)^2 = 1 - 2 = -1$

$f(0) = 0 - 0 = 0$, and $f(1) = 1^4 - 2 = -1$

Answer: A

7. What are the greatest lower bound and least upper bound of the sequence $\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}$, respectively?

A. –2 and 2

B. $-\frac{3}{2}$ and 2

C. –2 and $\frac{3}{2}$

D. –2 and $-\frac{3}{2}$

Solution:

i) For even numbers: n = 2, 4, 6, 8, …,∞, then:

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$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}=\left\{1+\frac{1}{n}\right\}=\left\{\frac{n+1}{n}\right\}=\left\{\frac{3}{2},\frac{5}{4},\frac{7}{6}...1\right\}$$

ii) For odd numbers: n = 1, 3, 5, 7, ...∞, then:

$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}=-\left\{\left(1+\frac{1}{n}\right)\right\}=\left\{-2,\frac{-3}{2},\frac{-5}{4},\frac{-7}{6},....-1\right\}$$

$$\therefore -2\le\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}\le\frac{3}{2}$$

Answer: C

8. Which one of the following is an equation of the circle whose endpoints of a diameter are (0, –2) and (2, 2)?

A. $x^2+y^2=4$ C. $(x-1)^2+y^2=4$

B. $x^2+y^2-2y-4=0$ D. $x^2+y^2-2x-4=0$

Solution:

✓ Diameter: $D=\sqrt{(2-(-2))^2+(2-0)^2}=\sqrt{20}=2\sqrt{5}$

✓ Radius $=\frac{D}{2}=\frac{2\sqrt{5}}{2}=\sqrt{5}$ and center $=\left(\frac{0+2}{2},\frac{-2+2}{2}\right)=(1,0)$

✓ Equation of circle: $(x-1)^2+(y-0)^2=\sqrt{5}^2$

$\Rightarrow(x-1)^2+y^2=5\Leftrightarrow x^2-2x+1+y^2=5$

$\therefore x^2+y^2-2x-4=0$ **Answer: D**

9. What is the equation of the line that passes through (1, 1) and is parallel to the line $3y-x=1$?

A. $x-3y+2=0$ C. $3y-x+2=0$

B. $x+3y=4$ D. $3x-y=2$

Solution: Any line parallel to the line $3y-x=1$ is of the form $3y-x=k$, where $k\in\mathbb{R}$. If $3y-x=k$, passes through (1, 1).

$\Rightarrow 3(1)-1=k\Rightarrow k=2$

∴ The required equation $3y-x=2\Leftrightarrow x-3y+2=0$ **Answer: A**

10. Which one of the following is true about signum, absolute value and greatest integer functions?

A. $\text{sgn}(x)=\pm|x|$, for all $x\in\mathbb{R}$

B. $\text{sgn}(x)\le[x]$, for all $x\le 0$

C. $|x|=x\ \text{sgn}(x)$, for all $x\in\mathbb{R}$

D. $\text{sgn}(x)\le[x]$, for all $x\ge 0$

Solution: Properties of signum; absolute value and greatest integer function

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A. $\text{sgn}(x)=\begin{cases}1 \text{ for } x>0\\0 \text{ for } x=0\\-1 \text{ for } x<0\end{cases}$ then $\text{sgn}(x)\neq\pm|x|$ for $x\in\mathbb{R}$

B. $\text{sgn}(x)=\begin{cases}-1 \text{ for } x<0\\0 \text{ for } x=0\end{cases}\Rightarrow \text{sgn}(x)\geq[x]$ for all $x\leq 0$

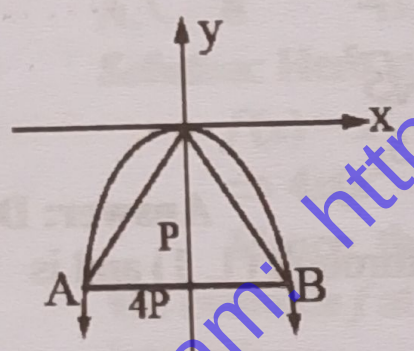
C. $|x|=\begin{cases}x \text{ for } x>0\\0 \text{ for } x=0\\-x \text{ for } x<0\end{cases}$ implies to $|x|=x\,\text{sgn}\,x$ for $x\in\mathbb{R}$

D. Let $x=\frac{1}{2}, \text{sgn}\left(\frac{1}{2}\right)\leq\left[\frac{1}{2}\right]\Rightarrow 1\leq 0$ is false **Answer: C**

11. What is the area of the triangle (in sq. units) formed by the lines joining the vertex of the parabola $x^2=-36y$ to the end points of the latus rectum?

A. 126 B. 261 C. 216 D. 162

Solution:



✓ The length of the latus rectum is $AB=|4P|=36$ unit

✓ The distance between the focus and the vertex is $P=\frac{36}{4}=9$

$\therefore \text{Area}=\frac{1}{2}(\text{base})(\text{height})=\frac{1}{2}(36)(9)=162$ **Answer: D**

12. What is the partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$?

A. $\frac{3}{5(x+2)}+\frac{2x+1}{5(x^2+1)}$ C. $\frac{2}{5(x+2)}+\frac{3x+1}{5(x^2+1)}$

B. $\frac{5}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$ D. $\frac{2}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$

Solution: Partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$ is

$$\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{A}{x+2}+\frac{Bx+C}{x^2+1}=\frac{Ax^2+A+Bx^2+Cx+2Bx+2C}{(x+2)(x^2+1)}$$

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$$\frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{Ax^2+Bx^2+(C+2B)x+A+2C}{(x+2)(x^2+1)}$$

$\Rightarrow A + B = 1$ ..... *

$C + 2B = 1 \Rightarrow C = 1 - 2B$ ..... **

$A + 2C = 1$ ..... ***

Substitute ** in to ***

$A + 2(1 - 2B) = 1$

$A + 2 - 4B = 1$

$-A + 4B = 1$ ..... ****

Solve simultaneously * and ****

$$\begin{cases} A+B=1 \\ -A+4B=1 \end{cases}$$

$5B = 2 \Rightarrow B = 2/5, A = 3/5$ and $C = 1 - \frac{4}{5} = \frac{1}{5}$

$\therefore$ Decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{3}{5(x+2)} + \frac{2x+1}{5(x^2+1)}$

Answer: A

13. What is the area of the region enclosed by the graph of $y^2 = x + 1$ and $y^2 = -x + 1$?

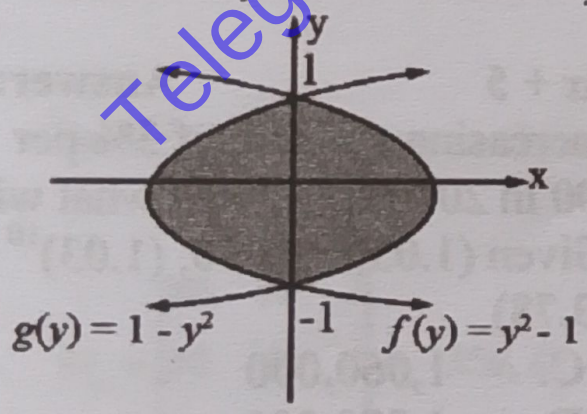
A. $\frac{3}{8}$ sq.units

B. $\frac{4}{3}$ sq.units

C. $\frac{8}{3}$ sq.units

D. $\frac{3}{4}$ sq.units

Solution: $y^2 = x + 1 \Rightarrow x = y^2 - 1$ and $y^2 = -x + 1 \Rightarrow x = 1 - y^2$



Let $f(y) = y^2 - 1$ and $g(y) = 1 - y^2$

The intersection of $f(y)$ and $g(y)$ is

$f(y) = g(y) \Rightarrow y^2 - 1 = 1 - y^2$

$\Rightarrow 2y^2 = 2 \Rightarrow y = 1$ or $y = -1$

$\therefore A = \int_{-1}^{1} [g(y) - f(y)]dy$

$$\therefore A = \int_{-1}^{1} [1 - y^2 - (y^2 - 1)]dy = \int_{-1}^{1} (-2y^2 + 2)dy$$

$$\therefore A = \frac{-2}{3}y^3 + 2y\Big|_{-1}^{1} = \left(\frac{-2}{3} + 2\right) - \left(\frac{2}{3} - 2\right) = 4 - \frac{4}{3} = \frac{8}{3}$$

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14. What is the value of $\int_1^2 \frac{x+4}{x(x+2)}dx$? **Answer: C**

A. 2 B. $\ln 3$ C. $\ln 2$ D. $\ln 4 - \ln 3$

Solution: $\int_1^2 \frac{x+4}{x(x+2)}dx$ but partial fraction decomposition of

$$\frac{x+4}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2} = \frac{Ax+2A+Bx}{x(x+2)}$$

$\Rightarrow A + B = 1$ and $2A = 4, A = 2, B = 1 - 2 = -1$

$$\int_1^2 \frac{x+2}{x(x+2)}dx = \int_1^2 \frac{2}{x}dx - \int_1^2 \frac{1}{x+2}dx = 2\ln|x| - \ln|x+2|$$

$$\ln\left|\frac{x^2}{x+2}\right| = \ln\frac{x^2}{x+2}\Big|_1^2 = \ln\frac{4}{4} - \ln\left(\frac{1}{3}\right)$$

$$= \ln 1 - (\ln 1 - \ln 3) = \ln 1 - \ln 1 + \ln 3$$

$$\therefore \int_1^2 \frac{x+4}{x(x+2)}dx = \ln 3$$ **Answer: B**

15. What is the slope of the tangent line to the graph of $f(x) = 3e^x + \sin x + 2$ at (0, 5)?

A. 2 B. 3 C. 5 D. 4

Solution:

$f(x) = 3e^x + \sin x + 2$ at (0, 5) then the slope of the tangent line is

$f'(x) = 3e^x + \cos x \Rightarrow f'(0) = 3e^0 + \cos 0 = 4$

The slope of tangent line is 4 and equation of the tangent line:

$$y - f(0) = f'(0)(x - 0)$$

$\Rightarrow y - 5 = 4(x - 0) \Rightarrow y = 4x + 5$ **Answer: D**

16. The population of certain city is increasing at a rate of 3% per year. If the population was 100,000 in 2010 E.C., then what will be the population in 2020 E.C? (Given $(1.03)^9 = 1.30$, $(1.03)^{10} = 1.34$, $(1.3)^9 = 10.60$, $(1.3)^{10} = 13.78$)

A. 134,000 C. 1,060,000

B. 130,000 D. 1,378,000

Solution: Population in certain country increasing at the rate of 3%

$P_0 = 100{,}000$ then the population in 2020

$P = P_0(1 + r)^{10} \Rightarrow P = 100{,}000\,(1.03)^{10} = 134{,}000$

Answer: A

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17. If $z=-3+4i$ and $w=1+2i$, then what is the value of $\frac{2z}{w}+\bar{w}$?

A. $2+3i$ B. $3+5i$ C. $3+2i$ D. $3-2i$

Solution: $z=-3+4i$ and $w=1+2i$

$$\frac{2z}{w}+\bar{w}=\frac{2(-3+4i)}{1+2i}+1-2i=\frac{-6+8i}{1+2i}+1-2i$$

$$=\frac{-6+8i+1-2i+2i+4}{1+2i}=\frac{-1+8i}{1+2i}$$

$$\therefore \quad \frac{2z}{w}+\bar{w}=\frac{-1+8i}{1+2i}\times\frac{1-2i}{1-2i}=\frac{-1+2i+8i+16}{5}=3+2i$$

Answer: C

18. A salesman sold items x_1, x_2 and x_3, with different rates of commissions as shown in the table below.

Months	Sales of units			Total Commission (in Birr)
	x_1	x_2	x_3	
February	90	100	20	800
March	130	50	40	900
April	60	100	30	850

What are the rates of commission on items x_1, x_2 and x_3, respectively?

A. 4, 2 and 11
B. 2, 4 and 11
C. 4, 11 and 2
D. 11, 2 and 4

Solution: From the above table $\begin{cases}90x_1+100x_2+20x_3=800\\130x_1+50x_2+40x_3=900\\60x_1+100x_2+30x_3=850\end{cases}$

Solve by using cramer's rule $D=\begin{vmatrix}9&10&2\\13&5&4\\6&10&3\end{vmatrix}=-175$

$$D_x=\begin{vmatrix}80&10&2\\90&5&4\\85&10&3\end{vmatrix}=-350, D_y=\begin{vmatrix}9&80&2\\13&90&4\\6&85&3\end{vmatrix}=-700$$

$$D_z=\begin{vmatrix}9&10&80\\13&5&90\\6&10&85\end{vmatrix}=-1925 \text{ then } x=\frac{D_x}{D}=\frac{-350}{-175}=2$$

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$$y = \frac{D_y}{D} = \frac{-700}{-175} = 4 \text{ and } z = \frac{-1925}{-175} = 11$$

The rates of common x_1, x_2 and x_3 are 2, 4 and 11 respectively.

Answer: B

19. If A is a square matrix of order 3 and det (A) = 5, then what is the value of det (A . adj (A))?

A. 3 B. 5 C. 125 D. 25

Solution: $A^{-1} = \frac{\text{adj} \cdot A}{|A|} \Rightarrow \text{adj} \cdot A = |A| A^{-1}$

$$A \cdot \text{adj} \cdot A = A \cdot |A| \cdot A^{-1} = A \cdot A^{-1} |A| = 5I$$

$$|A \cdot \text{adj} \cdot A| = |5I| = 5^3 \cdot I = 125$$

Answer: C

20. If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5$, what is $f(x)$?

A. $f(x) = e^{x-1} + 3x^2 + \frac{1}{x^2} + 2$

B. $f(x) = e^{x-1} + x^3 + \ln x + 5$

C. $f(x) = e^{x-1} + 3x^2 - \frac{1}{x} + 5$

D. $f(x) = e^{x-1} + x^3 - \ln x + 3$

Solution: If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5, f(x) =$ ____

$$f(x) = \int f'(x)dx \Rightarrow f(x) = e^{x-1} + x^3 - \ln x + c$$

$$f(1) = e^{1-1} + 1^3 - \ln 1 + C = 5 \Rightarrow C = 3$$

$$\therefore f(x) = e^{x-1} + x^3 - \ln x + 3$$

Answer: D

21. Suppose the following are the premises of an argument.

He is healthy and he is not angry.

He is angry or his plan fails.

His plan does not fail if he does not travel abroad.

Given that the premises are true, which one of the following can be a conclusion that makes the argument valid?

A. His plan fails and he is angry.

B. His plan does not fail.

C. He travels abroad.

D. His plant fails and he does not travel abroad.

Solution: First translate into symbolic form:

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Let P ≡ he is healthy and q ≡ he is angry
S ≡ his plan fails and r ≡ he travel abroad
Then the argument, is of the form: $p \wedge \neg q, q \vee s, \neg r \Rightarrow \neg s$

1. $p \wedge \neg q \equiv T$ if $p \equiv T$ and $\neg q \equiv T, q \equiv F$
2. $q \vee s \equiv T$, if $s \equiv T$, because $q \equiv F$ (from 1)
3. $\neg r \Rightarrow \neg s \equiv T$, if $\neg r \equiv F$, because $\neg s \equiv F$... (from 2)

$\therefore p \wedge \neg q, q \vee s, \neg r \Rightarrow \neg s \vdash r$ is valid **Answer: C**

22. A man running a race-course noted that the sum of the distance from the two flag posts from him is always 10 meters and the distance between the flag posts is 8 meters. What is the equation of the path traced by the man? (Take flag posts to be on the x-axis with origin at their midpoint)

A. $\frac{x^2}{9}+\frac{y^2}{25}=1$ C. $\frac{x^2}{100}+\frac{y^2}{64}=1$

B. $\frac{x^2}{25}+\frac{y^2}{9}=1$ D. $\frac{x^2}{64}+\frac{y^2}{100}=1$

Solution: Distance between two foci is 8m

$$2C = 8m \Rightarrow C = 4m$$

$$2b = 10m \Rightarrow b = \frac{10}{2}m = 5m$$

$$a = \sqrt{b^2 - c^2} = \sqrt{25-16} = \sqrt{9} = 3m$$

$\therefore$ The equation of the path traced by the man $\frac{x^2}{9}+\frac{y^2}{25}=1$

Answer: A

23. What is the value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$?

A. $e^3 - e$ B. $\frac{e^3}{3} - e$ C. $e\left(e^2 - \frac{1}{3}\right)$ D. $2(e^3 - e)$

Solution: The value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$ let $u = \sqrt{x}$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x}\,du = 2u du$$

$$2\int_1^3 e^u du = 2e^u\Big|_1^3 = 2e^3 - 2e = 2(e^3 - e)$$

Answer: D

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24. What is the value of $\int x\sqrt{1-x^2}\,dx$?

A. $\frac{-1}{3}(1-x^2)^{3/3}+C$ C. $(1-x^2)^{3/2}+C$

B. $-2x\sqrt{1-x^2}+C$ D. $\frac{1}{2}\sqrt{1-x^2}+C$

Solution: $\int x\sqrt{1-x^2}\,dx=\int\sqrt{1-x^2}\,xdx$

Let $u=1-x^2\Rightarrow du=-2xdx\Rightarrow xdx=\frac{du}{-2}$

$=\frac{-1}{2}\int u^{1/2}du=-\frac{1}{3}u^{3/2}+c$

$\therefore\int x\sqrt{1-x^2}\,dx=\frac{-1}{3}(1-x^2)^{3/2}+c$ **Answer: A**

25. Which one of the following is **NOT** true about the function $f(x)=3x^4-4x^3$?

A. (0, 0) is point of inflection of f.

B. 0 and 1 are critical number of f.

C. f is concave upward on $\left(0,\frac{2}{3}\right)$ and concave downward on $(-\infty, 0)$ and on $\left(\frac{2}{3},\infty\right)$.

D. f is decreasing on $(-\infty, 1)$ and increasing on $(1, \infty)$.

Solution: $f(x)=3x^4-4x^3$

$f'(x)=12x^3-12x^2=12x^2(x-1)=0$

$\Rightarrow x=0$ and $x=1$ are critical numbers

		0		1	
$12x^2$	− − −	0	+ + + +		+ + +
$x-1$	− −		− −	0	+ + +
$f'(x)=12x^2(x-1)$	+ + +		− − −	0	+ + +

$f'(x)>0$ on $(-\infty, 0)$ and $(1, \infty)$ is increasing

$f'(x)<0$ on $(0, 1)$ is decreasing

To test concavity

$f''(x)=36x^2-24x=12x(3x-2)=0\Rightarrow x=0$ or $x=\frac{2}{3}$

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$\therefore$(0, 0) and $\left(\frac{2}{3}, \frac{-16}{27}\right)$ are inflection points

		0			$\frac{2}{3}$	
$12x$	− − −	0	+ + + +		+ + + +	
$3x-2$	− −		− −	0	+ + + +	
$f''(x)=12x(3x-2)$	+ + +	0	− − −	0	+ + + +	

$f''(x) < 0$ on the intervals $\left(0, \frac{2}{3}\right)$ the graph is concave upward

$f''(x) > 0$ on $(-\infty, 0)$ and $\left(\frac{2}{3}, \infty\right)$ which is concave down ward

Answer: D

26. What is the value of $\lim_{x\to 0} \frac{1}{x^2} \sin^2\left(\frac{x}{2}\right)$?

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. 2 D. 4

Solution: $\lim_{x\to 0} \frac{1}{x^2} \sin^2\left(\frac{x}{2}\right)$

Let $u = \frac{x}{2} \Rightarrow x = 2u$ as $x \to 0, u \to 0$

$\lim_{u\to 0} \frac{\sin^2 u}{(2u)^2} = \frac{1}{4} \lim_{u\to 0} \frac{\sin^2 u}{u^2} = \frac{1}{4}$ **Answer: B**

27. If $f(x) = \frac{|x|}{x}$ and $g(x) = \frac{x+2}{x^3 - 4x}$ then what is the value of $\lim_{x\to -2} (f(x) + g(x))$?

A. $\frac{-9}{8}$ B. ∞ C. $\frac{9}{8}$ D. $\frac{-7}{8}$

Solution: $f(x) = \frac{|x|}{x}, g(x) = \frac{x+2}{x^3 - 4x}$

$\lim_{x\to -2} (f(x) + g(x)) = \lim_{x\to -2} f(x) + \lim_{x\to -2} g(x)$

$\lim_{x\to -2} (f(x) + g(x)) = \lim_{x\to -2} \frac{|x|}{x} + \lim_{x\to -2} \frac{x+2}{x(x+2)(x-2)}$

$\therefore \lim_{x\to -2} (f(x) + g(x)) = -1 + \frac{1}{8} = \frac{-7}{8}$ **Answer: D**

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28. Let $z=\left(\frac{1-i}{1+i}\right)^{18}$. Then what is the value of z?

A. -1 B. i C. $-i$ D. $1-i$

Solution: Let $z=\left(\frac{1+i}{1+i}\right)^{18}$

$$z=\left(\frac{1-i}{1+i}\times\frac{1-i}{1-i}\right)^{18}=\left(\frac{-2i}{2}\right)^{18}=(-i)^{18}=(-1)^{18}.((i)^2)^9=1.(-1)^9=-1$$

Answer: A

29. Let A be a 3×3 invertible matrix and B be any 3×3 matrix. If $|A|=a$ and, $|B|=b$, then which one of the following is **NOT** true?

A. $|A^TA|=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}B|=ab$

D. If $b=0$, then B is not invertible

Solution: A be non – singular 3×3 matrix and B be any matrix

If $|A|=a$ and $|B|=b$

$|A|=|A^T|=a$

A. $|A^TA|=|A^T||A|=a.a=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}||B|=\frac{1}{a}.b=\frac{b}{a}$

D. If $b=0$ then B is not invertible

Answer: C

30. The time needed to type a sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes. What are the mean $(\bar{x})$ and the standard deviation (s) of the data in minute?

A. $\bar{x}=7, s=\sqrt{1.5}$ C. $\bar{x}=8, s=\sqrt{2}$

B. $\bar{x}=7, s=\sqrt{2}$ D. $\bar{x}=8, s=\sqrt{1.5}$

Solution:

Sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes

$$\text{Mean}: \bar{x}=\frac{7+8+6+8+9+7+5+6}{8}=\frac{56}{8}=7$$

Standard deviation:

$$\delta=\sqrt{\frac{(5-7)^2+2(6-7)^2+2(7-7)^2+2(8-7)^2+1(9-7)^2}{8}}$$

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$$\delta = \sqrt{\frac{4+2+0+2+4}{8}} = \sqrt{\frac{12}{8}} = \sqrt{1.5}$$

$\therefore$ Mean $(\bar{x}) = 7$ and standard deviation $(\delta) = \sqrt{1.5}$ **Answer: A**

31. A private college has 1000 students. 60% of these students are males. 45% of these students pay their payment by credit card including 175 females. What is the probability that the student is a male or a credit card user?

A. 0.675 B. 0.225 C. 0.325 D. 0.775

Solution: Let s – students; M – male students; F – Female students; S_c – Credit card user students

n(S) = 1000

n(M) = 60% of 1000 = 600

n(F) = 40% of 1000 = 400

P(M) = 0.6, P(F) = 0.4, P (of credit card user) = 0.45

P(male or a credit card user) = 0.6 + 0.45 – 0.275 = 0.775

Answer: D

32. Which one of the following is true about the graph of

$$f(x) = \frac{2x^3 + 2x^2 + 3x}{x^2 + x}?$$

A. The vertical asymptote of the graph is only $x = -1$ and its oblique asymptote is $y = 2x$.

B. The graph has y-intercept at $(0, 3)$.

C. The graph has at least one x-intercept.

D. The vertical asymptotes of the graph are at $x = 0$ and $x = -1$ but it has not horizontal asymptote.

Solution: $f(x) = \frac{2x^3 + 2x^2 + 3x}{x^2 + x} = \frac{x(2x^2 + 2x + 3)}{x(x+1)}$

f has a hole at $x = 0$ and vertical asymptote $x = -1$ and oblique asymptote $y = 2x$

The graph has no x and y intercept. **Answer: A**

33. Let $f(x) = \frac{3x+1}{x-2}$. Then what is the range of $f(x)$?

A. $\mathbb{R}\setminus\{2\}$ B. $\mathbb{R}$ C. $\mathbb{R}\setminus\{3\}$ D. $\mathbb{R}\setminus\left(\frac{1}{3}\right)$

Solution: Let $y = \frac{3x+1}{x-2}$

$yx - 2y = 3x + 1$

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$$a_1 = \frac{6 \pm \sqrt{36 - 4\times 5}}{2} = \frac{6 \pm 4}{2} = 5 \text{ or } 1$$

$$a_3 = \frac{5}{5} = 1 \text{ or } a_3 = \frac{5}{1} = 5$$

$\therefore (a_1, a_2, a_3) = (1, 3, 5)$ or $(a_3, a_2, a_1) = (5, 3, 1)$

If $a_1 = 1, d = 2$ then $a_n = a_1 + (n-1)d$

$$a_n = 1 + (n-1)2 = 2n - 1$$

$$S_n = \frac{n}{2}(1 + 2n - 1) = n^2$$

$\therefore$ The sum of the first n – terms is $S_n = n^2$ **Answer: C**

38. Let A be a 3 × 3 matrix and $|A| = -2$. Then what is the value of $|\text{adj}(A)|$?

A. –8 B. –2 C. $-\frac{1}{2}$ D. 4

Solution:

$$A^{-1} = \frac{\text{adj} \cdot A}{|A|} \Rightarrow \text{adj} \cdot A = A^{-1}|A|$$

$$|\text{adj} \cdot A| = A\left|-2A^{-1}\right| = \frac{-2^3}{|A|} = \frac{-8}{-2} = 4$$ **Answer: D**

39. Let A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9} and C = {8, 9, 10}. If one of the numbers is deleted randomly from each of these sets, what is the probability that all the three deleted numbers are even or all are multiplies of 3?

A. $\frac{4}{5}$ B. $\frac{2}{21}$ C. $\frac{1}{9}$ D. $\frac{8}{63}$

Solution:

A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9}, C = {8, 9, 10}

A ∪ B ∪ C both even and multiplies of 3 = {6}

$$P(\text{even or multiple of 3}) = \frac{4}{10} + \frac{2}{10} + \frac{3}{10} - \frac{1}{10} = \frac{8}{10} = \frac{4}{5}$$

Answer: A

40. The age distribution of students in a certain class is given below:

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Age	10 – 14	15 – 19	20 – 24	25 – 29
No. of students	2	10	6	7

What is the modal value of the distribution?

A. 17.83 B. 17.38 C. 18.37 D. 18.73

Solution:

Age (Class boundaries)	9.5–14.5	14.5–19.5	19.5–24.5	24.5–29.5
No. of students	2	10	6	7

Modal class 14.5 – 19.5

$d_1 = 10 - 2 = 8$ and $d_2 = 10 - 6 = 4$ and $i = 5$

then $M_o = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$

$$M_o = 14.5 + \left(\frac{8}{3+4}\right)5 \Rightarrow M_o = 14.5 + \left(\frac{8}{3+4}\right)5$$

$$M_o = 14.5 + \frac{40}{12} = 17.83$$

Answer: A

41. If $f(x) = k \ln x + e^{\sin x}$ and $f''(\pi) = -1$, then what is the value of k?

A. $2\pi^2$ B. π^2 C. π D. 2π

Solution: $f(x) = k \ln x + e^{\sin x}$, $f''(\pi) = -1$ then $k =$ _____

$f'(x) = \frac{k}{x} + \cos x e^{\sin x}$ and let $u = e^{\sin x}$; $dv = \cos x\, dx$, $v = \sin x$

$f''(x) = \frac{-k}{x^2} - \sin x e^{\sin x} + \cos^2 x e^{\sin x}$

$f''(\pi) = \frac{-k}{\pi^2} - 0 + 1 = -1 \Rightarrow \frac{-k}{\pi^2} = -2 \Rightarrow k = 2\pi^2$ **Answer: A**

42. Let $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$. Then what is $g\left(f\left(\frac{1}{x}\right)\right)$ equals to

A. $x - x^2$ B. $\frac{x^2}{x-1}$ C. $\frac{1}{x^2 - x}$ D. $\frac{x-1}{x^2}$

Solution: $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$ then

$$g\left(f\left(\frac{1}{x}\right)\right) = g\left(\frac{1}{x} - \frac{1}{x^2}\right) = g\left(\frac{x-1}{x^2}\right) = \frac{1}{\frac{x-1}{x^2}} = \frac{x^2}{x-1}$$

Answer: B

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43. A cylindrical tank whose inner diameter is 2m contains $4\pi\ m^3$ oil. If the oil is discharged from the tank at the rate of $\frac{2\pi}{3} m^3/min$, then how long (in min) does it take for the tank to be empty?

A. $\frac{4}{3}$ B. 4 C. 12 D. 6

Solution: $V = \pi r^2 h$, radius: $r = \frac{D}{2} = \frac{2}{2} = 1$

$$\frac{dv}{dt} = \pi r^2 \frac{dh}{dt}$$

$$\frac{2\pi}{3} = \pi(1)\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{3} \Rightarrow dh = \frac{2}{3}dt$$

$$\int dh = \frac{2}{3}dt \Rightarrow h = \frac{2}{3}t \text{ and } h = \frac{4\pi}{\pi(1)} = 4m$$

$$\therefore t = \frac{3}{2}h = \frac{3}{2}(4) = 6 \text{ minute}$$ **Answer: D**

44. What are the values of a and b so that the function

$$f(x) = \begin{cases} x+1, x<1 \\ ax+b, 1 \le x < 2 \\ 3x, x \ge 2 \end{cases}$$ is continuous everywhere?

A. $a = 4, b = -2$ C. $a = 4, b = 2$

B. $a = -4, b = -2$ D. $a = -4, b = 2$

Solution: f is continuous everywhere so that, limit exists.

✓ $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x)$ and $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^-} f(x)$

$\Rightarrow \lim_{x\to 1^+}(ax+b) = \lim_{x\to 1^-}(x+1)$ and $\lim_{x\to 2^+} 3x = \lim_{x\to 2^-}(ax+b)$

$\Rightarrow a + b = 2$ and $6 = 2a + b$

Solving together: $\begin{cases} a+b=2 \\ 2a+b=6 \end{cases}$ (−)

$-a = -4$

$\therefore a = 4$ and $b = 2 - a = -2$ **Answer: A**

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45. Let U = ℕ (the set of natural numbers) be the universe. Which one of the following proposition is true?

A. $(\exists x)(x+x=x)$ C. $(\forall x)(\exists y)(x \div y = y \div x)$

B. $(\forall x)(\exists y)(y<x)$ D. $(\forall x)(\exists y)(x-y=x)$

Solution: u = the set of natural numbers

A. $(\exists x)(x+x=x)$ is false, because there is no natural number x, which is $x+x=x$

B. $(\forall x)(\exists y)(y<x)$. For every natural number x, there is a natural number y such that $y<x$ is false. For example, let x=1 and y<1.

There is no natural number less than 1.

C. $(\forall x)(\exists y)(x \div y = y \div x)$ is true

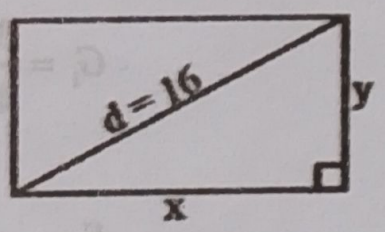
For every natural number x, there is a natural number y such that $x \div y = y \div x$ is True. For example, if $x=1, y=1$; if $x=3, y=3$, and so on.

D. For every natural number x, there is a natural number y such that $x-y=x$ is false. **Answer: C**

46. What is the maximum possible area of a rectangle in square units with diagonal of length 16 units?

A. 48 B. 128 C. 64 D. 256

Solution: Area = xy



- $x^2+y^2=16^2=256 \Rightarrow y=\sqrt{256-x^2}$
- Area: $A(x)=x\sqrt{256-x^2}=\sqrt{256x^2-x^4}$

Let $g(x)=256x^2-x^4$ and $g'(x)=512x-4x^3$

$g'(x)=0 \Rightarrow 512x-4x^3=0 \Leftrightarrow 4x(128-x^2)=0$

$\therefore x=0$ or $x=\sqrt{128}=8\sqrt{2}$

$\therefore$ The maximum area $=\left(8\sqrt{2}\right)\left(8\sqrt{2}\right)=128$ **Answer: B**

47. Let $f(x)=\ln\left(x\sqrt{x}\right)$. Then what is $f'(x)$ equal to?

A. $\frac{2x}{3}$ B. $\frac{\sqrt{x}}{2}$ C. $\frac{2}{x\sqrt{x}}$ D. $\frac{3}{2x}$

Solution: $f(x)=\ln\left(x\sqrt{x}\right)=\ln\sqrt{x^3}=\ln x^{3/2}$

$\Rightarrow f(x)=\frac{3}{2}\ln x$ and $f'(x)=\frac{3}{2x}$ **Answer: D**

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48. Which one of the following is equal to $\frac{d}{dx}\log_2^{\sqrt{6x}}$?

A. $\frac{1}{2x\ln(2)}$ C. $\frac{1}{6x\ln(2)}$

B. $\frac{3}{2x\ln(2)}$ D. $\frac{1}{6x\ln(2)}$

Solution: $\log_2\sqrt{6x} = \frac{\ln\sqrt{6x}}{\ln 2} = \frac{\ln(6x)^{1/2}}{\ln 2} = \frac{\ln 6x}{2\ln 2}$

$$\frac{d}{dx}\log_2\sqrt{6x} = \frac{d}{dx}\left(\frac{\ln 6x}{2\ln 2}\right) = \frac{6}{(6x)2\ln 2} = \frac{6}{12x\ln 2} = \frac{1}{2x\ln 2}$$

Answer: A

49. What is the sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n}$?

A. ∞ B. $\frac{3}{16}$ C. 3 D. 4

Solution: The sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n} = \frac{3}{4} + \frac{9}{16} + \ldots$

$$G_1 = \frac{3}{4}, r = \frac{3}{4}$$

$$S_{n\to\infty} = \frac{G_1}{1-r} = \frac{3/4}{1-3/4} = \frac{\frac{3}{4}}{1/4} = 3$$

Answer: C

50. If $z = (1+\sqrt{3}i)(1+i)$, then which one of the following is the polar representation of z?

A. $z = 4(\cos(105°) + \text{i}\sin(105°))$

B. $z = 2\sqrt{2}(\cos(105°) + \text{i}\sin(105°))$

C. $z = 2\sqrt{2}(\cos(15°) + i\sin(15°))$

D. $z = 4((\cos 75°) + \text{i}\sin(75°))$

Solution: Let $z_1 = 1+\sqrt{3}i$ and $z_2 = 1 + \text{i}$

✓ Let $\theta_1 = Arg(z_1) = Arg\left(1+\sqrt{3}i\right) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60°$

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✓ Let $\theta_2 = Arg(z_2) = Arg(1+i) = \tan^{-1}\left(\frac{1}{1}\right) = 45°$

✓ $|z_1| = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{4} = 2$ and $|z_2| = \sqrt{1^2 + 1^2} = \sqrt{2}$

✓ Polar form of $z = z_1 . z_2 = r_1.r_2 (\cos(\theta_1 + \theta_2) + i(\sin(\theta_1 + \theta_2))$

$\therefore z = 2\sqrt{2}[\cos(60° + 45°) + i \sin(60° + 45°)]$

$= 2\sqrt{2}(\cos 105° + i \sin 105°)$ **Answer: B**

51. The variance of 20 observations is 5. If each observation is multiplied by 2, then what is the variance of the resulting observations?

A. 5 B. 10 C. 20 D. 40

Solution: If a data multiplying by constant k:

New variance = k^2 (old variance)

$\therefore$ New variance = $(2)^2(5) = (4)(5) = 20$ **Answer: C**

52. What is the value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$?

A. 4 B. 14 C. 10 D. 18

Solution: $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$

$\lim_{n\to\infty} \frac{16}{n^4}\sum_{k=1}^{n} k^3 + \frac{20}{n^2}\sum_{k=1}^{n} k$

$\lim_{n\to\infty} \frac{16}{n^4}\left(\frac{n^2(n+1)^2}{4}\right) + \lim_{n\to\infty} \frac{20(n(n+1))}{2n^2}$

$\frac{16}{4}\lim_{n\to\infty} \frac{n^4 + 2n^3 + n^2}{n^4} + \frac{20}{2}\lim_{n\to\infty} \frac{n^2 + n}{n^2} = 4 + 10 = 14$ **Answer: B**

53. The grouped frequency distribution of data is given in the following table

Class interval	8 -12	13-17	18-22	23-27	28-32
Frequency	4	8	10	5	3

What is the median (m) and mean deviation (MD) about the median?

A. m = 19, MD = 4.2 B. M = 19, MD = 4.7

C. m = 20, MD = 4.7 D. D.M = 20, MD = 4.2

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Solution:

Median class $(m) = \left(\frac{30}{2}\right)^{th}$ item $= 15^{th}$ found in the 3^{rd} class

$$B_L = 17.5 \qquad m = B_L + \frac{\left(\frac{n}{2} - Cf_b\right)}{f_c}$$

$$f_c = 10 \qquad m = 17.5 + \frac{(15-12)5}{10} = 17.5 + \frac{15}{10}$$

$$Cf_b = 12 \qquad m = 17.5 + 1.5 = 19$$

$$i = 5$$

Mean deviation about median is

$$MD = \frac{f_1|xm_1 - M| + f_2|xm_2 - m| + f_3|xm_3 - m| + f_4|xm_4 - m| + f_5|xm_5 - m|}{f_1 + f_2 + f_3 + f_4 + f_5}$$

$$MD = \frac{4|10-19| + 8|15-19| + 10|20-19| + 5|15-19| + 3|30-19|}{4+8+10+5+3}$$

$$MD = \frac{4\times 9 + 8\times 4 + 10\times 1 + 5\times 6 + 3\times 11}{30}$$

$$MD = \frac{36+32+10+30+33}{30} = \frac{141}{30} = 4.7$$

∴ Median = 19 and Medivation of median = 4.7 **Answer: B**

54. What is the maximum value of $z = 3x - 2y$ Subject to: $-x + y \le 0$

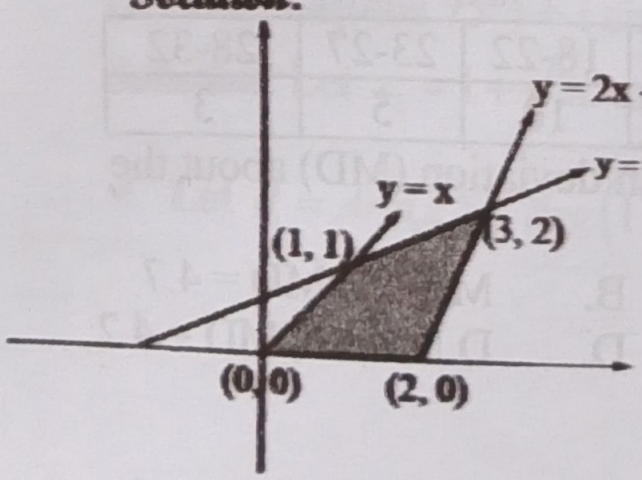
$$-x + 2y \le 1$$

$$2x - y \le 4$$

$$x \ge 0, y \ge 0$$

A. 6 B. 5 C. 8 D. 10

Solution:



Corner Parts	The value of $z = 3x - 2y$
(0, 0)	0
(2, 0)	6
(1, 1)	1
(3, 2)	5

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The maximum value is 6 **Answer: A**

55. A person is planning to make a regular monthly saving over the next five years in an account that pay no interest. If his monthly income is Birr 4000, then what percentage of his income should he save monthly so that his total saving will be birr 26,400 by the end of 5 years?

A. 9% B. 11% C. 10% D. 12%

Solution: 5 years = 60 months let r = be rate of deposition per month r of 4000 of 60 = 26, 400

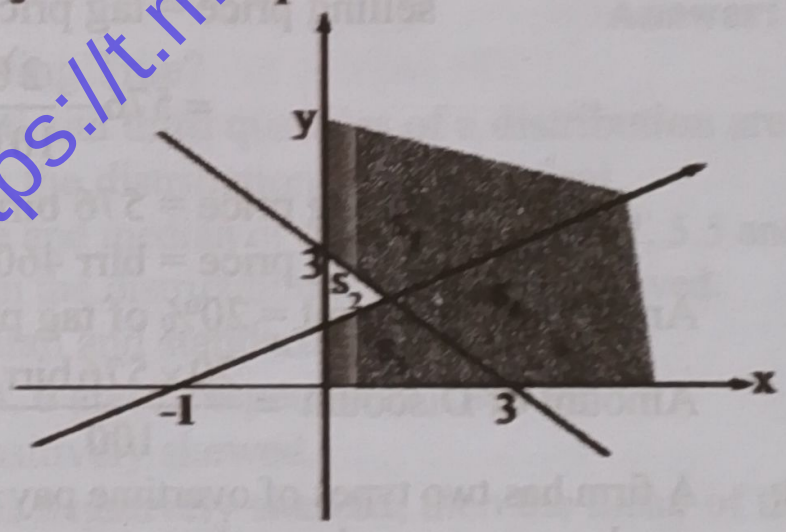
$$240,000\,r = 26,4000 \Rightarrow r = \frac{26,400}{240,000} = 0.11$$

$$r = 0.11 \times 100\% = 11\%$$

or $r = \frac{26,400}{240,000} \times 100\% = 11\%$ means the deposited 11% of 4000 per month = birr 440 **Answer: B**

56. Which are the regions shaded in the figure below is the solution region of the following system of inequalities?

$$\begin{cases} x + y \le 3 \\ -x + 3y \ge 1 \\ x \ge 0,\ y \ge 0 \end{cases}$$



A. S_2 B. S_1 C. S_3 D. S_4

Solution:

The solution set of $\begin{cases} x + y \le 3 \\ -x + 3y \ge 1 \\ x \ge 0, y \ge 0 \end{cases}$ is the intersection of the region

$x + y \le 3, -x + 3y \le 1$ and $x \ge 0, y \ge 0$ which is S_2

Answer: A

57. Birr 1,000 deposited in saving account that pays 6% annual interest compounded monthly. Which one of the following is the amount (in Birr) that will be in the account by the end of 3years.

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A. $1000 + 1000\,(1.005)^{36}$ C. $1000 \times (1.005)^{36}$
B. $1000 + 1000\,(1.05)^{3}$ D. $1000 \times (1.05)^{36}$

Solution: p = 1000 birr $A = P\left(1+\frac{r}{m}\right)^{mt}$

$r = 6\%$ $A = 1000\left(1+\frac{6\%}{12}\right)^{12(3)}$

$m = 12$ $A = 1000\left(1+\frac{1}{200}\right)^{36}$

$A = 1000\,(1.005)^{36}$ **Answer: C**

58. A jacket discounted by 20% for holiday has a price tag of Birr 576. What is the amount of discount?
A. Birr 154 C. Birr 135.6
B. Birr 115.2 D. Birr 144

Solution: Marked (bag) price = birr 576
rate of dis count = 20%
selling price = tag price – 20% of tag price

$$= 576 - \frac{20}{100} \times 576$$

selling price = 576 birr – 115.2 birr
selling price = birr 460.80
Amount of Discount = 20% of tag price

$$\text{Amount of Discount} = \frac{20 \times 576\,\text{birr}}{100} = 115.2\,\text{birr}$$ **Answer: B**

59. A firm has two types of overtime pay rates (payment per hour) for its employees: normal overtime pay rate and holiday over time rate (for Sundays and holidays) the holiday overtime pay rate is 1.5 times the normal over time pay rate; and the normal over time pay rate is 1.5 times the regular pay rate. The regular pay rate for an employee is given by his weekly basic wage divided by 40. If an employee whose weekly basic wage is birr 2400 has additionally, 6 hours of normal overtime and 3 hours of holiday overtime in a week, how much (in Birr) is his payment for the week?
A. 3250 B. 3300 C. 3525 D. 3345

Solution: Holiday over time rate = $\frac{3}{2}$ of normal overtime rate

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$$\text{normal over time rate} = \frac{3}{2} \text{ of regular pay rate}$$

$$\text{normal wage per hour} = \frac{2400}{40} = 60 \text{ birr per hour per week}$$

$$\text{Holiday overtime rate} = \frac{3}{2} \text{ of normal overtime rate}$$

$$\text{normal over time rate} = \frac{3}{2} \times 60 \times 6 = 540 \text{ birr}$$

$$\text{Holiday overtime per} = \frac{3}{2} \times 90 \times 3 = 405 \text{ birr per week}$$

Total wages per week = (regular pay rate) + (normal overtime rate) + (holiday over time rate)

$$\text{Total wages per week} = \frac{2400\text{birr}}{\text{week}} + \frac{540\text{birr}}{\text{week}} + \frac{405\text{birr}}{\text{week}}$$

$$\text{Total wages per week} = \frac{3345\text{birr}}{\text{week}} = 3345 \text{ birr per week}$$

Answer: D

60. Which one of the following True?

A. If the first, second and third quartiles of a distribution are 7, 10 and 13 then the distribution is symmetrical.

B. If the mean, mode and median of a distribution are 7, 5.5 and 6 respectively then the distribution is negatively skewed.

C. If the mean, median and standard deviation of a distribution are 4, 6 and 2 respectively, then the distribution is positively skewed.

D. If a distribution is negatively skewed; then the mean of the distribution is greater than its second quartile.

Solution: A. $Q_1 = 7$, Q_2 = median = 10 $Q_3 = 13$

By Bow ley's coefficient

$$B = \frac{Q_3 + Q_1 - 2(\text{median})}{Q_3 - Q_1}$$

$$= \frac{13 + 7 - 2(10)}{13 - 7} = \frac{20 - 20}{3} = 0$$

then the distribution is symmetrical.

B. $\bar{x} = 7$ $m_o = 5.5$ and $m_d = 6$

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$\bar{x} > m_d > m_o$ then the distributions is positively skewed

C. $\bar{x} = 4, m_d = 6, \quad 5 = 2$

then afficient of skiwness $(x) = \dfrac{3(\bar{x} - md)}{5}$

$x = 3\dfrac{(4-6)}{2} = -3 < 0$ the distributions

Negatively skewed $Q_2 = md$ then $\bar{x} < Q_2$ the distribution is positively skewed

Answer: A

61. A hospital wants to buy three x – ray machines from a supplier suppose the price of each machine including 15% VAT is birr 161,000. If the hospital wants to subtract a 2% with holding tax before VAT. What is the amount the hospital should pay (in birr) to supplier after with holding 2% is subtracted?

A. 473,340 B. 475,340 C. 474,600 D. 483,340

Solution: one x –ray machine coses birr 165,000 in chiding vats

2% holding tax before VAT

for one machine = 161, 000 – 2% of 161, 000

$= 161{,}0000 - \dfrac{-2}{100} \times 161{,}000 = 161{,}000 - 3220 =$ birr 157,780

Hence 3 x – ray machine costs = 3×157, 780 = birr 473,340

Answer: A

62. A firm deals with two kinds of fruit juices pine apple and orange juice. These are mixed and two types of mixtures are obtained which are sold as soft drinks A and B. One tin of A needs 4 litters pine apple juice and 1 litters of orange juice. One tin of B needs 2 liters of pine apple and 3 liters of orange juice. The firm has only 46 liters of pine apple juice and 24 liters of orange juice. Each tin of A and B is sold at a profit of 4 birr and 3birr, respectively. How many tins A and B should the firm produce to maximize its profit?

A. 6 tins of A and 9 tin of B C. 5 tin of A and 9 tin of B

B. 9 tins of A and 5 tin of B D. 9 tin of A and 6 tin of B

Solution: $Z = 4A + 3B$

Subject to: $4A + 2B = 46$ 1

$A + 3B = 24$ 2

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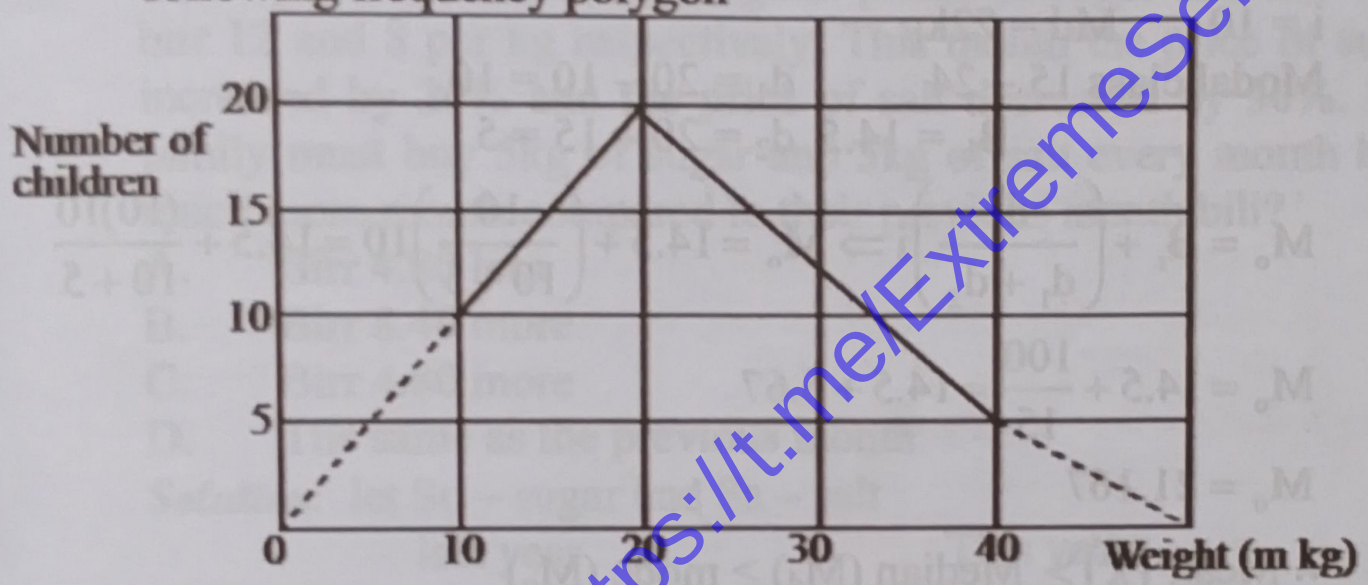
$$\begin{cases} 4A + 2B = 46 \\ 4A + 12B = 96 \end{cases}$$

$$-10B = -50$$

$$B = 5 \quad \text{and} \quad A = 9$$

∴ 9 tins of A and 5 tins of B used to maximize its profit **Answer: B**

63. The weights (in kg) of children in a certain nursary are grouped into four class intervals of equal width and represented by the following frequency polygon



Which one of the following is true about the data?

A. The distribution of the weight is positively skewed
B. The mode of the weights is 20 kg
C. The median weight is 25 kg
D. The mean weight is 25 kg

Solution:

For the above frequency polygon frequency distribution table is

Weight of children	5 – 14	15 – 24	25 – 34	35 – 44
Number of children	10	20	15	5

$$\text{Mean } (\bar{x}) = \frac{f_1 \times m_1 + f_2 \times m_2 + f_3 \times m_3 + f_4 \times m_4}{f_1 + f_2 + f_3 + f_4} = \frac{\sum_{i=1}^{4} f_i x m_i}{\sum_{i=1}^{4} f_i}$$

where x_m: class mark of each class

$$\bar{x} = \frac{10 \times 10 + 20 \times 20 + 15 \times 30 + 5 \times 40}{10 + 20 + 15 + 5} = \frac{1150}{50} = 23\text{kg}$$

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$$\text{Median} = \left(\frac{50}{2}\right)^{th} = 25^{th} \text{ lies in the second class}$$

$$B_L = 14.5 \qquad M_d = B_L + \frac{\left(\frac{n}{2} - f_b\right)}{fc} i$$

$$fc = 20$$

$$c\,fb = 100 \qquad M_d = 14.5 + \frac{(25-10)}{20} 10 = 14.5 + 7.5$$

$$i = 10 \qquad Md = 22kg$$

Modal class $15 - 24$ $\quad d_1 = 20 - 10 = 10$

$B_L = 14.5 \quad d_2 = 20 - 15 = 5$

$$M_o = B_r + \left(\frac{d_1}{d_1 + d_2}\right) i \Rightarrow M_o = 14.5 + \left(\frac{10}{10+5}\right) 10 = 14.5 + \frac{(10)10}{10+5}$$

$$M_o = 14.5 + \frac{100}{15} = 14.5 + 6.67$$

$$M_o = 21.167$$

$\therefore$ Mean $(\bar{x})$ > Median (M_d) > mode (M_o)

Hence the distribution positively skewed **Answer: A**

64. A man bought one bed room house for birr 250,000 and paid a down payment of 20%. The remaining is a mortage to be paid monthly for 15 years with 6% annual interest. What is the monthly payment in Birr?

Given $(1.005)^{-15} = 0.93$, $(1.005)^{-180} = 0.41$ $(1.006)^{-15} = 0.95$, $(1.006)^{-180} = 0.34$]

A. 1333 B. 1428 C. 1818 D. 1694

Solution: Mortage = Purchage price – down payment

Mortage = 250,000 – 20% of 250,000

$$\text{Mortage} = 750,000 - \frac{20 \times 250,000}{100}$$

Mortage = 250,000 – 50,000 = birr 200,000

Monthly payment mortage evaluated by amortization formula

$$\text{Periodic payment: } P \cdot P = P\left(\frac{\bar{i}}{1-(1+i)^{-n}}\right) \qquad i = \frac{6\%}{12} = \frac{1}{2}\%$$

$$i = 0.005$$

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n = number of payment
n = 15×12 = 180 month

$$P \cdot P = 200,000\left(\frac{0.005}{1-(1.005)^{-180}}\right)$$

$$P \cdot P = 200,000\left(\frac{0.005}{1-0.41}\right) = 200,000\left(\frac{0.005}{0.59}\right)$$

P.P = birr 1694.92

Thus, the monthly payment is birr 1694 **Answer: D**

65. In the previous month, the regular prices of sugar and salt were birr 12 and 8 per kg respectively. This month the price of sugar increased by 20% and the price of salt decreased by 30%. If a family must buy 5kg of sugar and 3kg of salt every month how much more month compared to their previous month bill?

A. Birr 4.80 less
B. Birr 8.40 more
C. Birr 4.80 more
D. The same as the previous month

Solution: let Su – sugar and Sa – salt

last year	This year
Su ⟶ 12birr/kg	Su = 12 birr + 20% of 12
1 kg Su = 12	1 kg = 14.4 birr/kg
5 kg Su = ? → 5kg of Su = 60 birr	5 kg = ?
	5kg = 72 birr

Salt (Sa)

Last year	This year
1kg = 8 birr	sal = 8birr – 30% of 8
3 kg =	1kg salt = 5.6 birr
⇒ 3kg = 24 birr	3kg = birr 16.8
Last year total = 60birr + 24birr = 84 birr	This year = 72birr+16.8 birr = 88.8 birr

This year compared to last year = 88. 8 – 84 = 4.8 birr more

Answer: C

66. The time needed to type a sample of 8 business letters in an office is 7,8,6,8,9,7,5,6 minutes. What are the mean $(\bar{x})$ and the standard deviation (S) of the data in minute?

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A. $\bar{x}=7,\ S=\sqrt{2}$ C. $\bar{x}=8,\ S=\sqrt{8}$

B. $\bar{x}=8,\ S=\sqrt{1.5}$ D. $\bar{x}=7,\ S=\sqrt{1.5}$

Solution:

t (minute)	5	6	7	8	9
f	1	2	2	2	1

$$\text{Mean } (\bar{x})=\frac{f_1x_1+f_2x_2+f_3x_3+f_4x_4+f_5x_5}{f_1+f_2+f_3+f_4+f_5}$$

$$\bar{x}=\frac{1x5+2x6+2x7+2x8+1x9}{1+2+2+2+1}=\frac{5+12+14+16+9}{8}$$

$$\bar{x}=\frac{56}{8}=7$$

Standard deviation

$$(S)=\sqrt{\frac{f_1(x_1-\bar{x})^2+f_2(x_2-\bar{x})^2+f_3(x_3-\bar{x})^2+f_4(x_4-\bar{x})^2+f_5(x_5-\bar{x})^2}{f_1+f_2+f_3+f_4+f_5}}$$

$$S=\sqrt{\frac{1(5-7)^2+2(6-7)^2+2(7-7)^2+2(8-7)^2+1(9-7)^2}{1+2+2+2+1}}$$

$$S=\sqrt{\frac{4+2+0+2+4}{8}}$$

$$S=\sqrt{\frac{2}{8}}=\sqrt{\frac{3}{2}}=\sqrt{1\cdot5}$$

$\therefore \bar{x}=7$ and $S=\sqrt{1\cdot5}$

Answer: D

67. The following histogram presents the number of HIV positive persons among five social groups coded as SG_1, SG_2, SG_3, SG_4, and SG_5 in certain town

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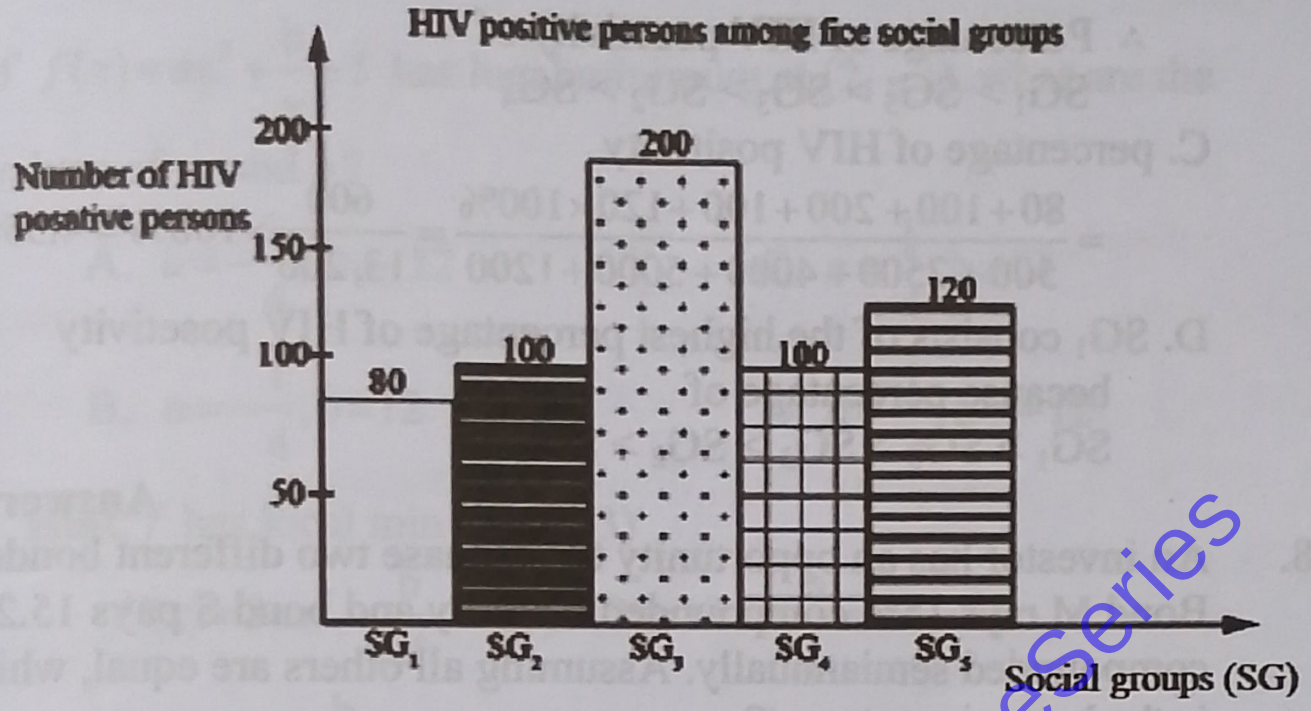

Moreover, the number of persons in each of these social groups SG_1, SG_2, SG_3, SG_4 and SG_5 are 500, 2500, 4000, 5000 and 1200 respectively. If so which one of the following can be deduced from the data?

A. 3% of those in SG_4 are HIV positive

B. SG_3 consists of the highest percentage of HIV positive persons

C. 7.4% of the entire population of the five social groups are HIV positive

D. SG_1 consists of the highest percentage of HIV positive persons followed by SG_5

Solution: A. $SG_4 = \frac{\text{number of HIV posetive}}{SG_4} \times 100\% = \frac{100}{5000} \times 100\%$

B. $SG4 = 2\% \neq 3\ \%$

$$SG_3 = \frac{\text{number of HIV posetive}}{SG_3} \times 100\% = \frac{200}{4000} \times 100\% = 5\%$$

$$SG_1 = \frac{80}{500} \times 100\% = 16\%$$

$$SG_2 = \frac{100}{2500} \times 100\% = 4\%$$

$$SG_5 = \frac{120}{1200} \times 100\% = 50\%$$

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∴ Percentage of HIV posetivity of
$SG_1 > SG_5 > SG_3 > SG_2 > SG_4$

C. percentage of HIV positivity

$$= \frac{80+100+200+100+120 \times 100\%}{500+2500+4000+5000+1200} = \frac{600}{13,200} \times 100\% = 4.55\%$$

D. SG_1 consists of the highest percentage of HIV posetivity because percentage of
$SG_1 > SG_5 > SG_3 > SG_2 > SG_4$

Answer: D

68. An investor has an opportunity to purchase two different bonds. Bond M rays 15% compounded monthly and bond S pays 15.2% compounded semiannually. Assuming all others are equal, which is the better investment?
Suppose that $[(1.0125)^{12} = 1.160755, (1.076)^{12} = 2.408503$
$(1.0425)^2 = 1.025156$ and $(1.076)^2 = 1.157776]$

A. Bond S is by far better than bond M
B. Bond S is lightly better than bond M
C. Bond M is the preferred investment than bond S
D. Both of them have the same value at the end of the year

Solution: let A_m – is an amount of bond M
A_s – is an amount of bonds

$$A_m = P\left(1+\frac{15}{12}\%\right)^{12} \Rightarrow A_m = P(1.0125)^{12} = 1.160755P$$

$$A_s = P\left(1+\frac{15.2\%}{2}\right)^{2} \Rightarrow A_s = P(1.076)^2 = 1.157776P$$

but $A_m \neq A_s$
$\Rightarrow$ $1.160755P \neq 1.157776\,P$ from this bond M is the preferred investment than bonds S

Answer: C

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1. If $f(x)=ax^3+\frac{b}{x}+5$ has local minimum at (2, - 3), what are the values of a and b?

A. $a=-\frac{1}{4},b=-12$ C. $a=\frac{1}{4},b=12$

B. $a=-\frac{1}{4},b=12$ D. $a=\frac{1}{4},b=-12$

Hint: f has local min . at (2,-3)

$f'(x)=3ax^2-\frac{b}{x^2}$ and $f'(2)=12a-\frac{b}{4}=0$

$f(2)=8a+\frac{b}{2}+5=-3$ and $16a+b=16$

Solving 48 a = b and 16 a + b = -16

$\Rightarrow$ 16a + 48a = -16 $\therefore$ $= a=-\frac{1}{4}$ and $b=-12$

Answer:- A

2. What is the simplified form of $\frac{a^{-1}b^{-1}}{a^{-3}-b^{-3}}$?

A. $\frac{a^2b^2}{b^2-a^2}$ B. $\frac{a^3-b^3}{ab}$ C. $\frac{a^2b^2}{b^3-a^3}$ D. $\frac{a^3-b^3}{a-b}$

Hint: Multiply the numerator and denominator by a^3b^3.

$\Rightarrow \frac{a^{-1}.b^{-1}}{a^{-3}-b^{-3}}\left(\frac{a^3.b^3}{a^3.b^3}\right)=\frac{a^2b^2}{b^3-a^3}$.

Answer: C

3. Consider the following system of equations: $\begin{cases} x-2y+z=1 \\ -x+y+z=3 \\ 3x-5y+z=k \end{cases}$ How much must be the value of k so that the system has a solution?

A. – 1 B. 1 C. 0 D. 7

Solution: Determinant of coefficient matrix is D = 0.

Since the system has solution, So that $D_x = D_y = D_z = 0$.

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$$\Rightarrow D_x=\begin{vmatrix}1 & -2 & 1\\ 3 & 1 & 1\\ k & -5 & 1\end{vmatrix}=k\begin{vmatrix}-2 & 1\\ 1 & 1\end{vmatrix}+5\begin{vmatrix}1 & 1\\ 3 & 1\end{vmatrix}+1\begin{vmatrix}1 & -2\\ 3 & 1\end{vmatrix}=0$$

$$\Rightarrow K(-2-1)+5(1-3)+1(1+6)=0\Leftrightarrow -3K-3=0$$

$$\therefore K=-1$$

Answer: A

4. If $\begin{vmatrix}-1 & 1 & 2\\ 3 & 2 & x\\ 2 & 4 & 1\end{vmatrix}=\begin{vmatrix}-x & 3 & 2\\ 2 & 2 & 3\\ 1 & -1 & -2\end{vmatrix}$, then what is the value of x?

A. $-\frac{2}{3}$ B. $-\frac{2}{5}$ C. $\frac{2}{3}$ D. $\frac{2}{5}$

Solution:

$$\begin{vmatrix}-1 & 1 & 2\\ 3 & 2 & x\\ 2 & 4 & 1\end{vmatrix}=\begin{vmatrix}-x & 3 & 2\\ 2 & 2 & 3\\ 1 & -1 & -2\end{vmatrix}$$

$$-1\begin{vmatrix}2 & x\\ 4 & 1\end{vmatrix}-1\begin{vmatrix}3 & x\\ 2 & 1\end{vmatrix}+2\begin{vmatrix}3 & 2\\ 2 & 4\end{vmatrix}=-x\begin{vmatrix}2 & 3\\ -1 & -2\end{vmatrix}-3\begin{vmatrix}2 & 3\\ 1 & -2\end{vmatrix}+2\begin{vmatrix}2 & 2\\ 1 & -1\end{vmatrix}$$

$$-1(2-4x)-1(3-2x)+2(12-4)=-x(-4+3)-3(-4-3)+2(-2-$$

$$4x-2-3+2x+16=x+21-8$$

$$6x+11=x+13\Rightarrow 5x=2\Rightarrow x=2/5$$

Answer:-D

5. What is the value of the left-hand side limit, $\lim_{x\to 0^-}\frac{\sin x}{x^2}\sqrt{4x^3+9x^2}$?

A. 3 B. -3 C. 2 D. Does not exist

Hint: $\lim_{x\to 0^-}\frac{\sin x}{x^2}\sqrt{4x^3+9x^2}=\lim_{x\to 0^-}\frac{\sin x}{x^2}\sqrt{x^2(4x+9)}$

$$\Rightarrow \lim_{x\to 0^-}\frac{\sin x.|x|}{x^2}.\sqrt{4x+9}=\lim_{x\to 0^-}\frac{\sin x}{-x}.\lim_{x-10^-}\sqrt{(4x+9)}$$

$$\Rightarrow (-1)\lim_{x\to 0^-}\frac{\sin x}{x}.\sqrt{\lim_{x\to 0^-}(4x+9)}=(-1)(1)\sqrt{9}=-3$$

Answer: B

6. Which one of the following is equal $f(x)=\sqrt{(x+4)^2}$, for every $x\in\Re$?

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A. $g(x)=x+4$ C. $g(x)=x+2$

B. $g(x)=|x+4|$ D. $g(x)=|x|+4$

Hint: $f(x)=\sqrt{(x+4)^2}=|x+4|$, for all $x\in\mathbb{R}$.

Answer: B

7. Let z be a complex number and $w=3+4i$. if $\frac{z^2+1}{z+i}=|w|z-\frac{1-}{i}w$, then what is the value of z?

A. $-1-i$ B. $4-2i$ C. $-1+i$ D. $-4-2i$

Hint: We know that $Z^2+1=(Z+i)(Z-i)$

and $|W|=|3+4i|=\sqrt{3^2+4^2}=\sqrt{25}=5$

$\Rightarrow \frac{(Z+i)(Z-i)}{Z+i}=5Z-\frac{1}{i}\cdot\frac{i}{i}(3-4i)=5Z+3i+4$

$\Rightarrow Z-i=5Z+3i+4 \Leftrightarrow -4Z=4i+4$

$\therefore Z=-i-1$

Answer: A

8. If the circle passing through the point (-1,0) touches the y-axis at (0,2), then what is the equation of the circle?

A. $x^2+y^2+5x+4y+4=0$ C. $x^2+y^2-5x-4y+4=0$

B. $x^2+y^2+5x-4y+4=0$ D. $x^2+y^2-5x+4y+4=0$

Hint: Let the center = (h,K) and the circle touches the y-axis (0,2),

So that the center is (h,2)

$\Rightarrow (h+1)^2+(2-0)^2=(h-0)^2+(2-2)^2$

$\Rightarrow (h+1)^2+4=h^2$ and $h^2+2h+1+4=h^2$

$\Rightarrow 2h=-5$ and $h=\frac{-5}{2}$

Equation of circle: $\left(x+\frac{5}{2}\right)^2+(y-2)^2=h^2=\frac{25}{4}$

$\Rightarrow x^2+5x+\frac{25}{4}+y^2-4y+4=\frac{25}{4}$

$\therefore x^2+y^2+5x-4y+4=0.$

Answer: B

9. What is the value of $\int_1^2 \frac{\ln x}{x^2}dx$?

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A. $-\frac{\ln 2}{2}-\frac{1}{2}$

B. $\frac{\ln 2}{2}-\frac{1}{2}$

C. $\frac{\ln 2}{2}+\frac{1}{2}$

D. $-\frac{\ln 2}{2}+\frac{1}{2}$

Hint: We use integration by parts $\int u\,dv = uv - \int v\,du$

Let $u = \ell nx$ and $dv = \frac{1}{x^2}dx$.

$du = \frac{1}{x}dx$ and $v = \frac{-1}{x}$

$$\therefore \int_1^2 \frac{\ell nx}{x^2}dx = -\frac{\ell nx}{x} + \int \frac{dx}{x^2} = -\frac{\ell nx}{x} - \frac{1}{x}\Big|_1^2 = -\frac{\ell n2}{2} + \frac{1}{2}$$

Answer:- D

10. Everyday a person saves 5 cents more than the amount he saved on the previous day. His target is to save the total amount of 3225 cents by the end of 30 days. How much must be the starting saving to meet the target?

A. 60cents B. 25 cents C. 50 cents D. 35cents

Hint: Use $S_{30} = \frac{30}{2}(2A_1 + 29d)$ and $d = 5$

$\Rightarrow 3225 = 15(2A_1 + 29(5))$ and $2A_1 = 215 - 145$ $\therefore A_1 = \frac{70}{2} = 35$ cents

Answer:- D

11. If there are two children in a family, what is the probability that there is at least one girl in the family?

A. $\frac{1}{2}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{2}{3}$

Hint: $E = (B+G)^2 = B^2 + 2BG + G^2 = \{BB, BG, GB, GG\}$

At least one girl = {BG, GB, GG}

$\therefore P(\text{at least one girl}) = \frac{3}{4}$

Answer: B

12. If $F(x) = \int_0^x e^{-t}\,dt$, then what is the value of $F'(x)$?

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A. $\frac{e^{-x+1}}{-x+1}$ B. $-e^{-x}-1$ C. e^{-x} D. $-e^{-x}$

Hint: Let u = - t and du= - dt

$$F(x) = -\int_0^x e^U du = -e^u = -e^{-t}\Big|_0^x = -e^{-x} - (-e^0) = -e^{-x}+1$$

$$F'(x) = -e^{-x}(-x)' = -e^{-x}(-1) = e^{-x}$$ **Answer: C**

13. Which one of the following is true about the pair of lines: $3x + 9y - 24 = 0$ and $4x + 12y + 32 = 0$?

A. perpendicular lines C. representing the same lines

B. intersecting lines D. parallel and distinct lines

Hint: Compare the coefficient ratio.

$$\frac{3}{4} = \frac{9}{12} \neq -\frac{24}{32} \Leftrightarrow \frac{3}{4} = \frac{3}{4} \neq \frac{-3}{4}$$ **Answer: D**

14. If A is a 3 x 3 matrix and det (A) = 5, then how much is det $(2A^TA)$?

A. 50 B. 100 C. 200 D. 20

Hint: $\det\left(2A^TA\right) = 2^3\left|A^T\right||A|$ and $\left|A^T\right| = |A| = 8(5)(5) = 200$

Answer: C

15. Which one of the following is the multiplicative inverse of $z = \frac{3+4i}{4-5i}$?

A. $-\frac{8}{25} - \frac{31}{25}i$ C. $\frac{8}{25} - \frac{31}{25}i$

B. $-\frac{8}{25} + \frac{31}{25}i$ D. $\frac{8}{25} + \frac{31}{25}i$

Hint: Multiplicative inverse of Z is $\frac{1}{Z}$

$$\frac{1}{Z} = \frac{4-5i}{3+4i}\cdot\frac{3-4i}{3-4i} = \frac{4(3-4i)-5i(3-4i)}{9+16}$$

$$\frac{1}{Z} = \frac{12-16i-15i-20}{25} = \frac{-8}{25} - \frac{31i}{25}$$ **Answer: A**

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16. What is the value of $\lim_{h\to 0}\dfrac{8\left(\frac{1}{2}+h\right)^2-8\left(\frac{1}{2}\right)^2}{h}$?

A. 0 B. 8 C. 4 D. The limit does not exist

Hint: $8\lim_{h\to 0}\left(\frac{1}{2}+h+\frac{1}{2}\right)\left(\frac{1}{2}+h-\frac{1}{2}\right)=8\lim_{h\to 0} h\frac{(h+1)}{h}\Rightarrow 8\lim_{h\to 0}(h+1)=8$

Answer: B

17. What is the value of $\lim_{n\to\infty}\left(\dfrac{3^n+2^n}{6^n}\right)$?

A. 1 B. ∞ C. 0 D. $\frac{5}{6}$

Hint: $\lim_{n\to\infty}\frac{3^n}{6^n}+\lim_{n\to\infty}\frac{2^n}{6^n}=\lim_{n\to\infty}\left(\frac{1}{2}\right)^n+\lim_{n\to\infty}\left(\frac{1}{3}\right)^n=0+0=0$

Answer: A

Extreme Tip: What is the value of $\lim_{n\to\infty}\sum_{k=0}^{n}\left(\dfrac{3^k+2^k}{6^k}\right)$?

Solution: Let $a_n=\sum_{k=0}^{n}\left(\frac{1}{2}\right)^k=\frac{G_1\left[1-r^n\right]}{1-r}=1\frac{\left[1-(\frac{1}{2})^n\right]}{1-\frac{1}{2}}$

$\therefore a_n=2(1-(\frac{1}{2})^n)$, as $n\to\infty, a_n=2$

Let $b_n=\sum_{k=0}^{n}\left(\frac{1}{3}\right)^k=1\frac{\left[1-\left(\frac{1}{3}\right)^n\right]}{1-\frac{1}{3}}=\frac{3}{2}\left[1-\left(\frac{1}{3}\right)^n\right]$, as $n\to\infty, b_n=\frac{3}{2}$

$\lim_{n\to\infty}\sum_{k=0}^{n}\left(\dfrac{3^k+Z^k}{6^K}\right)=\lim_{n\to\infty}(a_n+b_n)=2+\frac{3}{2}=\frac{7}{2}$

18. What is the greatest lower bound of the sequence $\left\{(-1)^n\dfrac{1}{n+1}\right\}_{n=0}^{\infty}$?

A. −1 B. −½ C. 0 D. ½

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Hint:

✓ *If* n = 0,2,4,6, …then $\left\{(-1)^n \frac{1}{n+1}\right\}_{n=0}^{\infty} = 0 < \left\{\frac{1}{n+1}\right\}_{n=0}^{\infty} \leq 1$

✓ *If* n = 1,3,5,7, …then $\left\{(-1)^n \frac{1}{n+1}\right\}_{n=1}^{\infty} = -\frac{1}{2} \leq \left\{\frac{-1}{n+1}\right\}_{n=1}^{\infty} < 0$

$\therefore -\frac{1}{2} \leq \left\{(-1)^n \frac{1}{n+1}\right\}_{n=0}^{\infty} \leq 1$ **Answer: B**

19. What is the value of $\lim_{x \longrightarrow 2}\left(\frac{x^3+8}{x^2-4}\right)$?

A. 2 B. The limit does not exist. C. −2 D. 3

Hint: $\lim_{x \to 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} = \frac{4+4+4}{2+2} = 3$

20. Among 2000 students who took a regional exam, the percentile of certain student's score is 90. Which of the following is correct about the student's score?

A. The student's score is greater than or equal to that of 1800 students.
B. The student have answered 90% of the questions correctly.
C. The student's score is the same as the top 10% of the scores.
D. The score of the student is as good as that of 90% of the students.

Answer: C

21. Let $f(x) = 2e^x - k\sin x + 1$. If the equation of the tangent line to the graph of f at (0,3) is $y = 5x+3$, then what is the value of k?

A. 3 B. − 5 C. − 3 D. 2

Hint: $f'(x) = 2e^x - k\,Cos\,x$ and $f'(0) = 5$

$f'(0) = 2 - k\,Cos(0) = 5 \Leftrightarrow 2 - K = 5 \quad \therefore K = -3$ **Answer: C**

22. If $a_1 = 2, a_2 = 6, a_3 = 10, \quad a_4 = 14, ...,$ then $\sum_{n=1}^{100} a_n =$ _____

A. 20,000 B. 20,200 C. 20,020 D. 22,000

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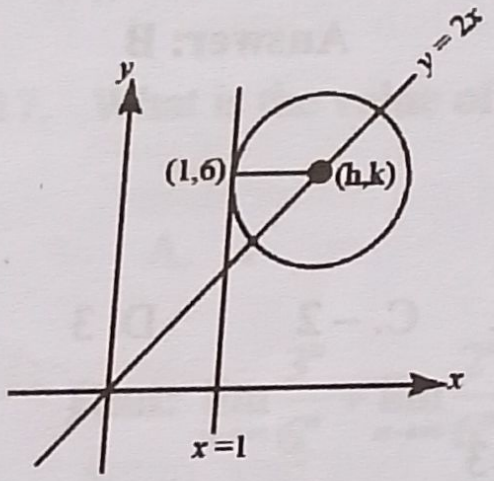
Hint: $S_n = \frac{n}{2}(2a_1 + (n-1)d)$

$$\Rightarrow S_{100} = \frac{100}{2}[2(2)+(99)(4)] = 20{,}000$$ **Answer: A**

23. The center of a circle on the line $y=2x$ and the line $x=1$ is tangent to the circle at (1,6). How long is the radius of the circle?

A. 2 B. 4 C. 3 D. 5



Center: (h,k) = (h,2h)

Since the line passing through the center is perpendicular to the tangent line at the point of tangency, then $y=6$

$\therefore 2h=6$

$h = 3$

$\therefore$ center: (3,6)

$\therefore r = 3 - 1 = 2$ **Answer: A**

24. Which one of the following is the set of critical numbers of $f(x) = \frac{3}{8}x^{8/3} - 6x^{2/3}$?

A. {-2, 0, 2} B. {-1, 1} C. {-1, 0, 1} D. {-2, 2}

Hint: $f'(x) = \frac{8}{3}\cdot\frac{3}{8}x^{5/3} - \frac{4}{x^{1/3}} = \frac{x^2-4}{x^{1/3}}$ and $f'(x) = 0$

$x^2 - 4 = 0 \Rightarrow x = 2$, and $x = -2$

$\therefore$ A number $x_0 = 2, -2$ and 0 in the domain of f such that $f'(2) = f'(-2) = 0$ *and* $f'(0)$ is not defined is called critical number **Answer: A**

25. If $\neg p \Rightarrow r$ is False and $p \Leftrightarrow q$ is True, then which of the following is True?

A. $p \vee (\neg q \wedge r)$ C. $\neg p \Rightarrow (q \vee r)$

B. $p \Leftrightarrow (\neg q \vee r)$ D. $\neg p \wedge (q \Rightarrow r)$

Hint: $\neg p \Rightarrow r$ is False, means $r \equiv F$ and $\neg p \equiv T$ and $P \equiv F$

$P \Leftrightarrow q$ is True if $q \equiv F$ and $P \equiv F$

A. $PV(\neg q \wedge r) \equiv FV(T \wedge F) \equiv FVF \equiv F$

B. $P \Leftrightarrow (\neg qvr) \equiv F \Leftrightarrow (TVF) \equiv F \Leftrightarrow T \equiv F$

C. $\neg p \Rightarrow (qvr) \equiv T \Rightarrow (FVF) \equiv T \Rightarrow F \equiv F$

D. $\neg p \wedge (q \Rightarrow r) \equiv T \wedge (F \Rightarrow F) \equiv T \wedge T = T$ **Answer:- D**

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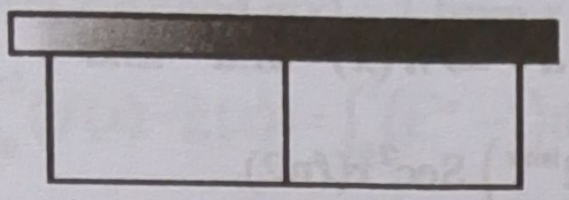
26. A man wants to fence two identical rectangular enclosures in a field alongside a straight river as shown in the following figure.



What is the maximum area of each enclosure that he can make with 192 meter fencing material if the side along the river does not need a fence?

A. 1536 m^2 B. 1564 m^2 C. 1664 m^2 D. 1530 m^2

Hint: Perimeter: $P(x,y) = 2x+3y = 192 \Rightarrow y = 64 - \frac{2}{3}x$

Area: $A(x,y) = (2x)(y)$ and $A(x) = 2x\left(64 - \frac{2}{3}x\right)$

$A'(x) = 128 - \frac{8}{3}x$ and $A'(x) = 0$, if $x = 48$ $y = 64 - \frac{2}{3}(48) = 32$

$\therefore$ Area of each region will be $xy = (48)(32) = 1536$

Answer: A

27. If $f(x) = x^3 - 2x + 1$, then which one of the following is **NOT** true?

A. $f(c) = 0$ for some $c \in [-2,0]$. C.

B. $f(c) = \frac{1}{4}$ for some $c \in [-1,0]$.

C. $f(c) = \frac{1}{2}$ for some $c \in [-0,1]$.

D. $f(c) = 3$ for some $c \in [1,2]$.

Hint: Refer "Intermediate value theorem"

A. $f(-2) = -3$ and $f(0) = 1 \Rightarrow f(-2) < f(c) < f(o)$

B. $f(-1) = 2$ and $f(0) = 1 \Rightarrow \frac{1}{4} \notin [f(-1), f(0)]$..False

C. $f(0) = 1$ and $f(1) = 1 - 2 + 1 = 0 \Rightarrow f(1) < f(c) < f(0)$

$\Rightarrow 0 < \frac{1}{2} < 1$ and $c \in [0,1]$.

D. $f(1) = 0$ and $f(2) = 5 \Rightarrow f(1) < f(c) < f(2) C \in [1,2]$ *i.e* $0 < 3 < 5$

Answer: B

28. If $f(x) = \ln(2^{\tan x})$, then what is the value of $f'(0)$?

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A. $\frac{\ln 2}{2}$ B. $-2\ln 2$ C. $\ln 2$ D. 1

Hint: Recall: $h(x) = a^{mx} \Rightarrow h'(x) = m.a^{mx}\,\ell na$

$$\therefore f'(x) = \left(\frac{1}{2^{\tan x}}\right)\left(2^{\tan x}\right).\text{Sec}^2 x(\ell n 2).$$

$$\Rightarrow f'(0) = Sec^2(0).\ell n 2 = \ell n 2.$$

Answer: C

29. If $f(x) = ax - b$ and $f^{-1}(x+1) = \frac{1}{2}x + 2$, for each $x \in \Re$, then what must be the values of a and b?

A. $a = ½, b = -2$
B. $a = 2, \ b = 2$
C. $a = 2, \ b = 3$
D. $a = 1, \ b = 1$

Hint: $f^{-1}(x+1) = \frac{1}{2}x + 2 \Rightarrow f(\frac{1}{2}x + 2) = x + 1$

$f(x) = ax - b$ and $f(\frac{1}{2}x + 2) = a(\frac{1}{2}x + 2) - b = x + 1$

$$\Rightarrow \frac{a}{2}x + 2a - b = x + 1 \Leftrightarrow \frac{a}{2} = 1 \text{ and } 2a - b = 1$$

$$\Rightarrow a = 2 \text{ and } 2(2) - b = 1 \Rightarrow b = 3$$

Answer: C

30. If $2\begin{pmatrix} 2x & x \\ -5 & -3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & 2 \\ -5 & -4 \end{pmatrix}$, then what is the value of x?

A. -2 B. $-\frac{1}{2}$ C. $\frac{1}{2}$ D. 2

Hint: Let $A = \frac{1}{2}\begin{pmatrix} 3 & 2 \\ -5 & -4 \end{pmatrix} = \begin{pmatrix} 1.5 & 1 \\ -2.5 & -2 \end{pmatrix}$ and $|A| = -3 + 2.5 = -\frac{1}{2}$

$$\text{Adj.A} = \begin{pmatrix} -2 & -1 \\ 2.5 & 1.5 \end{pmatrix} \text{ and } A^{-1} = \frac{\text{adj.A}}{|A|} = \begin{pmatrix} 4, & 2 \\ -5 & -3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2x & x \\ -5 & -3 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ -5 & -3 \end{pmatrix} \Leftrightarrow 2x = 4 \text{ and } x = 2$$

Answer: D

31. What is the area of the region bounded by the lines $x = 0$, $x = 2$, and $y = 1$ and the curve $y = e^{2x}$?

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A. $\frac{e^3}{3}-\frac{1}{3}$ B. $\frac{e^2}{2}-\frac{1}{2}$ C. $\frac{e^4}{2}-\frac{5}{2}$ D. $-\frac{e^4}{2}+\frac{5}{2}$

Hint: Let $f(x)=e^{2x}$ and $g(x)=1$ from $x=0$ to $x=2$

$$\therefore A=\int_0^2\left(f(x)-g(x)\right)=\int_0^2\left(e^{2x}-1\right)dx=\frac{1}{2}e^{2x}-x\Big|_0^2$$

$$\therefore A=\left(\frac{1}{2}e^4-2\right)-\left(\frac{1}{2}e^0-0\right)=\frac{1}{2}e^4-2-\frac{1}{2}=\frac{e^4-5}{2}$$

Answer: C

32. If f is the greatest integer function and g is the absolute value function, then what is the value of $(fog(\frac{3}{2})+(g\,of)(-\frac{4}{3})$?

A. 3 B. 1 C. -1 D. 2

Hint: Let $f(x)=[x]$ and $g(x)=|x|$.

$$(f\,og)\left(\frac{3}{2}\right)+(g\,of)\left(\frac{-4}{3}\right)$$

$$\Rightarrow f\left(g\left(\frac{3}{2}\right)\right)+g\left(f\left(\frac{-4}{3}\right)\right)=f\left(\frac{3}{2}\right)+g(-2) \quad =1+2=3$$

Answer: A

33. Which one of the following is true about the graph of $f(x)=\frac{x^2+5x+6}{x^2-4}+3$?

A. The graph has a hole at $x=2$.
B. A horizontal asymptote of the graph is $y=4$
C. The vertical asymptotes of the graph are $x=-2$ and $x=2$.
D. The graph has $y-$ intercept at $(0,-\frac{3}{2})$

Hint: $\frac{x^2+5x+6}{x^2-4}+3\Leftrightarrow\frac{(x+2)(x+3)}{(x-2)(x+2)}+3=\frac{4x-3}{x-2}$

✓ H.A, $y=4$ and V.A is $x=2$ and the graph has hole at $x=-2$

✓ $y-$ intercept is $(0,\frac{3}{2})$

Answer: B

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34. What is the solution of $\int 4x(\ell n\, x+\frac{1}{x^2})dx$?

A. $x^2(2\ell nx-1)+4\ell nx+c$
B. $x^2(4\ell nx-1)+2\ell nx+c$
C. $x^2(2\ell nx+1)+4\ell nx+c$
D. $4x^2(\ell nx+1)-2x^2+c$

Hint: $\int 4x\left(\ell nx+\frac{1}{x^2}\right)dx=\int 4x\,\ell nxdx+\int\frac{4}{x}\,dx.$

$=4\left(\frac{x^2\ell nx}{2}-\frac{x^2}{4}\right)+4\ell nx+C \;=2x^2\,\ell nx-x^2+4\ell nx+c$

$=x^2\,(2\ell nx-1)+4\ell nx+c$ **Answer: A**

35. The earth's orbit has a semi-major axis $a\approx 149.6$ Gm (gigameters) and an eccentricity of $e\approx 0.017$. What is the approximate value of the semi – minor axis?

A. 152.14 Gm
B. 149.58 Gm
C. 149.06 Gm
D. 145.32 GM

Hint: $C=ea$ and $b=\sqrt{a^2-C^2}=\sqrt{a^2-e^2a^2}$

$b=a\sqrt{1-e^2}=149.6\sqrt{1-0.017}=149.58$ **Answer: B**

36. A water tank is a rectangular parallelepiped with base length 3m, width 2m and height 2.5m. If water is flowing into the tank at the rate of 0.12 m^3/sec, then how fast does the level of water rise up in the tank?

A. 0.06 m/sec
B. 0.03 m/sec
C. 0.04 m/sec
D. 0.02 m/sec

Hint: $V=\ell Wh\Rightarrow\frac{dv}{dt}=\ell w\left(\frac{dh}{dt}\right)$

$\Rightarrow 0.12m^3/_{sec}=(3)(2)\frac{dh}{dt}\Leftrightarrow\frac{dh}{dt}=\frac{0.12}{6}=0.02$ **Answer: D**

37. What is the sum of the infinite series $\sum_{k=0}^{\infty}5\left(\frac{2k+5k}{10k}\right)$?

A. 15
B. $\frac{65}{4}$
C. 5
D. $\frac{25}{4}$

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Hint: $\sum_{K=0}^{\infty} 5\left(\frac{2^k+5k}{10^k}\right) = 5\sum_{k=0}^{\infty}\left(\frac{2}{10}\right)^k + 5\sum_{k=0}^{\infty}\left(\frac{5}{10}\right)^k$

$$= 5\sum_{k=0}^{\infty}\left(\frac{1}{5}\right)^k + 5\sum_{k=0}^{\infty}\left(\frac{1}{2}\right)^k = 5\left[\frac{1}{1-\frac{1}{5}}\right] + 5\left[\frac{1}{1-\frac{1}{2}}\right] = \frac{25}{4} + 10 = \frac{65}{4}$$

Answer:- B

38. The marks of 50 students is given below:

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of students	5	8	f_1	10	f_2

If the median of the data is 26, what are the values of f_1 and f_2?

A. $f_1 = 7, f_2 = 20$
B. $f_1 = 12, f_2 = 15$
C. $f_1 = 20, f_2 = 7$
D. $f_1 = 15, f_2 = 12$

Solution

V	0 - 10	10 -20	20 -30	30 - 40	40 -50
f	5	8	f_1	10	f_2
cf	5	13	$13 + f_1$	$23 + f_1$	$23 + f_1 + f_2$

Md = 26

∴ 20 – 30 is the median class.

$\Sigma f = n = 50$

$23 + f_1 + f_2 = 50$

$f_1 + f_2 = 27$

$$Md = B_L + \left(\frac{\frac{n}{2} - cfb}{f_c}\right) i \Rightarrow 26 = 20 + \left(\frac{\frac{50}{2} - 13}{f_1}\right).10 \qquad 6 = \frac{12}{f_1}.10$$

$6f_1 = 120 \Rightarrow f_1 = 20$

$f_1 + f_2 = 27 \Rightarrow f_2 = 7$

Answer: C

39. There are three children in a room with, ages four, five, and six. If a five –year – old child enters the room, then which of the following statement is correct?

A. mean age and standard deviation will stay the same.
B. mean age will stay the same but the standard deviation will increase.

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C. mean age and standard deviation will increase.
D. mean age will stay the same but the standard deviation will decrease.

Solution:

age	4	5	6
f_1	1	1	1
f_2	1	2	1

$$\bar{x}_1 = \frac{4+5+6}{3} = \frac{15}{3} = 5$$

$$\bar{x}_2 = \frac{4+2\times5+6}{4} = \frac{20}{4} = 5$$

$\therefore$ Mean age will stay the same.

$$\sigma_1 = \sqrt{\frac{\sum f_1(x-\bar{x})^2}{n}} = \sqrt{\frac{(4-5)^2+(5-5)^2+(6-5)^2}{3}} = \sqrt{\frac{2}{3}}$$

$$\sigma_2 = \sqrt{\frac{\sum f_2(x-\bar{x})^2}{n}} = \sqrt{\frac{(4-5)^2+2(5-5)^2+(6-5)^2}{3}} = \sqrt{\frac{1}{2}}$$

$\therefore \sigma_1 > \sigma_2$

Thus, the standard deviation will decrease. **Answer:- D**

40. For real numbers x and y, which one of the following is **NOT** true?

A. $(\exists x)(\forall y)(y \geq x^2+1)$
B. $(\forall x)(\exists y)(y \geq x^2+1)$
C. $(\forall x)(\forall y)(y^2+x^2 \geq -1)$
D. $(\exists x)(\exists y)(y \geq x^2+1)$

Solution:

$U = \mathbb{R}$

A. $\exists x \forall y: y \geq x^2+1 \longleftarrow$ There is one fixed x for any y which satisfy $y \geq x^2+1$ - F.

B. $\forall x \exists y: y \geq x^2+1 \longleftarrow$ For any x there is a corresponding y Such that $y \geq x^2+1$. – T.

C. $\forall x \forall y = y^2+x^2 \geq -1 \longleftarrow$ For any arbitrary x and y, $y^2+x^2 \geq -1$ – T.

D. $\exists x \exists y: y \geq x^2+1 \longleftarrow$ There is at least one x and one y such that $y \geq x^2+1$ – T **Answer: A**

41. If the second and fifth terms of a geometric progression are $\frac{1}{2}$ and $\frac{1}{16}$ respectively, what is the sum of the first eight terms of the sequence?

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A. $\frac{255}{256}$ B. $\frac{85}{128}$ C. $\frac{128}{85}$ D. $\frac{256}{255}$

Solution:

$G_2=-\frac{1}{2}$, $G_5=\frac{1}{16}$. Then $G_n=r^{n-1}G_1$ and $G_5=r^3G_2$

$\frac{1}{16}=r^3.(-\frac{1}{2}) \Rightarrow r^3=-\frac{1}{8} \Rightarrow r=-\frac{1}{2}$ and $G_1=\frac{1}{r}.G_2=1$

$$S_n=G_1\left(\frac{1-r^n}{1-r}\right) \Rightarrow S_8=G_1\frac{\left(1-\left(-\frac{1}{2}\right)^8\right)}{1-\left(-\frac{1}{2}\right)}=\frac{2\left(2^8-1\right)}{3\,256}=\frac{85}{128}$$

Answer: B

42. Which one of the following is a valid argument?
   A. If I don't change my oil regularly, my engine will die. My engine died, Thus, I didn't change my oil regularly.
   B. If it rains or snows, then my roof leaks. My roof is leaking. Thus, it is raining and snowing.
   C. If you do every problem in the book, then you will learn the subject. You learned the subject. Thus, you did every problem in the book.
   D. If I am literate, then I can read and write. I can read but I can't write. Thus, I am not literate.

***Solution*:**

Let P : I am literate
q: I can read
r: I can write
$p \Rightarrow q \wedge r$
$\underline{q \wedge \neg r}$
$\neg p$
$q \wedge \neg r = T$ (premise)

Thus, $q = T$ and $\neg r = T$
($\because$ rule of $\wedge$) $\Rightarrow r = F$
$P \Rightarrow q \wedge r = T$ (Premise
$P \Rightarrow F = T$
$\therefore p = F$ ($\because$ rule of $\Rightarrow$)
$\therefore \neg p = T$ (Valid argument)

Answer: D

43. What is the value of $\int \frac{x^{e-1}+e^{x-1}}{x^e+e^x}dx$?

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A. $\frac{1}{e}\ln(x^e+e^x)+c$

B. $\frac{1}{e}\ln(x^e+e^{x+1})+c$

C. $\frac{1}{e}\ln\left((x+1)^e+e^{x+1}\right)+c$

D. $\frac{1}{e}\ln(x^{e+1}+e^{x+1})+c$

Solution:

Let $u=x^e+e^x$ and $\frac{du}{dx}=ex^{e-1}+e^x$

$$\int\frac{x^{e-1}+e^{x-1}}{x^e+e^x}dx=\int\frac{x^{e-1}+e^{x-1}}{u}\cdot\frac{du}{ex^{e-1}+e^x}=\frac{1}{e}\int\frac{x^{e-1}+e^{x-1}}{u}\cdot\frac{du}{x^{e-1}+e^{x-1}}$$

$$=\frac{1}{e}\int\frac{1}{u}du=\frac{1}{e}\ln|u|+c=\frac{1}{e}\ln|x^e+e^x|+c$$

Answer: A

44. Suppose 2,500 items are produced by a machine and 2% of the product are randomly selected and tested. If 5 of the tested items have a defect, then what is the probability that an item produced by the machine has NO defect?

A. 0.80 B. 0.85 C. 0.95 D. 0.90

Solution:

No of items = 2500

No of tested items = 2% of $2500=\frac{2}{100}\times 2500=50$

No of defected items = 5

Non - defected item will be $50-5=45$ and P (no defect) $=\frac{45}{50}=0.90$

Answer: D

45. If $f(x)=\frac{1}{x}$, then what is the value of $f^{(n)}(x)$?

A. $f^{(n)}(x)=\frac{(-1)^n n!}{x^{n+1}}$

B. $f^{(n)}(x)=\frac{(-1)^{n+1}(n+1)!}{x^{n+1}}$

C. $f^{(n)}(x)=\frac{(-1)^n n!}{x^{n-1}}$

D. $f^{(n)}(x)=\frac{(-1)^n n!}{x^n}$

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Solution:

$$f(x)=\frac{1}{x}=x^{-1}$$

$$f'(x)=-1x^{-2}$$

$$f''(x)=(-2)(-1)x^{-3}=2x^{-3}$$

$$f'''(x)=(-3)(-2)(-1)=-6x^{-4}$$

$$f^{(4)}(x)=(-4)(-3)(-2)(-1)x^{-5}=24x^{-5}$$

$$f^{n}(x)=(-1)^{n}.n!.x^{-(n+1)}=\frac{(-1)^{n}n!}{x^{n+1}}$$

Answer: A

46. If the region enclosed by the graphs of $f(x)=x^2$ and $f(x)=x^3$ from $x=0$ to $x=1$ rotates about the *x- axis*, what is the volume of the solid of revolution?

A. $\frac{2\pi}{27}$ cubic units

B. $\frac{2\pi}{35}$ cubic units

C. $\frac{2\pi}{5}$ cubic units

D. $\frac{2\pi}{25}$ cubic units

Solution:

Using the formula, $V=\pi\int_a^b\left(\left(f(x)\right)^2-\left(g(x)\right)^2\right)dx$, we have

$$V=\pi\int_0^1\left(\left(x^2\right)^2-\left(x^3\right)^2\right)dx=\pi\int_0^1\left(x^4-x^6\right)dx$$

$$=\pi\left[\frac{x^5}{5}-\frac{x^7}{7}\Big|_0^1\right]=\pi\left(\frac{1}{5}-\frac{1}{7}\right)=\frac{2}{35}\pi \text{ cubic unit}$$

Answer: B

47. What is the limit of the sequence $1, \frac{2}{2^2-1^2}, \frac{3}{3^2-2^2}, \frac{4}{4^2-3^2}, \ldots$?

A. 0 B. 1 C. $\frac{3}{2}$ D. $\frac{1}{2}$

Solution:

$$a_1=1,\ a_2=\frac{2}{2^2-1^2},\quad a_3=\frac{3}{3^2-2^2}, a_4=\frac{4}{4^2-3^2},\ldots,$$

Then. $a_n=\frac{n}{n^2-(n-1)^2}=\frac{n}{n^2-(n^2-2n+1)},=\frac{n}{2n-1}=\frac{1}{2-\frac{1}{n}}$

$$\lim_{n\to\infty}a_n=\lim_{n\to\infty}\frac{1}{2-\frac{1}{n}}=\frac{1}{2}$$

Answer: D

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48. At what value of x does the function $f(x) = \frac{4x^3}{3} - x^4$ attains its maximum value?

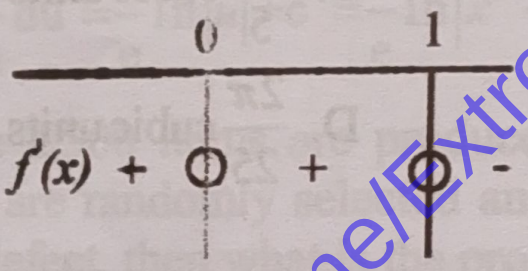
A. 1 B. 0 C. -1 D. $\frac{4}{3}$

Solution:

$f(x)=\frac{4}{3}x^3 - x^4 \Rightarrow f'(x)=4x^2 - 4x^3 = 4x^2(1-x)$

$f'(x)=0 \Rightarrow 4x^2(1-x)=0$

$\Rightarrow x=0$ or $x=1$



$\therefore f$ has a local maximum at $x = 1$.

Since $f'(x)$ is changing the sign from (+) to (-1) at $x = 1$ then f has a maximum value at $x = 1$ **Answer: A**

49. If $h(x) = \sqrt{1+\sqrt{x}}$, then which of the following is equal to $h'(x)$?

A. $\frac{1}{2\sqrt{1+\sqrt{x}}}$ C. $\frac{1}{4\sqrt{x+x\sqrt{x}}}$

B. $\frac{x}{2\sqrt{1+\sqrt{x}}}$ D. $\frac{x}{4\sqrt{x+x\sqrt{x}}}$

Solution:

$h(x)=\sqrt{1+\sqrt{x}}$ using the chain rule,

$h'(x)=\frac{1}{2\sqrt{1+\sqrt{x}}}.(1+\sqrt{x})'$

$=\frac{1}{2\sqrt{1+\sqrt{x}}}.\left(0+\frac{1}{2\sqrt{x}}\right)=\frac{1}{4\sqrt{x}.\sqrt{1+\sqrt{x}}}=\frac{1}{4\sqrt{x+x\sqrt{x}}}$

50. Fatuma can solve 90% of the problems given in a book and Mesfin can solve 70%. What is the probability that at least one of them will solve the problem, selected at random from the book? **Answer: C**

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A. 0.77 B. 0.87 C. 0.67 D. 0.97

Solution:

P(F) = 0.9 and P(F') = 0.1

P(M = 0.7 and P(M') = 0.3

Then, P (at least one of them can solve) = 1 – P(F' and M')

= 1 – P(F') . P(M')

= 1 – 0.1 . 0.3

= 1 – 0.03 = **0.97**

Answer:- D

51. What is the polar form of $\frac{7-i}{3-4i}$?

A. $\sqrt{2}(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4})$

B. $\sqrt{2}(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2})$

C. $2(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2})$

D. $2(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4})$

Solution:

$$\frac{7-i}{3-4i} = \frac{7-i}{3-4i}\cdot\frac{3+4i}{3+4i} = 1+i$$

Then, $r=\sqrt{2}$, $Arg(1+i) = \tan^{-1}\left(\frac{1}{1}\right)=\frac{\pi}{4}$

$\therefore$ Polar form: $r(\cos\theta+i\sin\theta) = \sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$

Answer:- A

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52. If birr 2000 is deposited in a Bank with interest rate 10% per annum compounded semi-annually, then how much birr will it be by the end of one year?

A. 2100 B. 2200 C. 2205 D. 2210

Solution: P = 2000 r = 10 %

m = 2 (compounded semi - annually) and t = 1

$$A = p\left(1+\frac{r}{m}\right)^{mt} = 2000\left(1+\frac{10}{100\times 2}\right)^{2\times 1} = 2000\left(1+\frac{1}{20}\right)^{2}$$

$$= 2000\left(\frac{21}{20}\right)^{2} = \frac{20\cancel{00}\times 21^{2}}{4\cancel{00}} = 5(21)^{2} = 2205$$

Answer: C

53. Hagos bought a bike from the manufacturer by 10,000 birr and sold it to Abraha by making a profit of 10%. Abraha also sold the bike by making a profit of 15%. What was the sale price of Abraha?

A. 12,500 birr B. 12,650birr C. 11,500 birr D. 11,550bir

Solution:

Hagos: Cost (buying) price = 10,000 Birr

Selling price = 10,000 + 10% × 10,000 = 11,000 Birr

Abraha: Cost (Buying) price = 11,000 Birr

Selling price = 11,000 + 15% × 11,000 = 12,650Birr

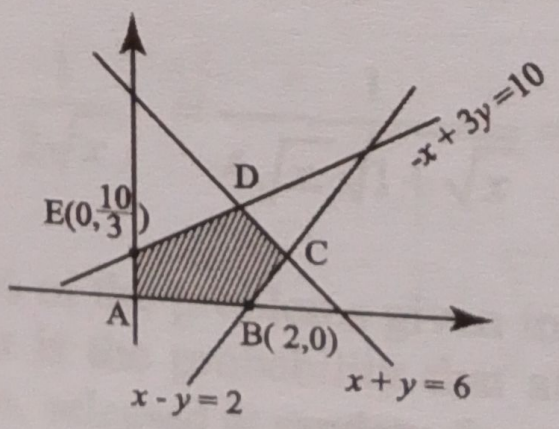
Answer: B

54. Which one of the following points is in the solution region of the linear inequalities $-x+3y\le 10, x+y\le 6, x-y\le 2, x, y\ge 0$?

A. (3,4) B) (1,1) C. (3,0) D. (0,4)

Solution: Constraints: $-x+3y\le 10,\ x+y\le 6, x-y\le 2, x, y\ge 0.$

The solution region (feasible region) is the region determined by the constraints.



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ABCDE is the feasible region. Any points in this region satisfies the constraints.

Point A: $-1+3 \leq 10$ (T)

$1+1 \leq 6$ (T)

$1-1 \leq 2$ (T)

$\therefore$ The point (1,1) belongs to the feasible region.

Check that the pts (3,4), (3,0) and (0,4) do not belongs to the feasible region.

Answer : A

55. A sweatshirt with a regular price of 340 Birr is first discounted by 20% and then by 10%. What is the final selling price of the sweatshirt after the discount?

A. 238 Birr B. 102 Birr C. 244.80 Birr D. 95.20 Birr

Solution: Regular price = 340. Birr

First discount rate = 20%

$\therefore$ Discount = r.p = 20%. 340 = 68 Birr

Discount sales price = 340 – 68 = 272 Birr

Second discount rate = 10%

$\therefore$ Discount = r.p = 10%. 272 = 27.20 Birr

Discount sales price = 272 – 27.20 = 244.80 Birr

Answer: C

56. A young student borrows a certain amount of money from a money lender. The money lender charges interest at a rate of 20% compounded yearly. If the money lender sends a letter of demand of 400,000 birr after 2 years, how much did the student borrow?

A. 320,000 birr C. 384,467.50 birr

B. 240,000 birr D. 277,777.80 birr

Solution: $r = 20\%$, $m = 1$, $A = 4 \times 10^5$, $t = 2$ $A = p(1+\frac{r}{m})^{mt}$

$$4\times10^5 = P\left(1+\frac{20}{100\times1}\right)^{1\times2} = P\left(1+\frac{2}{10}\right)^2 = P\left(\frac{12}{10}\right)^2$$

$$P = \frac{4\times10^5\times10^2}{12^2} = 2,77,777.80 \text{Birr}$$ **Answer: D**

57. The following table presents the mean weight (in kg) of four group of student.

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Group	I	II	III	IV
Mean weight	72	80	75	80
Number of students in the group	10	12	8	10

What is the mean weight (in kg) of all the students?

A. 77.4 B. 77 C. 76.7 D. 76

Solution: $\text{Mean Weight} = \dfrac{72\times10+80\times12+75\times8+80\times10}{10+12+8+10}$

$$= \frac{72\cancel{0}+96\cancel{0}+600+80\cancel{0}}{4\cancel{0}} = 18 + 24 + 15 + 20 = 77$$

Answer: B

58. What is the maximum value of $Z = -3x + 2y$,

Subject to $-x + y, \leq 1$

$x - y \leq 1$

$x + y \geq 3$

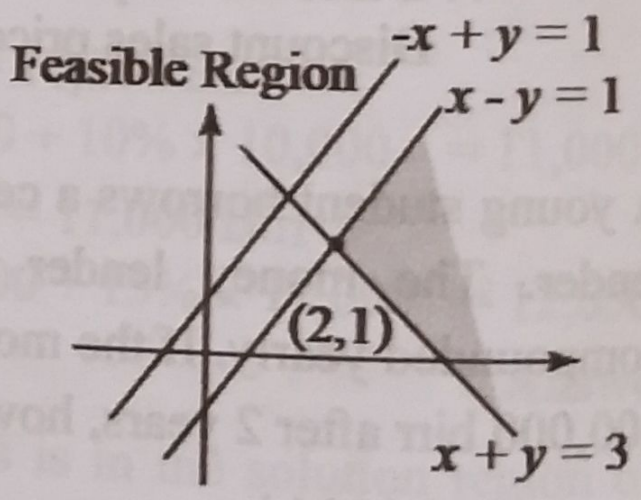
A. 1 B. 4 C. 6 D. It has No maximum

Solution: $Z = -3x + 2y$

s.t. $-x + y \leq 1$

$x - y \leq 1$

$x + y \geq 3$



From the graph, we see that

$Z = -3x + 2y$ has no maximum value. and

$Z = -3(2) + 2(1) = -4$ is the minimum value. **Answer: A**

59. The following table shows a simple frequency distribution of the mark that students scored in a test.

Score (Marks)	4	5	6	7	8	9	10
Frequency	5	8	9	14	7	5	3

Which of the following is **NOT** true about the frequency distribution?

A. The inter – quartile range of the mark is 3.

B. The 25^{th} percentile of the mark is 5.

C. The top 25% of the scores are 8 and above.

D. The 75^{th} percentile of the mark is 7.

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Solution:

M	4	5	6	7	8	9	10
f	5	8	9	14	7	5	3

$n = \sum f = 51$

$$P_t = \left(\frac{t(n+1)}{100}\right)^{th} \text{ item.}$$

$$P_{25} = \left(\frac{25(51+1)}{100}\right)^{th} \text{ item}, \qquad P_{75} = \left(\frac{75(51+1)}{100}\right)^{th} \text{ item}$$

$= 13^{th}$ item $= 39^{th}$ item

$Q_1 = 5$ $= 8 = Q_3 = 8$

$IQR = Q_3 - Q_1 = 8\text{-}5 = 3$

Top. 25% of the scores $\geq P_{75} = 8$ **Answer: B**

60. You are the process of buying a house, and you need a mortgage loan of Birr 200,000. You got approved for a 15-year fixed rate loan at a rate of 4.5%. How much, in total, will you pay for this house?

Given: $(1.00375)^{180} = 1.96, (1.00375)^{-180} = 0.51, \frac{0.00375}{0.49} = 7.65 \times 10^{-3}$

A. Birr 275,000
B. Birr 275,400
C. Birr 250,000
D. Birr 250,400

Solution: Mortgage (A) = Birr 200,000

t = 15 years.

r = 4.5% compounded monthly.

Then, monthly mortgage payment, $R = A\left[\frac{\frac{r}{12}}{1-(1+\frac{r}{12})^{-n}}\right]$

Where $n = 12 \times t$.

$$R = 2 \times 10^5 \frac{\left[\frac{4.5}{100 \times 12}\right]}{1-\left(1+\frac{4.5}{100 \times 12}\right)^{-12 \times 15}}$$

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$$= 2\times10^5\left[\frac{0.00375}{1-(1+0.00375)^{-180}}\right] \qquad = 2\times10^5\left[\frac{0.00375}{1-(1.00375)^{-180}}\right]$$

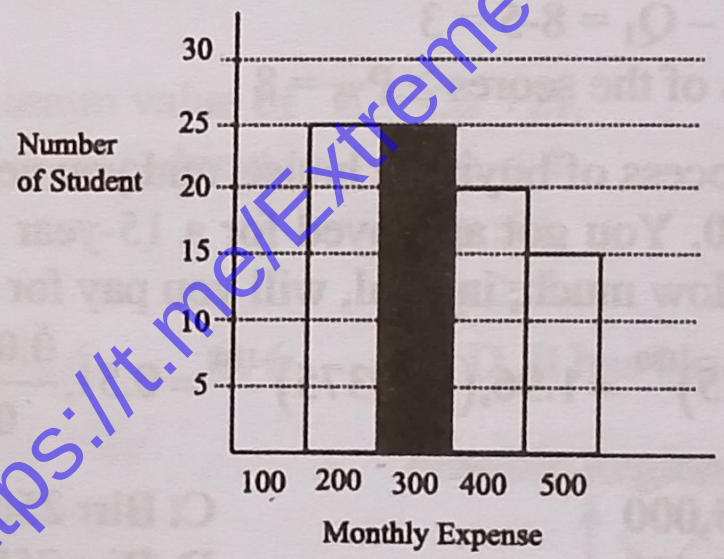
$$= 2\times10^5\left(\frac{0.00375}{1-0.5i}\right) = 2\times10^5\left(\frac{0.00375}{0.49}\right)$$

$$= 2\times10^5\left(7.65\times10^{-3}\right) = 15.3\times10^2 = 1530\text{Birr}$$

$\therefore$ The total payment for this house is $1530 \times 12 \times 15 = 275{,}400$

Answer: B

61. The following histogram gives the monthly expense of 100 students.



Which one of the following is true?

A. Only 40 students have monthly expense less than Birr 400.
B. 35% of the students have monthly expense more than 300.
C. 60% of the students have monthly expense less than 400.
D. 50 students have monthly expense more than Birr 200.

Solution:

Expense	100	200	300	400	500
No of Students	15	25	25	20	15

Total no of students = 100.

35% of 100 $= \frac{35}{100}\times100 = 35$ Students have monthly expense

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more than 300. $= 20 + 15$

Answer: C

62. A company wants to buy two cars from Marathon Motors. The price of each car including VAT (15%) is Birr 402,500. If the company wants to subtract a 2%withholding tax before VAT, what is the amount to be subtracted?

A. 16,000 Birr
B. 14,250 Birr
C. 14,000 Birr
D. 16,250 Birr

Solution: Consider the two cars, A and B.

Price of A + 15% of price of A = 402,500

$P_A + 0.15\,P_A = 402{,}500$

$1.15\,P_A = 402{,}500$

$P_A =$ Birr 350, 000

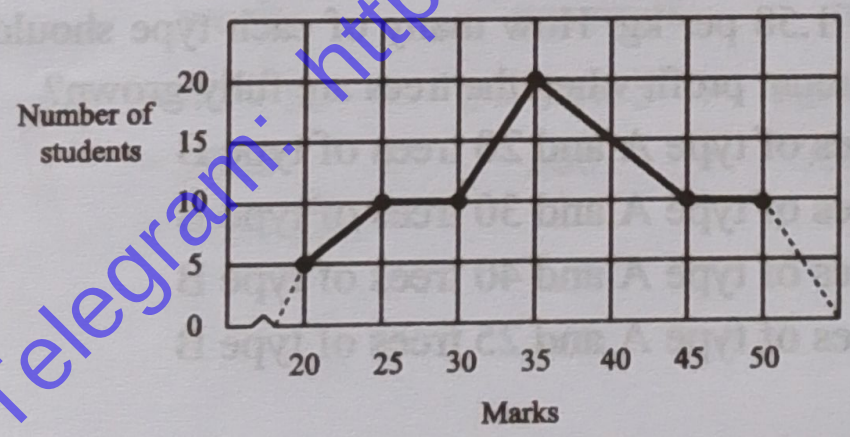
Withholding tax = 2% of P_A $= \frac{2}{100} \times 350{,}000 = 7000$Birr

Similar for car B.

∴ The total amount to be subtracted from the two cars will be birr 14,000

Answer: C

63. Marks scored by students in exam are grouped into seven class intervals of equal width and represented by the following frequency polygon.



Which of the following is true about the marks?

A. The mode of the mark is 35.
B. The mean and mode of the marks are equal.
C. The distribution of the mark is positively skewed.
D. 75% of the students scored greater than 30.

M	20	25	30	35	40	45	50
f	5	10	10	20	15	10	10

Solution: The modal value is 35.

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$$\text{Mean} = \frac{20\times5+25\times10+30\times10+35\times20+40\times15+45\times10+50\times10}{5+10+10+20+15+10+10}$$

$$= \frac{2900}{80} = \underline{\underline{36.25}}$$

$$\text{Median} = \frac{40^{th}\text{ item} + 41^{th}}{2}\text{ item} = \frac{35+35}{2} = \underline{\underline{35}}$$

Mean > median = Mode

$\therefore$ The distribution is not positively skewed.

75% of 80 $= \frac{75}{10\cancel{0}} \times 8\cancel{0} = 60$. But 55 students scored greater than 30.

Answer: C

64. A man owns a field of area 1000 *sq.m*. He wants to plant fruit trees in it. He has a sum of Birr 1400 to purchase young trees. He has the choice of two types of trees. Type a requires 10 *sq.m* of ground per tree and cost Birr 20 per tree and type **B** requires 20 *sq.m* of ground per tree and cost Birr 25 per tree. When fully grown, type A produces an average of 20kg fruit which can be sold at a profit of 2 Birr per kg and type B produces an average of 40 kg of fruit which can be sold at a profit of Birr 1.50 per kg. How many of each type should he plant to achieve maximum profit when the trees are fully grown?

A. 40 trees of type A and 20 trees of type B
B. 30 trees of type A and 30 trees of type B
C. 20 trees of type A and 40 trees of type B
D. 35 trees of type A and 25 trees of type B

Solution:

Type of tree	Area covered by a tree	Cost price	Average production	Profit
A	10 m^2/tree	20 birr/tree	20kg	2 birr/kg
B	20 m^2/tree	25 birr/tree	40kg	1.5birr/kg
Total	1000 m^2	1400 birr		

From the table, we see that

$10A + 20B = 1000$

$20A + 25B = 1400$

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Solving simultaneously we get,

$A = 20$, $B = 40$ **Answer: C**

65. In a company, a regular work time is 40 hours per week. For a service above the regular work time, the normal overtime pay rate (per hour) is 1.5 times the regular rate and the holyday overtime rate is twice of the regular rate. If an employee, whose regular pay is Birr 50 per hour, has worked 55 hours in a week among which 5 hours are holyday overtime, how much (in Birr) is his payment for the week?

A. 2500 B. 3125 C. 3250 D. 3500

Solution: Regular work time per week = 40 hrs.

Regular pay per hour = Birr 50.

Worked time per week = 55 hrs.

Therefore, $55 - 40 = 15$hrs becomes overtime.

Since, 5 hrs is a holyday overtime, Then 10 hrs will be a normal overtime.

Total Payment Per week	=	Regular Pay per week	+	Holyday over time per week	+	Norml over time per week
		40×50	+	$2 \times 50 \times 5$	+	$1.5 \times 50 \times 10$
		2000	+	500	+	750

$= 3250$ Birr **Answer:- C**

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1. A system of linear equations is given as:

$$\begin{cases} x+3y=9 \\ -y+4z=-2 \\ x-5z=0 \end{cases}$$

Which one of the following is the augmented matrix for the above system?

A. $\begin{pmatrix} 1 & 3 & 0 & 9 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{pmatrix}$

C. $\begin{pmatrix} 1 & 3 & 9 \\ -1 & 4 & -2 \\ 1 & -5 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 & 3 & 9 & 0 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{pmatrix}$

D. $\begin{pmatrix} 1 & 3 & 0 \\ 0 & -1 & 4 \\ 1 & 0 & -5 \end{pmatrix}$

Explanation: An augmented matrix is a matrix obtained by appending the columns of two matrices, usually the coefficient matrix and the constants for the given system of linear equations.

$$\begin{cases} x+3y=9 \\ -y+4z=-2 \\ x-5z=0 \end{cases}$$

Coefficient matrix : $\begin{pmatrix} 1 & 3 & 0 \\ 0 & -1 & 4 \\ 1 & 0 & -5 \end{pmatrix}$

The Augmented Matrix$(A|B$ $\left(\begin{array}{ccc|c} 1 & 3 & 0 & 9 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{array}\right)$

Answer: A

2. Let $E=\begin{pmatrix} 2 & -2 & -4 \\ -4 & 8 & 6 \\ -6 & 10 & -8 \end{pmatrix}$ and $F=\begin{pmatrix} -1 & 1 & 2 \\ 2 & -4 & -3 \\ 3 & -5 & 4 \end{pmatrix}$

Which one of the following is true about E and F?

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A. $F = -2E$ B. $F = -\frac{1}{2}E$ C. $F = 2E$ D. $F = \frac{1}{2}E$

Solution:

$$E = \begin{pmatrix} 2 & -2 & -4 \\ -4 & 8 & 6 \\ -6 & 10 & -8 \end{pmatrix}, \quad F \begin{pmatrix} -1 & 1 & 2 \\ 2 & -4 & -3 \\ 3 & -5 & 4 \end{pmatrix}$$

$$\Rightarrow E = -2\begin{bmatrix} -1 & 1 & 2 \\ 2 & -4 & -3 \\ 3 & -5 & 4 \end{bmatrix} = -2F$$

$$\Rightarrow F = -\frac{1}{2} \rightarrow E$$

Answer: B

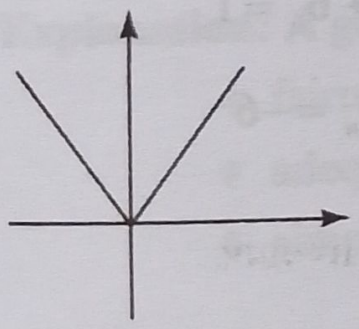
3. What is the range of function f given by $f(x) = \left|x - \frac{1}{2}\right| + 3$?

A. $[-3, \infty)$ B. $\left[-\frac{1}{2}, \infty\right)$ C. $\left[\frac{1}{2}, \infty\right)$ D. $[3, \infty)$

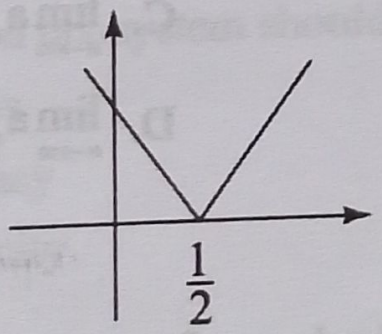
Solution: $f(x) = \left|x - \frac{1}{2}\right| + 3$

Given a function g. A new function $f(x) = g(x - h) + k$ where h and k are constants, is a horizontal shift of h units (right for a positive h and left for a negative) and a vertical shift of k units (up for a positive k and down for a negative k), of the function g.

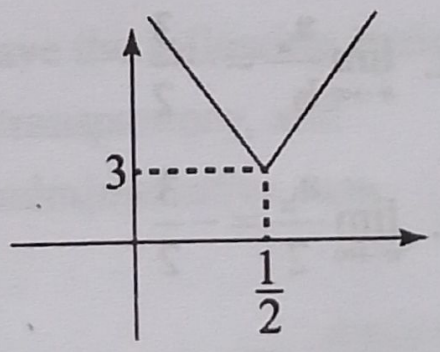
Thus,shift $|x|, \frac{1}{2}$ units to the right and 3 units up.



$s(x) = |x|$



$g(x) = |x - \frac{1}{2}|$



$f(x) = |x - \frac{1}{2}| + 3$

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$\therefore$ Range of $f = [3, \infty)$

Answer: D

4. Which one of the following is equivalent to $(p \Rightarrow q) \wedge (q \Rightarrow p)$?

A. $(p \vee q) \wedge (q \vee p)$ C. $p \wedge q$

B. $(p \Rightarrow q) \vee (q \Rightarrow p)$ D. $p \Leftrightarrow q$

Solution: $(p \Rightarrow q) \wedge (q \Rightarrow p) \equiv p \Leftrightarrow q$

Answer: D

5. If z_1 and z_2 are complex numbers, then which one of the following statement is **NOT** true about them?

A. $z_1 + z_2$ can be a real number.

B. $z_1 - z_2$ is always a real number.

C. $z_1 + z_2$ is a complex number.

D. $z_1 - z_2$ is a complex number.

Solution: Let $z_1 = a + bi$ and $z_2 = x + yi$

Then $z_1 + z_2 = (a + bi) + (x + yi)$

$= (a + x) + (b + y)\,i \leftarrow$ complex number.

if $b = -y$, (for instance, $z_1 = 3 + 2i$ and $z_2 = 4 - 2i$),

$z_1 + z_2$ is real number

$z_1 - z_2 = (a + bi) - (x + yi)$

$= (a - x) + (b - y)i \leftarrow$ complex number

Answer: B

6. If $\{a_n\}$ and $\{b_n\}$ are two given sequences so that $\lim_{n\to\infty} a_n = 3$ and $\lim_{n\to\infty} b_n = -2$, then which of the following is **NOT** true about them?

A. $\lim_{n\to\infty} \frac{a_n}{b_n} = -\frac{3}{2}$ C. $\lim_{n\to\infty} a_n + b_n = 1$

B. $\lim_{n\to\infty} \frac{a_n}{2} = -\frac{3}{2}$ D. $\lim_{n\to\infty} a_n b_n = -6$

Solution: $\lim_{n\to\infty} a_n = 3$ and $\lim_{n\to\infty} b_n = -2$

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Then, $\lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = 3 + (-2) = 1$

$\lim_{n\to\infty}(a_n \cdot b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n = 3 \cdot -2 = -6$

$\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} = \frac{3}{-2} = \frac{-3}{2}$

$\lim_{n\to\infty}\frac{a_n}{2} = \lim_{n\to\infty}\frac{1}{2}a_n = \frac{1}{2}\cdot\lim_{x\to\infty} a_n = \frac{1}{2}\cdot 3 = \frac{3}{2}$

Answer: B

7. A function f is continuous on an interval if and only if
   A. for exactly one point c is the interval $f(x) \neq f(c)$.
   B. for some point c in the interval $\lim_{x\to c} f(x) = 0$.
   C. it is continuous at every point of the interval.
   D. for some point c in the interval $f(x) \neq f(c)$.

   Explanation: A function f is continuous on an interval if and only if f is continuous at every point of the interval.

   Answer: C

8. Which one of the following is NOT principle of taxation?
   A. Compliance costs to business and administration cots for government should be minimized as far as possible.
   B. Tax rules should be clear and simple to understand so that taxpayers know where they stand.
   C. Exemption of taxes are made to ensure revenue to cover government cost of administration.
   D. Taxation should seek to be neutral and equitable between forms of business activities.

   Explanation: A good tax system should have the following basic
   - fairness
   - adequacy
   - simplicity
   - transparency, and
   - administrative ease.

   Answer: C

9. Let A and B be two sets and f be a function from A in to B($f: A \to B$). In order to say f is an on-to function which one of the following con-

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ditions should be fulfilled?

A. If you find an element y in B that is not an image of any one of the elements of A, then f is not onto function.

B. If you can find two distinct elements in A that range pre-image of exactly one elements of B, then f cannot be onto function.

C. If any two elements in B are the images of exactly one element of A, then f is onto function.

D. If each element x in A is the pre-image of at least one element y in B under f, then f is onto function.

Explanation: A function f from A to B ($f: A \longrightarrow B$) is called an onto function if for all y in B,there is at least one x in A such that $f(x) = y$. If there is at least one y in B has no pre-image in A, then f is not an onto function.

Answer: A

10. The frequency distribution table of a grouped data X is given as shown below.

X	10 –14	15 –19	20 –24	25 –29
Frequency	2	3	5	4

Which one of the following is true about this data?

A. The lower class limit of the 4th class is 24.5

B. The class mid-point of the 3rd class is 22.5

C. The class mark of the 2nd class is 17

D. The upper class boundary of the 3rd class is 24

Solution

X	f
10 –14	2
15 – 19	3
20 – 24	5
25 – 29	4

The distribution is in limit form.

Therefore, the lower class limit of the 4th class is 25.

- Class mark (Midpoint) is the number in the middle of

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the class. $CM = \frac{LCL + UCL}{2}$

$\therefore$ The class mid-point of the 3rd class $= \frac{20+24}{2} = 22$

The class mark of 2nd class $= \frac{15+19}{2} = 17$

- The upper class boundary of the 3rd class $= 24 + 0.5 = 24.5$

Answer: C

11. Which one of the following is a discrete variable?
    A. Number of road accidents in Ethiopia.
    B. Profit earned by a company.
    C. The speed of a moving object.
    D. The temperature of an object.

    Explanation: A discrete variable is a variable whose value is obtained by counting.

Answer: A

12. Which one is the standard form of the equation of the parabola with vertex $(-2, 3)$ and focus $(2, 3)$?
    A. $(y-3)^2 = -16(x+2)$
    B. $(y-3)^2 = 16(x+2)$
    C. $(x+2)^2 = -16(y-3)$
    D. $(x+2)^2 = 16(y-3)$

    Solution: V$(-2, 3)$, F $(2, 3)$

    - The parabola is horizontal and opening to the right.
    - Its equation has a form $(y-k)^2 = 4p(x-h)$

    $P = 2 - (-2) = 4$

    $\therefore$ Its is equation is $(y-3)^2 = 16(x+2)$

Answer: B

13. For a given ellipse, what do you call the ratio of the distance between the two foci to the length of the major axis?
    A. Length of minor axis
    B. Directrix
    C. Latus rectum
    D. Eccentricity

    Solution: $\frac{\text{Distance between two foci}}{\text{Length of the major axis}} = \frac{2c}{2a} = \frac{c}{a} = \text{Eccentricity}$

Answer: D

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14. When $3.1\overline{25}$ is written in the form of $\frac{a}{b}$, it is equal to

A. $\frac{3090}{1009}$ B. $\frac{3100}{990}$ C. $\frac{3094}{1000}$ D. $\frac{3094}{990}$

Solution

$$\text{Let } x = 3.1\overline{25}$$
$$\Rightarrow 10x = 31.\overline{25}$$
$$\Rightarrow 1000x = 3125.\overline{25}$$
$$1000x - 10x = 3125..\overline{25} - 31.\overline{25}$$
$$990x = 3094$$
$$x = \frac{3094}{990}$$
$$\therefore 3.1\overline{25} = \frac{3094}{990}$$

Answer: D

15. If $\{a_n\}$ is sequence with $a_1 = 3$ and $a_{n+1} = a_n + 5$ for all $n \geq 1$, then which one of the following is the 100^{th} term of the sequence?

A. 503 B. 508 C. 498 D. 500

Solution: $a_1 = 3$

$$a_{n+1} = a_n + 5, \ \forall n \geq 1$$

This implies the sequence is an arithmetic sequence with common difference d is 5.

So, $a_{100} = a_1 + 99d$
$= 3 + 99 \times 5$
$= 3 + 495 = 498$

Answer: C

16. Let $\{a_n\}_{n=1}^{\infty}$ be sequence with $a_1 = 2$ and $a_{n+1} = na_n$ for $n \in \mathbb{N}$.

Then which one of the following expression correctly describes a_n?

A. $2(n-1)!$ B. $2(n+1)!$ C. $2n!$ D. $(2n)!$

Solution: $a_1 = 2, a_{n+1} = na_n \ \forall n \in \mathbb{N}$

The first few terms are:

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$a_1 = 2$

$a_2 = 1 \cdot a_1 = 1 \cdot 2 = 2$

$a_3 = 2 \cdot a_2 = 2 \cdot 2 = 4$

$a_4 = 3 \cdot a_3 = 3 \cdot 4 = 12$

Thus, 2, 2, 4, 12 — $= 2(1), 2(1), (2(2), 2(2), —$

$= 2(0!), 2(1!), 2(2!)\ 2(3!), —$

$= 2(n-1)!$, where $n \in \mathbb{N}$

Answer: A

17. If fourth term of a geometric sequence with positive term is 125. If the tenth term is $\frac{125}{64}$, then which one of the following is the first term this sequence?

A. 750 B. 1000 C. 250 D. 500

Solution: $G_4 = 125$, $G_{10} = \frac{125}{64}$

$\Rightarrow G_{10} = r^6 G_4$

$\Rightarrow \frac{125}{64} = r^6 \cdot 125$

$\Rightarrow r^6 = \frac{1}{64}$

$\Rightarrow r = \frac{1}{2}$ (since the terms are positive)

So, $G_4 = r^3 G_1$

$125 = \left(\frac{1}{2}\right)^3 G_1$

$G_1 = 125 \times 8 = 1000$

Answer: B

18. Consider the function $f(x) = x^3 + x^2 - 4x - 3$ whose graph is shown below.

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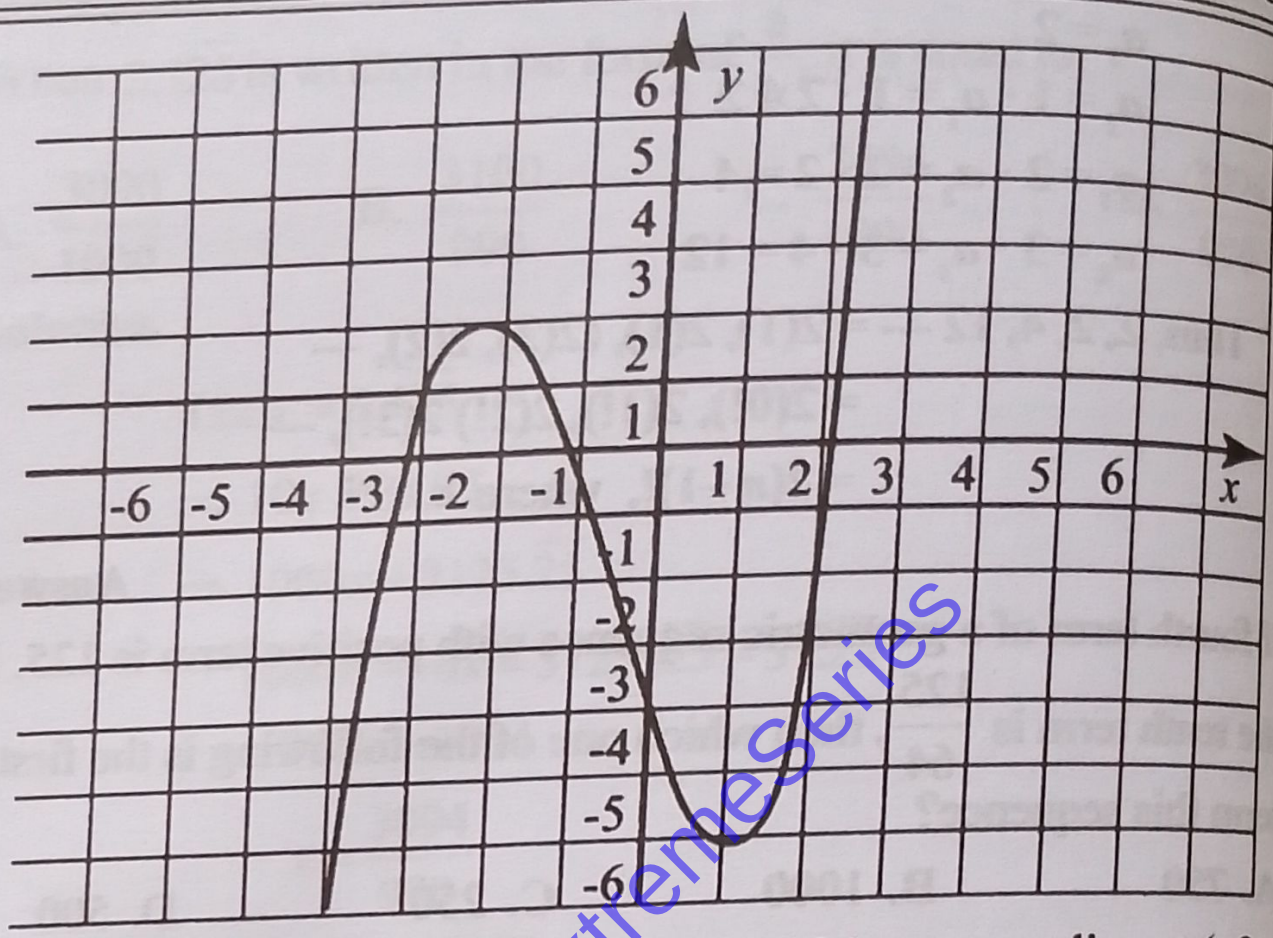

Which one of the following is the slope of the tangent line at (–2, 1)?

A. 3 B. –3 C. 4 D. –4

Solution: $f(x) = x^3 + x^2 - 4x - 3$

Slope of the tangent line at (–2, 1) = f ' (–2)

$$= \left(3x^2 + 2x - 4\right)\Big|_{x=-2}$$

$$= 3(-2)^2 + 2(-2) - 4$$

$$= 12 - 8 = 4$$

Answer: C

19. Which of the following is an equation of the line tangent to the graph of $y = x^2 + e^x$ at $x = 0$?

A. $y = x + 1$ B. $y = 2x$ C. $y = x$ D. $y = x - 1$

Solution: $y = x^2 + e^x$, at $x = 0$

$$m_t = \frac{dy}{dx}\Big|_{x=0} = 2x + e^x\Big|_{x=0} = 2(0) + e^0 = 1$$

Equation of the tangent line : $y - y_1 = m_t\left(x - x_1\right)$

$$\Rightarrow y - f(0) = 1(x - 0)$$

$$\Rightarrow y - e^0 = 1(x - 0)$$

$$\Rightarrow y - 1 = x$$

$$\Rightarrow y = x + 1$$

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Answer: A

20. If $f(x) = 4 - x^2 \sin x$, then which of the following expressions describes $f'(x)$?

A. $1 - \cos x - 2 \sin x$

B. $x(x \cos x - 2\sin x)$

C. $1 + \cos x + 2\sin x$

D. $-x(x \cos x + 2\sin x)$

Solution: $f(x) = 4x - x^2 \sin x$

Recall that

- $k' = 0$ where k is a constant.
- $(f \cdot g)' = f' g' + g' f$
- $(f \pm g)' = f' \pm g'$

So, $(4 - x^2 \sin x)' = 4' - (x^2 + \sin x)'$

$= 0 - [(x^2)' \sin x + x^2 (\sin x)']$

$= -[2x \sin x + x^2 \cos x]$

$= x[2 \sin x + x \cos x]$

Answer: D

21. A sum on money is invested at a compound interest calculated annually which amounts to 21,632 Birr at the end of the second year and to 22,497.28 Birr at the end of the third year. Which one of the following pairs of values gives the rate of interest and the principal, respectively?

A. 0.05 and 21, 000 Birr

B. 0.06 and 21,550 Birr

C. 0.03 and 18,550 Birr

D. 0.04 and 20,000 Birr

Solution

$A = \left(1 + \frac{r}{n}\right) nt$ where A = final amount

P = initial principal balance

n = number of times interst applied per time period

r = interest rate

n = number of time periods elapsed

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Then, $21,632 = P\left(1+\frac{r}{1}\right)^2$ and $22,497.28 = P\left(1+\frac{r}{1}\right)^3$

$\Rightarrow 22,497.28 = P(1+r)^2.(1+r)$

$\Rightarrow 22,497.28 = 21,632.(1+r)$

$\Rightarrow \frac{22,497.28}{21,632} = 1+r$

$\Rightarrow 1.04 = 1+r$

$\Rightarrow r = 0.04$

$\Rightarrow 21,632 = P(1+0.04)^2$

$\Rightarrow P(1.04)^2$

$P = \frac{21,632}{1.0816} = 20,000$

Answer: D

22. If 20,000 Birr is divided among three persons in the ratio 5:2:3, then which one of the following is the maximum share?

A. 4,000 Birr
B. 10,000 Birr
C. 6,000 Birr
D. 3,000 Birr

Solution: $5x + 2x + 3x = 20,000$ where x is unit share

$10x = 20,000$

$\therefore x = 2000$

So, Maximum share $= 5x = 5 \times 2000 = 10,000$

Answer: B

23. An object has a circular bas of radius r cm. Its volume (V) is directly proportional to its height (h), the constant π and the square of its base radius (r). When $r = 5$cm and $h = 12$cm, its volume $V = 100\pi cm^3$. If $r = 9$cm and $h = 10$cm, then which of the following equals to its volume?

A. $150\pi cm^3$ B. $120\pi cm^3$ C. $270\pi cm^3$ D. $210\pi cm^3$

Solution: $V \alpha h$, $V \alpha \pi$, $V \alpha r^2$

$\Rightarrow V = k\pi r^2 h$, where r is proportionality cons tan t.

$\Rightarrow 100\pi cm^3 = k\pi \cdot (5cm)^2 \cdot 12cm$

$\Rightarrow 100 = k \cdot 25 \cdot 12$

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$\Rightarrow k = \frac{1}{3}$

Let $r = 9m$, and $h = 10cm$

Then, $V = k\pi r^2 h$

$$= \frac{1}{3} \cdot \pi \cdot (9cm)^2 \cdot (10cm)$$

$$= \frac{1}{3} \cdot \pi \cdot 81cm^2 \cdot 10cm = 270\pi cm^3$$

Answer: C

24. Which one of the following pairs contains the modulus and argument of the complex number $1 + \sqrt{3}i$ respectively?

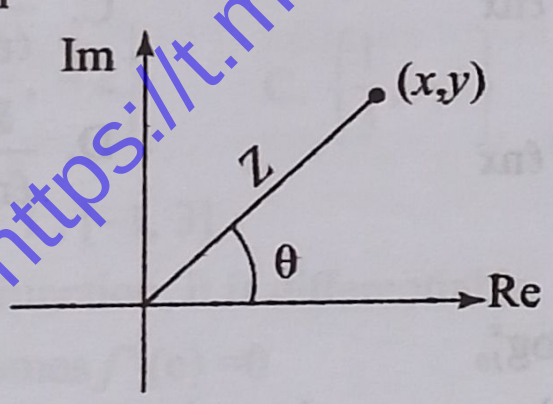
A. 1 and $\frac{\pi}{6}$ rad

B. 2 and $\frac{\pi}{3}$ rad

C. 1 and $\frac{\pi}{3}$ rad

D. 2 and $\frac{\pi}{6}$ rad

Solution: Let $z = x + yi$



Then, Modulus $= |z| = \sqrt{x^2 + y^2}$

$$\text{Argument} = \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

So, $z = 1 + \sqrt{3}i$

$$|z| = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{4} = 2$$

$$\text{Arg } z = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \tan^{-1}\sqrt{3} = \frac{\pi}{3} \text{ rad}$$

Answer: B

25. If $z_1 = 5 - 3i$ and $z_2 = 3 - 4i$, then $\frac{z_1}{z_2}$ is equal to _____.

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A. $\frac{27}{25}+\frac{29}{25}i$ B. $\frac{27}{25}-\frac{11}{25}i$ C. $\frac{27}{25}+\frac{11}{25}i$ D. $\frac{3}{25}+\frac{11}{25}i$

Solution

$$z_1 = 5-3i, \quad z_2 = 3-4i$$

$$\text{Then } \frac{z_1}{z_2} = \frac{5-3i}{3-4i} = \frac{5-3i}{3-4i}\times\frac{3+4i}{3+4i}$$

$$= \frac{15+20i-9i-12i^2}{9+15}$$

$$= \frac{15+11i+12}{25} \quad (\text{since } i^2=-1)$$

$$= \frac{27}{25}+\frac{11}{25}i$$

Answer: C

26. If $f(x) = x^3\log_{10}^x$, then which one of the following is equal to $f'(x)$?

A. $\frac{x^2}{\log_{10}^x} - 3x^2\ell nx$ C. $\frac{x^3}{\ell n10} + 3x^2\log_{10}^x$

B. $\frac{x^3}{\log_{10}^x} - 3x^2\ell nx$ D. $\frac{x^2}{\ell n10} + 3x^2\log_{10}^x$

Solution:

$$f(x) = x^3\log_{10}^x$$

$$f'(x) = (x^3)'\log_{10}^x + x3(\log_{10}^x)'$$

$$= 3x^2\log_{10}^x + x^3\left(\frac{1}{x\ell n10}\right)\left(\begin{array}{l}\text{since } \log_{10}^x = \frac{\log_e^x}{\log_e^{10}} = \ell nx \\ (\log_{10}^x) = \frac{1}{\ell n10}(\ell nx)' = \frac{1}{x\ell n10}\end{array}\right)$$

$$= 3x^2\log_{10}^x + \frac{x^2}{\ell n10}$$

Answer: D

27. Which one of the following events explains the meaning of derivative?

A. It gives the amount of money at the end of n years, if someone deposit P birr in a bank that pays r % annual interest compound

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monthly.

B. It is the slope of secant line to graph of a function f where this secant line passes through two distinct points (x_0, y_0) and (x_1, y_1) of f

C. It is the marginal cost of a product which is the rate of change of cost with respect to the level of production.

D. It gives the position of an that is moving along a coordinate line at a certain given time t.

Explanation

Derivative

$\Rightarrow$ The rate of change of a certain quantity

$\Rightarrow$ The slope of the tangent line to the graph of a function f at a point (x_0, y_0)

Answer: C

28. Which one of the following is the set of all critical numbers of $f(x) = x^3 - 2x^2 + x + 1$ on the interval $[-1, 3]$?

A. $\{1\}$ B. $\left\{\frac{1}{3}, -2\right\}$ C. $\left\{\frac{1}{3}, -1\right\}$ D. $\left\{\frac{1}{3}, 1\right\}$

Solution: $x^3 - 2x^2 + x + 1, [-1, 3]$

Since f is a polynomial function, it is differentiable.

So, critical number becomes $f'(c) = 0$

$\Rightarrow f'(x) = 3x^2 - 4x + 1$

$\Rightarrow f'(c) = 0$

$\Rightarrow 3c^2 - 4c + 1 = 0$

$\Rightarrow 3c^2 - 3c - c + 1 = 0$

$\Rightarrow 3c(c-1) + 1(c-1) = 0$

$\Rightarrow (3c-1)(c-1) = 0$

$\Rightarrow c = 1/3$ or $c = 1$

Answer: D

29. Which one of the following statements gives the geometric interpretation of the Rolle's theorem stated for a continuous function f on $[a, b]$ that is differentiable in (a, b)?

A. There is a point on the graph of f where the tangent line through

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$$=\frac{10\times9\times8}{3\times2}=\frac{720}{6}=120$$

Answer: C

33. If $f(x)=\ell n(2+\sin x)$, then $f'(x)$ is equal to

A. $\frac{\sec x}{1+\sin x}$ B. $\frac{\csc x}{1+\sin x}$ C. $\frac{\cos x}{2+\sin x}$ D. $\frac{-\cos x}{1+\sin x}$

Solution:

$f(x)=\ell n(2+\sin x)\leftarrow$ composition function

Then $f'(x)=\frac{1}{2+\sin x}\cdot(2+\sin x)'$ $\left[\because (\ell nk)'=\frac{1}{x}\right]$

$$=\frac{1}{2+\sin x}\cdot(2'+(\sin x)')$$

$$=\frac{\cos x}{2+\sin x}$$

Answer: C

34. What is the third derivative of $f(x)=\frac{1}{3}x^6+\frac{1}{2}x^4-3x^3+2x^2+4x-1$?

A. $40x^3+12x^2-18x+4$
B. $40x^3+12x-18$
C. $20x^3+12x^2-18x-4$
D. $10x^4+6x^2-18x+4$

Solution

$$f(x)=\frac{1}{3}x^6+\frac{1}{2}x^4-3x^3+2x^2+4x-1$$

$$f'(x)=\frac{1}{3}\cdot6x^5+\frac{1}{2}\cdot4x^3-3\cdot3x^2+2\cdot2x+4$$

$$=2x^5+2x^3-9x^2+4x+4$$

$$f''(x)=2\cdot5x^4+2\cdot3x^2-9\cdot2x+4$$

$$10x^4+6x^2-18x+4$$

$$f'''(x)=10\cdot4x^3+6\cdot2x-18$$

$$=40x^3+12x-18$$

Answer: B

35. Which one of the following is the contra positive of the statement:

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"If 9 is divisible by 2, then 5 is an even number"?

A. If 5 is an even number, then 9 is divisible by 2.

B. If 5 is not an even number, the 9 is not divisible by 2.

C. If 9 is not divisible by 2, then 5 is not an even number.

D. If 5 is not an even number, then 9 is divisible by 2.

Solution: Let p: q is divisible by 2

q: 5 is an even number

So, if is q divisible by 2, then 5 is an even number $\equiv p \Rightarrow q$.

The contrapositive of $p \Rightarrow q$ is $\neg q \Rightarrow p$.

i.e If 5 is not an even number, then q is not divisible by 2.

Answer: B

36. If $p \Rightarrow \neg q$ is False, then for any proposition r which one of the following statements is True?

A. $(p \vee q) \Rightarrow (p \wedge r)$

B. $(\neg p \vee r) \Leftrightarrow (\neg q \wedge r)$

C. $(p \wedge \neg q) \Leftrightarrow (\neg r \wedge r)$

D. $(p \wedge q) \Rightarrow (r \wedge \neg r)$

Solution

$p \Rightarrow \neg q \equiv$ F is p is T and $\neg q$ is F.

Since, for any proposition r, $\neg r \wedge$ T is F.

Therefore, $(p \wedge \neg q) \Leftrightarrow (\neg r \wedge r) \equiv (T \wedge F) \Leftrightarrow F$

$\equiv F \Leftrightarrow F = T$

Answer: C

37. A textile factory produces shirts and calculates the revenue r, cost c and profit p as functions of the number of shirts. The functions are as follows:

$r(x)$ = the revenue from selling × shirts,

$c(x)$ = the cost of producing the × shirts,

$p(x) = r(x) - c(x)$ = the profit from producing and selling × shirts.

If $r(x) = 9x$ and $c(x) = x^3 - 6x^2 + 15x$, where x represents thousands of shirts produced, then which one of the following statements is correct about the loss or profit gained at the given production level?

A. Maximum loss occurs at about $x = 1586$

B. Maximum profit occurs at about $x = 6828$

C. Maximum loss occurs at about $x = 2510$

D. Maximum profit occurs at about $x = 3414$

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Solution

$$p(x) = r(x) - c(x)$$
$$= 9x - (x^3 - 6x^2 15x)$$
$$= -x^3 + 6x^2 - 6x$$

It is a polynomial function of the form left up and right down. (И).
This implies the function has number maximum or minimum value.

Answer:

38. Let a function f be continuous on $[a, b]$ and differentiable in (a, b). Which one of the following is correct about f?

A. If $f'(c) = 0$ and f'' is continuous at c such that $f''(c) > 0$, then f has a local maximum at c.

B. If $f(a) = f(b)$ and $f(x) > f(a)$ for all $x \in (a,b)$, then f has got a minimum value at $c \in (a,b)$ where $f'(c) = 0$.

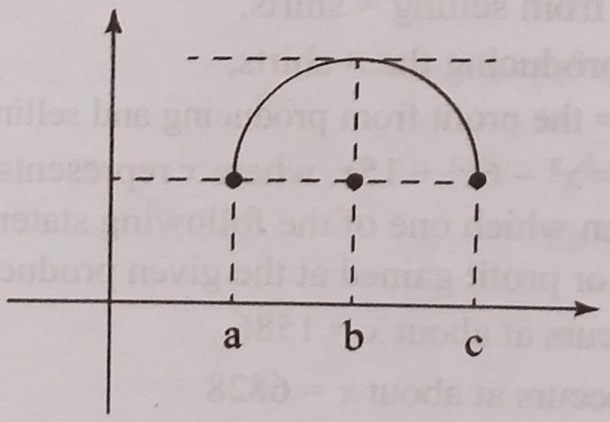
C. If $f'(x) \leq 0$ for all $x \in (a,b)$, then f is increasing on (a, b).

D. $f'(x) = 0$ for all $x \in (a,b)$, then f is a constant on $[a, b]$

Solution

Let f continuous on $[a, b]$ and differentiable in (a, b),

- If $f'(x) = 0 \ \forall \ x \in (a,b)$, then f is constant on (a, b).
- If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.
- If $f'(x) \leq \ \forall x \in (a,b)$, then f is decreasing on (a, b).
- If $f(a) = f(b)$ and $f(x) > f(a) \ \forall x \in (a,b)$ then f has got a maximum value at $c \in (a,b)$ where $f'(c) = 0$



Answer: D

39. Which one of the following is a possible solution for

$$\frac{2x^2 - 5x + 2}{2x^2 - x} = 1 - \frac{1 + x}{2x - 1}$$

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A. $\frac{1}{2}$ B. -1 C. 2 D. 0

Solution

$$\frac{2x^2-5x+2}{2x^2-x}=1-\frac{1+x}{2x-1}$$

$$\Rightarrow \frac{2x^2-5x+2}{x(2x-1)}=\frac{2x-1-(1+x)}{2x-1}$$

$$\Rightarrow \frac{2x^2-5x+2}{x(2x-1)}=\frac{x-2}{2x-1}$$

$$\Rightarrow \frac{2x^2-5x+2}{x}=x^2-2x,\quad x\neq\frac{1}{2}$$

$$\Rightarrow 2x^2-5x+2=x-2,\quad x\neq 0$$

$$\Rightarrow x^2-3x+2=0$$

$$\Rightarrow (x-2)(x-1)=0$$

$$\Rightarrow x-2=0\ ,x-1=0$$

$$\Rightarrow x=2\ ,\ x=1$$

$$S.S=\{1,2\}$$

Answer: C

40. Which of the following is true the graph of the function given by $f(x)=\frac{-8}{x^2-4}$?

A. Its horizontal asymptote is $y=0$
B. Its vertical asymptote is $x=4$
C. Its is symmetric about the origin.
D. (2, 0) is the x-intercept.

Solution: $f(x)=\frac{-8}{x^2-4}=\frac{-8}{(x-2)(x^2)}$ The graph of f has

- ✓ a horizontal asymptote $y=0$
- ✓ a vertical asymptote $x=2$ and $x=-2$
- ✓ no x-intercept.

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i.e $0 = \dfrac{-8}{x^2 - 4}$

$\Rightarrow 0 = -8$ which is false.

Since, $f(-x) = \dfrac{-8}{(-x)^2 - 4} = \dfrac{-8}{x^2 - 4} = f(x)$,

the graph is symmetric w.r.t the y- axis.

Answer: A

41. Which one of the following functions is **NOT** invertible?

A. $h = [(x, y): y = \log_3 2x\}$, where $h : (0, \infty) \to \mathbb{R}$

B. $k = \{(x, y) : y = e^{2x}\}$, where $k : \mathbb{R} \to (0, \infty)$

C. $f = \{(x, y) : y = 4x - 3\}$, where $f : \mathbb{R} \to \mathbb{R}$

D. $f = \{(x, y) : y = x^2 + 1\}$ where $g : \mathbb{R} \to \mathbb{R}$

Solution

Let f be a function whose domain is the set A and whose co-domain is the set B (i.e $f : A \to B$). Then f is invertiable if there exits a function g with domian B and codomain A, with the property: $f(x) = y \Leftrightarrow = x$.

So, g: $\{(x, y): y = x^2 + 1\}$, where $g : \mathbb{R} \to \mathbb{R}$ has no inverse.

Answer: D

42. If $A = \begin{pmatrix} a & 1 & -1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}$, then det (A)= _____.

A. $a^3 - 1$ B. $-a(a^2 - 1)$ C. $a^2 - a$ D. $a^3 - a$

Solution

$$A = \begin{pmatrix} a & 1 & -1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}$$

$$\det A = a\begin{vmatrix} a & 1 \\ 1 & a \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & a \end{vmatrix} + (-1)\begin{vmatrix} 1 & a \\ 1 & 1 \end{vmatrix} \left(1^{st} \text{ row expansion}\right)$$

$$= a\left(a^2 - 1\right) - 1(a - 1) - 1(1 - a)$$

$$= a^3 - a - a + 1 - 1 + a = a^3 - a$$

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Answer: D

43. Let $A=\begin{pmatrix}-4 & 2 & 0\\ 1 & 3 & 6\\ 5 & 7 & 9\end{pmatrix}$. Which of the following is **NOT** true about this matrix?

A. Minor of a_{13} = Cofactor of a_{31}
B. Cofactor of $a_{23} = 38$
C. Minor of $a_{23} = -38$
D. Cofactor of $a_{13} = -8$

Solution

$$A=\begin{pmatrix}-4 & 2 & 0\\ 1 & 3 & 6\\ 5 & 7 & 9\end{pmatrix}$$

Minor of $a_{13} = |A_{13}| = \begin{vmatrix}1 & 3\\ 5 & 7\end{vmatrix}$ (A_{13} – a sub matrix obtained by deleting the 1st row and 3rd column.)

$= 1\times 7 - 5\times 3$

$= 7 - 15 = -8$

Cofactor of $a_{13} = (-1)^{1+3}|A_{13}|$, Cofactor of $a_{31} = (-1)^{3+1}|A_{31}|$

$= (+1)4\cdot -8$ $= (-1)^4\begin{vmatrix}2 & 0\\ 3 & 6\end{vmatrix}$

$= 2\times 6 - 3\times 0 = 12$

Minor of $a_{23} = |A| = |A_{23}| = \begin{vmatrix}-4 & 2\\ 5 & 7\end{vmatrix}$

$= -4\times 7 - 5\times 2$

$= -28 - 10 = -38$

Cofactor of $a_{13} = (-1)^{2+3}|A_{23}|$

$= (-1)^5\cdot -38 = 38$

Answer: A

44. Which one of the following sequence has no upper bound?

A. $\{2^n - 1\}$ B. $\left\{\frac{4n-1}{2n}\right\}$ C. $\left\{\frac{(-1)^n}{n}\right\}$ D. $\{1-n)$

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sequence $\{a_n\}$. One way of determining the upper bound of a given sequence is list the 1st few terms and examine their nature.

For instance,

C) $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \frac{1}{6}$

$\Rightarrow -1, -\frac{1}{3}, -\frac{1}{5}, \frac{1}{6}, \cdots\cdots \frac{1}{6}, \frac{1}{4}, \frac{1}{2}$ ← Bounded

$\therefore$ The set of upper bounds : $\left[\frac{1}{2}, \infty\right)$

D) $0, -1, -2, -3, -4,$

$\Rightarrow -\infty, \leftarrow\text{---}, -4, -3, -2, -1, 0.$

$\therefore$ The set of upperbounds : $[0, \infty)$

B) The set of upperbounds : $[2, \infty)$, because $\frac{3}{2} \leq \frac{4n-1}{2n} < 2$

A) The first few terms are 1, 3, 7, 15, ← Increasing without bound.

$\therefore$ $\{2^n - 1\}$ has no upper bound

Answer: A

45. Which one of the following matrics is the inverse of $\begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix}$?

A. $\begin{pmatrix} \frac{1}{5} & \frac{-1}{3} \\ -1 & \frac{1}{2} \end{pmatrix}$

B. $\begin{pmatrix} \frac{5}{7} & \frac{-3}{7} \\ \frac{-1}{7} & \frac{2}{7} \end{pmatrix}$

C. $\begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ 1 & \frac{1}{5} \end{pmatrix}$

D. $\begin{pmatrix} \frac{5}{13} & \frac{-3}{13} \\ \frac{-1}{13} & \frac{2}{13} \end{pmatrix}$

Solution

Let $A = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix}$

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Then $|A| = \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix} = 2\times5 - 3\times1 = 7 \neq 0$

Thus A is invertible; and

$$A^{-1} = \frac{1}{|A|}\begin{pmatrix} 5 & -3 \\ -1 & 2 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 5 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{7} & \frac{-3}{7} \\ \frac{-1}{7} & \frac{2}{7} \end{pmatrix}$$

Answer: B

46. Which one of the following is equal to $\lim_{x\to5} \frac{x-5}{\sqrt{x+4}-3}$?

A. 1 B. 6 C. ∞ D. 0

Solution

$$\lim_{x\to\infty} \frac{x-5}{\sqrt{x+4}-3}\left(\frac{0}{0} \text{ form}\right)$$

$$\lim_{x\to\infty} \frac{(x-5)}{\sqrt{x+4}} \frac{\left(\sqrt{x+4}+3\right)}{\left(\sqrt{x+4}+3\right)} \text{(Rationalize the demon int or)}$$

$$= \lim_{x\to5} \frac{(x-5)\left(\sqrt{x+4}+3\right)}{x-5}$$

$$= \lim_{x\to5} \sqrt{x+4}+3 = \sqrt{5+4}+3$$

$$= 3+3=6$$

Answer: B

47. If two $\ell_1 = 2x+3y-5=0$ and $\ell_2 : kx+2y+p=0$, are perpendicular at point (–2, 3), then which one of the following pairs gives the correct values of k and p?

A. k = –3 and p = –12
B. k = 3 and p = 15
C. k = 3 and p = –6
D. k = –3 and p = 9

Solution

$\ell_1 : 2x+3y-5=0$ and $\ell_2 : kx+2y+p=0$

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$$\Rightarrow 3y = -2x + 5$$

$$y = \frac{-2}{3}x + \frac{5}{3}$$

$$\Rightarrow m_1 = \frac{-2}{3}$$

$$2y = -kx - p$$

$$y = \frac{-k}{2} - \frac{p}{2}$$

$$\Rightarrow m_2 = \frac{-k}{2}$$

Since $\ell_1 \perp \ell_2$, $m1 \cdot m_2 = -1$

$$\frac{-2}{3} \cdot \frac{-k}{2} = -1$$

$$\frac{k}{3} = -1, \quad k = -3$$

Since $(-2, 3)$ is the *point* of *intersection*, then

$$kx + 2y + p = -3(-2) + 2(3) + p = 0 \Rightarrow p = -12$$

Answer:

48. Which one of the following is the value of $\lim_{x \to 2} \frac{\sqrt{x+2} - 2}{x - 2}$

A. 2 B. $\frac{1}{2}$ C. 4 D. $\frac{1}{4}$

Solution

$$\lim_{x \to 2} \frac{\sqrt{x+2} - 2}{x - 2} \qquad \left(\frac{0}{0} \text{ form}\right)$$

$$= \lim_{x \to 2} \frac{\sqrt{x+2} - 2}{x - 2} \cdot \frac{\sqrt{x+2} + 2}{\sqrt{x+2} + 2}$$

$$= \lim_{x \to 2} \frac{(x+2) - 4}{(x - 2)\left(\sqrt{x+2} + 2\right)}$$

$$= \lim_{x \to 2} \frac{x - 2}{(x - 2)}$$

$$= \frac{1}{\sqrt{2+2} + 2} = \frac{1}{4}$$

Answer:

49. For all $n \in \mathbb{N}$, which one of the following sequences is a null se-

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quence?

A. $\left\{\left(\frac{7}{5}\right)^n\right\}$ B. $\left\{\frac{n^2}{n^2+1}\right\}$ C. $\left\{\left(\frac{1}{8}\right)^n\right\}$ D. $\left\{\frac{n}{n+1}\right\}$

Solution: $\{a_n\}$ is a null sequence if and only if $\lim_{n\to\infty} a_n = 0$.

So, $\lim_{n\to\infty}\left(\frac{1}{8}\right)^n = 0$

Answer: C

50. The nth term of a sequence $\{a_n\}$ is given by $a_n = \frac{1-2n}{1+2n}$, then which of the following statements is true about this sequence?

A. It is unbounded sequence and hence diverges.
B. It is a monotonic sequence and converges to -1.
C. It coverages to its least upper bound 0.
D. All its terms of positive numbers and hence diverges.

Solution

$$a_n = \frac{1-2n}{1+2n}$$

$$\lim_{n\to\infty} an = \lim_{n\to\infty} \frac{1-2n}{1+2n} = \lim_{n\to\infty} \frac{\frac{1}{2n} - \frac{2n}{2n}}{\frac{1}{2n} + \frac{2n}{2n}} = \lim_{n\to\infty} \frac{\frac{1}{2n} - 1}{\frac{1}{2n} + 1} = -1$$

$\therefore$ The sequence converges

The 1st few terms are: $\frac{-1}{3}, \frac{-3}{5}, \frac{-5}{7}, \frac{-7}{9}$

$\Rightarrow \frac{-1}{3} > \frac{-3}{5}, > \frac{-5}{7}, > \frac{-7}{9} \leftarrow$ Decreasing

$\therefore$ The sequence is monotonic and coverages to -1.

Answer: B

51. If $f(x) = x^{3/2} + \frac{8}{x} + 7$, then $\lim_{x\to 4} \frac{f(x) - f(4)}{x-4} =$ _____.

A. $\frac{5}{2}$ B. $\frac{7}{2}$ C. 1 D. $\frac{3}{2}$

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Solution

$$f(x)=x^{\frac{3}{2}}+\frac{8}{x}+7$$

Since, $\lim_{x\to 4}\frac{f(x)-f(4)}{x-4}=f'(4)$

$$\Rightarrow f'(x)=\frac{3}{2}x^{\frac{3}{2}-1}+8\left(-\frac{1}{x^2}\right)\left[\because (x^n)'=nx^{n-1}\right]$$

$$=\frac{3}{2}x^{\frac{1}{2}}-\frac{8}{x^2}$$

$$\text{So, } f'(4)=\frac{3}{2}(4)^{\frac{1}{2}}-\frac{8}{4^2}$$

$$=\frac{3}{2}\cdot 2-\frac{8}{16}$$

$$=3-\frac{1}{2}=\frac{5}{2}$$

Answer: A

52. If $f\cdot g$ and h be differentiable functions such that $f(x)=g(x)+c$, for some $c\in\mathbb{R}$. If $f'(x)=e^{-x}-2$ and $\frac{d}{dx}(f(x)+g(x)-h(x))e^{-x}$, then $h'(x)=$ _____.

A. $2-g'(x)$

B. $e^{-x}-2+g'(x)$

C. $e^{-x}-4$

D. $2e^{-x}+2$

Solution: $f(x)=g(x)+c$

$$\Rightarrow f'(x)=g'(x)\quad(\because c'=0)$$

Let $f'(x)=e^{-x}-2$

$$\frac{d}{dx}(f(x)+g(x)-h(x))=e^{-x}$$

$$\Rightarrow f'(x)+g'(x)-h'(x)=e^{-x}$$

$$\Rightarrow e^{-x}-2+e^{-x}-2-h'(x)=e^{-x}$$

$$\Rightarrow 2e^{-x}-4-h'(x)=e^{-x}$$

$$\Rightarrow 2e^{-x}-e^{-x}-4=h'(x)$$

$$\Rightarrow h'(x)=e^{-x}-4$$

Answer: C

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53. Consider the following linear programming model:

objective function $Z = 3x + 5y$

subject $3x + y \leq 6$

$x + y \geq 1$

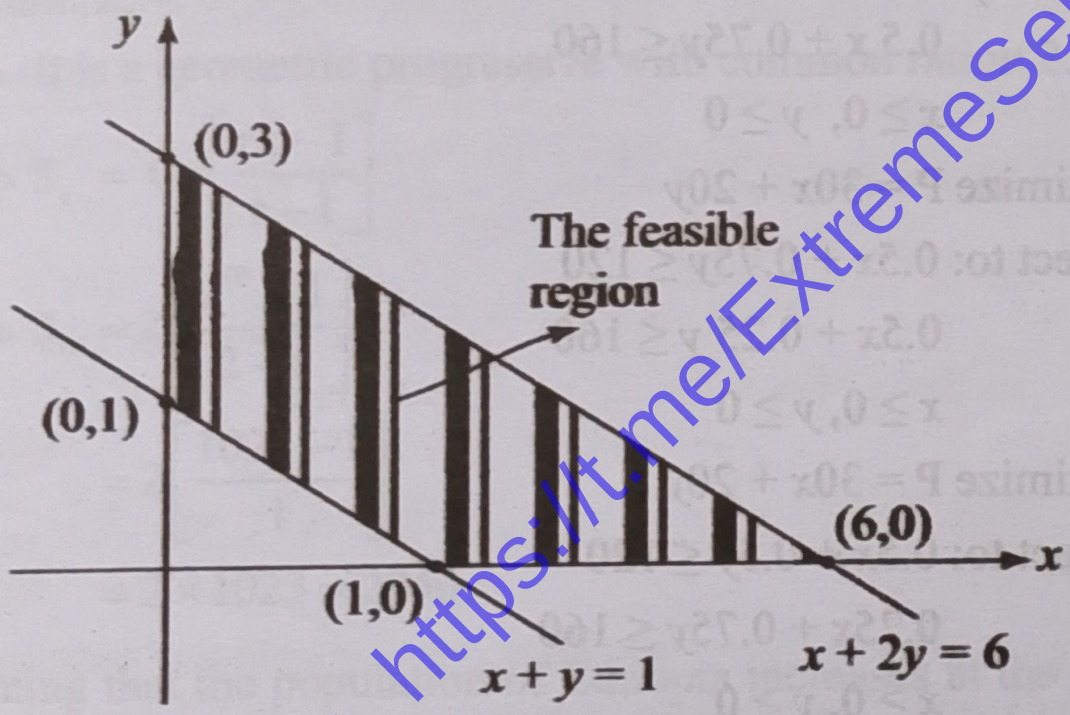
$x + 2y \leq 6$

$x \geq 0,\ y \geq 0$

Which one of the following the minimum value of Z?

A. 5 C. 0

B. Its has no minimum value D. 3



The maximum or minimum value of an objective function lies at the corner points.

So, $Z(0.1) = 3(0) + 5(1) = 5$

$Z(0.3) = 3(0) + 5(3) = 15$

$Z(1.0) = 3(1) + 5(0) = 3$

$Z(6.0) = 3(6) + 5(0) = 18$

∴ The minimum value of Z is 3 and lies at (1, 0)

Answer: D

54. A problem to be solved using linear programming is given as follows: "A coffee packer blends coffee from Harar and from Sidama to prepare two kinds products, viz., "Super" and "Deluxe" brands. Each kilogram of Super coffee contains 0.5 kg of Harar coffee and 0.5 kg of Sidama coffee, whereas Deluxe coffee contains 0.25 kg of Harar coffee and 0.75 kg of Sidama coffee. The packer has 120 kg of Harar coffee and 160 kg

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Sidama coffee in the store. Moreover, the plan is to get 20 Birr profit on each kilogram of Supper coffee and a profit of 30 Birr on each kilogram of Deluxe coffee. In order to maximize profit the packer needs to know the amount (in kg) of each brand of coffee that should be blended.

If x = the amount of Super coffee in kilogram and

y = the amount of Deluxe coffee in kilograms, then which one of the following standard forms of linear programming models should be used to solve the above problem?

A. Maximize $P = 20x + 30y$

subject to: $0.5x + 0.25y \leq 120$

$0.5x + 0.75y \leq 160$

$x \geq 0,\ y \geq 0$

B. Maximize $P = 30x + 20y$

subject to: $0.5x + 0.75y \leq 120$

$0.5x + 0.25y \leq 160$

$x \geq 0, y \geq 0$

C. Maximize $P = 30x + 20y$

subject to: $0.5x + 0.5y \leq 120$

$0.25x + 0.75y \leq 160$

$x \geq 0, y \geq 0$

D. Maximize $P = 20x + 30y$

subject to: $0.75x + 0.25y \leq 120$

$0.5x + 0.5y \leq 160$

$x \geq 0, y \geq 0.$

Solution

	Brand		
	Super	**Deluxe**	**Total**
Harar	0.5kg	0.25kg	120kg
Sidam	0.5kg	0.75kg	160kg
Profit	20 Birr/kg	30 Birr/kg	

Let x = the amount of super coffee in kgs.

y = the amount of Deluxe coffee in kgs.

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Then $Z = 20x + 30y$

constraints: $0.5x + 025y \leq 120$

$0.5x + 0.75y \leq 160$

$x \geq 0,\ y \geq 0$

Answer: A

55. A person has 2 parents, 4 grandparents, 8 great grandparents and so on. What is the total number of his ancestors during the ten generations preceding his own?

A. 5048 B. 2064 C. 1024 D. 4092

Solution: 2, 4, 8 ----

$\Rightarrow$ It is a geometric progressive with common ration 2.

$$\Rightarrow S_n = G_1\left[\frac{rn-1}{r-1}\right]$$

$$\Rightarrow S_{10} = 2\left[\frac{2^{10}-1}{2-1}\right]$$

$$= 2\left[\frac{1024-1}{1}\right]$$

$$= 2 \times 1023 = 2046$$

Answer: B

56. Assuming that the population of Ethiopia increases at the rate of 3% per year and the current population is 102 million, which one of the following is the best approximate of the population of Ethiopia 10 years from now?

A. 133 million C. 125 million

B. 141 million D. 137 million

Solution: r = 3%

Current population = 102 million

After one years = 102M + 3% × 102 M

= 102M + 0.03 × 102M

= 102M (1 + 0.03)

After two years = 102M + 3% (102M + 3 %102M)

= 102M [1 + 0.03)3

After ten years = 102M [1+0.03]10

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$= 102M\,(10.3)10$

$= 102M \times 1.344$

$= 137.088M$

Answer: D

57. What is the number that satisfies the conclusion of the Mean Value Theorem for $f(x) = x^3 - 1$ on the interval [0, 1]?

A. $\frac{2}{3}\sqrt{3}$ B. $\frac{\sqrt{2}}{3}$ C. $\frac{2}{3}$ D. $\sqrt{\frac{2}{3}}$

Solution

MVT: Let f is continuous on [a, b] and differentiable in (a, b) then there is at least one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

Thus, $f'(c) = 3c^2$ and $f(2) = 2^3 - 1 = 7, f(0) = -1$

$$\Rightarrow 3c^2 = \frac{7 - (-1)}{2 - 0}$$

$$\Rightarrow 3c^2 = 4$$

$$\Rightarrow c = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

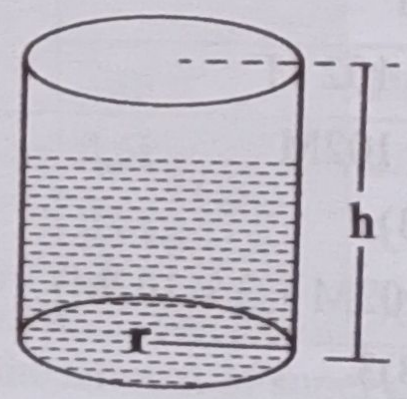
Answer: A

58. A water tank is in the form of a right circular cylinder with base radius 2m and height 3m. If water is pumped into the tank at the rate of $0.5m^3/min$, what is the rate at which the water level is rising?

A. $\frac{3\pi}{8}$ m / min C. $\frac{1}{8\pi}$ m / min

B. $\frac{4\pi}{3}$ m / min D. $\frac{8}{\pi}$ m / min

Solution



$r = 2m$

$h = 3m$

$\frac{d_v}{d_t} = 0.5m^3 / min$

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$$v = \pi r^2 h$$

$$\frac{d_v}{d_t} = \pi r^2 \cdot \frac{d_h}{d_t}$$

$$0.5m^3/\min = \pi.(2m)^2 \frac{d_h}{d_t}$$

$$\frac{0.5m^{\frac{3}{\min}}}{4\pi} = \frac{d_h}{d_t}$$

$$\Rightarrow \frac{d_h}{d_t} = \frac{1}{8\pi} m/\min$$

Answer: C

59. What is the tangent of the angle formed by the lines $\ell_1 : 2x+3y=5$ and $\ell_2 : x-2y=4$?

A. $\frac{2}{3}$ B. $-\frac{1}{4}$ C. $\frac{1}{3}$ D. $-\frac{7}{4}$

Solution

$$\ell_1 : 2x+3y=5$$

$$\Rightarrow 3y = -2x+5$$

$$\Rightarrow y = \frac{-2}{3}x + \frac{5}{3}$$

$$\therefore m_1 = \frac{-2}{3}$$

$$\ell_2 = x-2y=4$$

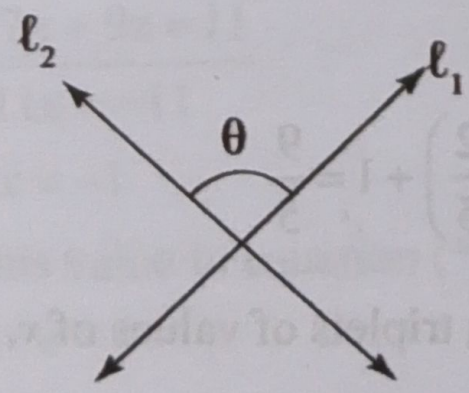
$$\Rightarrow x-4=2y$$

$$\Rightarrow y = \frac{1}{2}x - 2$$

$$\therefore m_2 = \frac{1}{2}$$

From their slopes, ℓ_1 and ℓ_2 are neither parallel ($m_1 \neq m_2$) nor perpendicular ($m_1 m_2 \neq -1$)

Let θ be the angle between ℓ_1 and ℓ_2 (where θ may be acute or obtuse)



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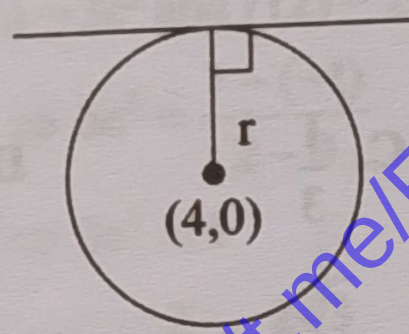
$$\tan\theta = \pm\frac{m_2 - m_1}{1 + m_1 + m_2} = \pm\frac{\frac{1}{2} - \left(\frac{-2}{3}\right)}{1 + \frac{1}{2}\cdot\left(\frac{-1}{2}\right)} = \pm\frac{\frac{1}{2} + \frac{2}{3}}{1 - \frac{1}{3}} = \pm\frac{\frac{7}{6}}{\frac{2}{3}} = \pm\frac{7}{4}$$

Answer: D

60. If the line $y = 2x + 1$ is tangent to a circle centered at (4, 0), then what are the coordinates for the intersection point of the line and the circle?

A. $\left(\frac{2}{5}, \frac{9}{5}\right)$ B. $\left(\frac{2}{5}, \frac{8}{5}\right)$ C. $\left(\frac{3}{4}, \frac{5}{2}\right)$ D. $\left(\frac{4}{5}, \frac{8}{5}\right)$

Solution



Equation of the circle $= (x-4)^2 + y^2 = r^2$ and the distance between the line and the center is given by

$$r = \left|\frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}\right| = \left|\frac{2\cdot 4 + (-1)\cdot 0 + 1}{\sqrt{2^2 + (-1)^2}}\right| = \frac{9}{\sqrt{5}}.$$

$$\Rightarrow (x-4)^2 + (2x+1)^2 = \left(\frac{9}{\sqrt{2}}\right)^2$$

$$\Rightarrow x^2 - 8x + 16 + 4x^2 + 4x + 1 = \frac{81}{5}$$

$$\Rightarrow (5x^2 - 4x + 17)\times 5 = 81$$

$$\Rightarrow 25x^5 - 20x + 4 = 0$$

$$\Rightarrow x = \frac{2}{5} \text{ and } y = 2\left(\frac{2}{5}\right) + 1 = \frac{9}{5}$$

Answer: A

61. Which one of the following triplets of values of x, y and z contains all solutions for

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$$\begin{cases} 4x - y + z = -5 \\ 2x + 2y + 3z = 10 \\ 5x - 2y + 6z = 1 \end{cases}$$

A. $x = 2, y = -1, z = 3$
B. $x = 3, y = 2, z = -1$
C. $x = 3, y = -1, z = 2$
D. $x = -1, y = 3, z = 2$

Solution

$$\begin{cases} 4x - y + z = -5 \\ 2x + 2y + 3z = 10 \\ 5x - 2y + 6z = 1 \end{cases}$$

Solve the 1st two equations.

$$2\begin{cases} 4x - y + z = -5 \\ 2x + 2y + 3z = 10 \end{cases}$$

$$+\begin{cases} 8x - 2y + 2z = -10 \\ 2x + 2y + 3z = 10 \end{cases}$$

$$10x + 5z = 0$$

$$2x + z = 0 \quad ---*$$

Then solve the last two equations together

$$\begin{cases} 2x + 2y + 3z = 10 \\ 5x - 2y + 6z = 1 \end{cases}$$

$$7x + 9z = 11 \quad ---(**)$$

Solve equation (*) and (**) together

$$9\begin{cases} 2x + z = 0 \\ 7x + 9z = 11 \end{cases}$$

$$-\begin{cases} 18x + 9z = 0 \\ 7x + 9z = 11 \end{cases}$$

$$11x = -11$$

$$x = -1$$

Substitute this value in equation (*).

$$2(-1) + z = 0$$

$$z = 2$$

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Substitute there values either of the ooriginal systme.

$$A(-1) - y + 2 = -5$$

$$\Rightarrow -y - 2 = -5$$

$$\Rightarrow -y = -3$$

$$\Rightarrow y = 3$$

$$\therefore S.S = \{(-1, 3, 2)\}$$

Answer: D

62. The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{1}{(k+2)(k+3)} \right)$ is equal to _____.

A. $\frac{2}{3}$ B. 0 C. $\frac{1}{3}$ D. $\frac{4}{3}$

Solution

First decompare $\frac{1}{(k+2)(k+3)}$

$$\frac{1}{(k+2)(k+3)} = \frac{A}{k+2} + \frac{B}{k+3}$$

Solve for A and B, we get $A = 1$ and $B = -1$

Therefore,

$$\frac{1}{(k+2)(k+3)} = \frac{1}{k+2} - \frac{1}{k+3}$$

Then, $\sum_{k=1}^{n} \left(\frac{1}{(k+2)(k+3)} \right) = \sum_{k=1}^{n} \left(\frac{1}{k+2} - \frac{1}{k+3} \right)$

$$= \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + - - - - + \left(\frac{1}{n+2} - \frac{1}{n+3}\right)$$

$$\frac{1}{3} - \frac{1}{n+3}$$

So, $\lim_{x\to\infty} \sum_{k=1}^{n} \frac{1}{(k+2)(k+3)} = \lim_{n\to\infty} \left(\frac{1}{3} - \frac{n}{n+3} \right) = \frac{1}{3}$

Answer: C

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63. Let $f(x)=\begin{cases}\dfrac{x^3+8}{x+2}, & \text{if } x\neq -2,\\ 12, & \text{if } x=-2\end{cases}$,

What is the maximum possible interval on which f continuous?

A. $(-\infty,\infty)$

B. $(-\infty,-2)\cup(-2,\infty)$

C. $(-\infty,-2)$

D. $(-2,\infty)$

Solution:

$$f(x)=\begin{cases}\dfrac{x^3+8}{x+2}, & \text{if } x\neq -2,\\ 12, & \text{if } x=-2\end{cases}$$

WTS f is continuous at -2. That is, $\lim_{x\to -2} f(x)=f(-2)$

$$\lim_{x\to -2}=f(x)=\lim_{x\to -2}\frac{x^3+8}{x+2}=\lim_{x\to -2}\frac{x^3+x^2}{x+2}=\lim_{x\to -2}\frac{(x+2)(x^2-2x+4)}{x+2}$$

$$=\lim_{x\to -2}x^2-2x+4=(-2)^2-2(2)+4=12=f(-2)$$

$\therefore f$ is continuous at -2.

Since $\lim_{x\to a} f(x)=f(a)$, where $a\neq -2$, f is continuous at any real number $a\neq -2$.

This implies f is continuous at every real number.

Answer: A

64. Among a group of 200 students who are asked whether or not they like Mathematics and/or History, it is found that 80of them like Mathematics only, 20 like History and 40 like both. If a student is randomly selected from the group, what is the probability that the likes neither Mathematics nor History?

A. 0.7 B. 0.8 C. 0.3 D. 0.5

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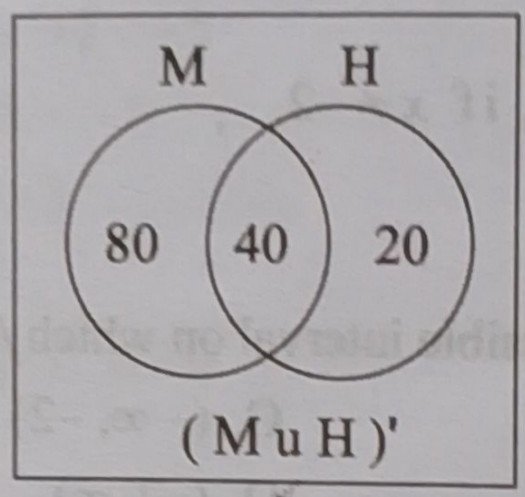
Solution



$$n(M \cup H) = n(H) + n(H) - n(H \cap H)$$

$$120 + 60 - 40 = 140$$

$$n(H \cup H)' = 200 - 140 = 60 \leftarrow \text{neither mathematics nor History likes}$$

$$p(\text{neither}) = \frac{60}{200} = 0.3$$

Answer: C

65. What is the sixth term in the expanded form of $(x + 2y)^8$?

A. $1024xy^7$ B. $448x^5y^5$ C. $1792x^3y^3$ D. $1120x^4y^4$

Solution

In the expansion of $(x + y)^n$, the $(r + 1)^{th}$ term is determined by

$$T_{r+1} = \binom{n}{r} x^{n-r} y^r$$

$$\text{So, } T_6 = \binom{8}{5} x^{8-5} (2y)^5 \quad (\because r+1 = 6 \text{ and } n = 8 \quad r = 5)$$

$$= \frac{8!}{3!5!} \times x^3 \cdot 2^5 \cdot y^5 \quad \left(\because \binom{n}{r} = \frac{n!}{r!(n-r)!}\right)$$

$$= \frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 5!} x^3 \cdot 32 \cdot y^5$$

$$= 56 \times 32 \times x^3 y^5 = 1792 x^3 y^5$$

Answer: C

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1. What is the determinant of the matrix $\begin{bmatrix} 1 & -7 \\ 6 & 5 \end{bmatrix}$?

A. 47 B. 37 C. –37 D. –47

Solution:

$$\text{Det}(A) = |A| = \begin{pmatrix} 1 & -7 \\ 6 & 5 \end{pmatrix} = (1)(5) - (6)(-7) = 5 + 42 = 47$$

Answer: A

2. The additive inverse of the complex number $3 - 4i$ is

A. $3 - 4i$ B. $-3 + 4i$ C. $3 + 4i$ D. $-3 - 4i$

Solution:-Additive inverse of $\not{Z}$ is

$$-\not{Z} = -(3 - 4i) = -3 + 4i$$

Answer: B

3. What is the modulus of the complex number $-6 + 8i$?

A. 10 B. 8 C. 5 D. 2

Solution:- Modulus of $\not{Z}$ is

$$|\not{Z}| = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = 10$$

Answer: A

4. What are the modulus and argument of the complex number $2 + 2i$ respectively?

A. $2\sqrt{2}$ and $\frac{\pi}{4}$ C. $2\sqrt{2}$ and $\frac{\pi}{4}$

B. 4 and $\frac{\pi}{4}$ D. 2 and $\frac{\pi}{4}$

Solution:

The modulus of $\not{Z}$ is $|\not{Z}| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$

$$\text{Arg}(\not{Z}) = \tan^{-1}\left(\frac{2}{2}\right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

Answer: C

5. Store-**I** sells a dozen of pens for 54 Birr and Store-**II** sells 5 pens of the same brand for 23 Birr. If a clerk wants to buy 100 pens from economical sales, then how much can he save in Birr?

A. 9.00 B. 8.50 C. 10.00 D. 12.50

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Solution:

Store I: Price of each pen $= \frac{54}{12} = 4.5$ Birr

Store II: Price of each pen $= \frac{23}{5} = 4.6$ Birr

Total price of 100 pens in store I is $4.5 \times 100 = 450$ Birr

Total price of 100 pens is store II is $4.6 \times 100 = 460$ Birr

When a clerk buys 100 pens from store II, he save Birr 10.

Answer: C

6. Which of the following is an advantage of using the techniques of random sampling in statistical methods?

A. The sample collected may not be a good representative.

B. The investigator personally cannot be a source of bias.

C. A person with limited know-how can apply the techniques.

D. Each techniques of sampling is applicable to any type of population.

Solution:- The major advantages of random sampling are:

- ✓ Its simplicity to use.
- ✓ Its lack of bias.
- ✓ Equal chance of selection.
- ✓ It requires less knowledge to complete.
- ✓ Less risk of carrying an error etc.
- ✓ The disadvantage is that it is very difficult to achieve (i.e time, effort and money).

Answer: B, C

7. If f is a continuous function on $[a, b]$ and F is its anti-derivative, then which of the following is true?

A. $\int_a^b f(x)dx = F(b) - F(a)$

B. $\int_a^b f'(x)dx = F(b) - F(a)$

C. $\int_a^b F'(x)dx = f(b) - f(a)$

D. $\int_a^b F(x)dx = f(b) - f(a)$

Solution:- By fundamental theorem of calculus

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

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Answer: A

8. On the average, which of the following is less significant as a factor to save money in a saving institution in our context?

A. Save for medical cases.
C. Save for emergency needs.
B. Save for retirement.
D. Save for charity and donations.

Answer: D

9. A Cottage industry produces goods either by capital or labour intensive method according to:

i) capital intensive using 1 unit of labour and 4 units of capital.

ii) labour intensive using 5 units of labour and 2 units of capital.

The industry manages to arrange up to 200 units of labour and 130 units of capital. If the owner can sell goods at a constant price and get a profit of 100 Birr per unit of the two quantities of goods produced by the processes **i)** and **ii)**, what will be the maximum profit generated.

A. 5,500 Birr B. 5,000 Birr C. 4,500 Birr D. 4,000 Birr

Solution:

	C	L	Maximum unit
CI	4	1	130
LI	2	5	200

Profit = 100/unit.

$$Z = 100x + 100y \text{ and s.t: } \begin{cases} 4x + y \le 130 \\ 2x + y \le 200 \\ x \ge 0,\ y \ge 0 \end{cases}$$

To solve for x and y

$$-2\begin{cases} 4x + y = 130 \\ 2x + 5y = 200 \end{cases} \Rightarrow \quad + \begin{cases} 4x + y = 130 \\ -4x - 10y = -400 \end{cases}$$

$$-9y = -270 \Rightarrow y = 30$$

Also, $4x + y = 130 \Rightarrow 4x + 30 = 130 \Rightarrow x = 25$

Maximum Z: $100x + 100y = 100(25) + 100(30) = 5500$

Answer: A

10. Consider the data given below:

10, 12, 10, 15, 12, 16, 13, 11, 12, 15

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The mean deviation about the mode of this data is equal to

A. 1.6 B. 1.5 C. 2.0 D. 1.8

Solution:

v	10	11	12	13	15	16
f	2	1	3	1	2	1

Mode = 12

The mean deviation is the mean of the absolute deviations of the observation

$$\text{MD (about mode)} = \frac{\sum(f|v-\text{mode}|)}{\sum f}$$

$$= \frac{2|10-12|+1|11-12|+3|12-12|+1|13-12|+2|15-12|+1|16-12}{2+1+3+3+1+2+1}$$

$$= \frac{4+1+1+6+4}{10} = \frac{16}{10} = 1.6$$

Answer: A

11. What are the domain and range of the function $f(x) = 2x^{\frac{3}{4}}$ respectively?

A. $[0,\infty)$ and $\mathbb{R}$ C. $\mathbb{R}$ and $[0,\infty)$

B. $[0,\infty)$ and $[0,\infty)$ D. $[0,\infty)$ and $(0,\infty)$

Solution:

$$f(x) = 2x^{\frac{3}{4}} = 2\left(x^{\frac{1}{4}}\right)^3 = 2\sqrt[4]{x^3}$$

Domain $= [0,\infty)$ and Range $= [0,\infty)$

Answer: B

12. Which one of the following defines a one - to - one function?

A. $f = \{(x, y) : y = x^2 + 1\}$ C. $f = \{(x, y)$: x is a sister of $y\}$

B. $f = \{(x, y)$: y is a father of $x\}$ D. $f = \{(x, y)$: $y = 3x - 1)\}$

Solution:- Linear function are one-to-one function.

Answer: D

13. The equation of a circle is given by $(x-3)^2 + (y+2)^2 = 9$. What are the center and the radius of the circle representatively?

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A. (3, –2) and 9
B. (3, 2) and 9
C. (3, –2) and 3
D. (3, 2) and 3

Solution:- For equation of a circle

$(x-3)^2+(y+2)^2=9$ has center (3, –2) and radius 3.

Answer: C

14. Which of the following sentences has a truth value?

A. Man is mortal.
B. Tomorrow is a beautiful day.
C. Man is selfish.
D. We shall win!

Answer: A

15. Which one of the following is the standard form of the equation of the parabola with vertex (4, 6) and focus (–8, 6)?

A. $(y-6)^2=48(x-4)$
B. $(y-6)^2=-48(x-4)$
C. $(x-4)^2=-12(y-6)$
D. $(x-4)^2=12(y-6)$

Solution:

The standard equation of parabola with vertex at (h, k) and directrix parallel to the y - axis given by $(y-k)^2=4p(x-h)$, where p is distance from vertex to focus p is $p=4-(-8)=12$, but the parabola opens to left, so that

$$(y-k)^2=-4p(x-h)$$
$$\Rightarrow (y-6)^2=-4(12)(x-4)$$
$$\Rightarrow (y-6)^2=-48(x-4)$$

Answer: B

16. Which of the following is true about the hyperbola given by $\frac{x^2}{4}-y^2=1$?

A. Its center is at (2, 1).
B. An asymptote to the hyperbola is $y=2x$.
C. Its foci are at (–5, 0) and (5, 0).
D. Its vertices are at (–2, 0) and (2, 0).

Solution:

$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ has center (0, 0) vertices $(\pm a, 0)$, with asymptote

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$y = \pm\frac{b}{a}x$. Now, for equation: $\frac{x^2}{4} - y^2 = 1$ is

center (0, 0) vertices (2, 0) and (–2, 0)

asymptote $y = \pm\frac{1}{2}x$ focus $\left(\sqrt{5},0\right)$ and $\left(-\sqrt{5},0\right)$

Answer: D

17. Two statements p and q are given by:

p: $\frac{22}{7}$ is an irrational number.

q: The sum of two odd integers is odd.

Then which of the following has truth value T?

A. $\neg p \wedge q$ B. $p \wedge q$ C. $p \wedge \neg q$ D. $p \vee \neg q$

Solution:

$p \equiv F$ and $q \equiv F$

$\neg p \equiv T$ and $\neg q \equiv T$

Then $p \vee \neg q \equiv F \vee T \equiv T$

Answer: D

18. Which one of the following is the contrapositive of the statement:

If 21 is a prime number, then 4 is an odd number?

A. If 4 is not an odd number, then 21 is not a prime number.

B. If 21 is not a prime number, then 4 is an even number.

C. If 4 is an odd number, then 21 is not a prime number.

D. If 4 is an odd number, then 21 is a prime number.

Solution:- The contrapositive of $p \Rightarrow q$ and $\neg q \Rightarrow \neg p$

Answer: A

19. Consider the following frequency distribution table.

Value	5	3	7	6	8
Frequency	3	4	5	3	5

What is the sixth decile?

A. 7.5 B. 7 C. 6.5 D. 5

Solution:- First arrange the data in increasing order

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v	3	5	6	7	8
f	4	3	3	5	5

$\Sigma f = 4+3+3+5+5 = 20 = n$

$$D_i = \left[\frac{\frac{in}{10}+\left(\frac{in}{10}+1\right)}{2}\right]^{th} \text{ item}$$

$$D_6 = \left[\frac{\frac{6(20)}{10}+\left(\frac{6(20)}{10}+1\right)}{2}\right]^{th} \text{ item} = 12.5^{th} \text{ item} = 7$$

Answer: B

20. In a certain health center there are 3 doctors, 8 nurses and 2 physicians. In how many ways one can form a group of 5 members consisting of 1 doctor, 3 nurses and 1 physician?

A. 336 B. 330 C. 168 D. 112

Solution: It is given that 3 doctors, 8 nurses and 2 physicians.

$$\text{Number of ways} = {}^3C_1 \cdot {}^8C_3 \cdot {}^2C_1$$

$$= 3 \cdot \left(\frac{8 \cdot 7 \cdot 6 \cdot 5!}{3 \times 2 \times 1 \cdot 5!}\right) \cdot 2$$

$$= 3 \cdot (56) \cdot 2 = 336$$

Answer: A

21. Which one of the following forms represents an exhaustive set of events if a die is thrown once?

A. {1, 2}, {2}, {4, 5}, {5, 6} C. {1, 2}, {4}, {4, 5, 6}

B. {1, 2, 3}, {3, 4}, {6} D. {1, 2, 3}, {4, 5}, {6}

Solution:

Experiment: Throwing a die once

Out comes: 1, 2, 3, 4, 5, 6.

$\therefore$ $\{1,2,3\}, \{4,5\}, \{6\}$ are exhaustive events.

Answer: D

22. A bag contains 2 red , 3 yellow and 5 green balls. One ball is drawn at

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random. What is the probability that the drawn ball is **NOT** yellow?

A. $\frac{4}{5}$ B. $\frac{7}{10}$ C. $\frac{3}{10}$ D. $\frac{1}{5}$

Solution:- $n(S) = 2 + 3 + 5 = 10$

$$P(\text{yellow}) = \frac{n(\text{yellow})}{n(S)} = \frac{3}{10}$$

$$P(\text{not yellow}) = 1 - \frac{3}{10} = \frac{7}{10}$$

Answer:

23. Which of the following is the product AB of the two matrices

$$A = \begin{bmatrix} -2 & 3 \\ 1 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 5 \\ 3 & 0 \end{bmatrix}?$$

A. $\begin{bmatrix} 11 & -10 \\ 11 & 5 \end{bmatrix}$ B. $\begin{bmatrix} -11 & 10 \\ 11 & -5 \end{bmatrix}$ C. $\begin{bmatrix} -11 & -10 \\ 11 & 5 \end{bmatrix}$ D. $\begin{bmatrix} -11 & 10 \\ 11 & 5 \end{bmatrix}$

Solution:

$$AB = \begin{pmatrix} -2 & 3 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} -1 & 5 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

$$C_{11} = \begin{pmatrix} -2 & 3 \end{pmatrix}\begin{pmatrix} -1 \\ 3 \end{pmatrix} = (-2)(-1) + (3)(3) = 11$$

$$C_{12} = \begin{pmatrix} -2 & 3 \end{pmatrix}\begin{pmatrix} 5 \\ 0 \end{pmatrix} = (-2)(5) + 3(0) = -10$$

$$C_{21} = \begin{pmatrix} 1 & 4 \end{pmatrix}\begin{pmatrix} -1 \\ 3 \end{pmatrix} = (1)(-1) + (4)(3) = 11$$

$$C_{22} = \begin{pmatrix} 1 & 4 \end{pmatrix}\begin{pmatrix} 5 \\ 0 \end{pmatrix} = 5 + 0 = 5$$

Answer: A

24. Which one of the following is **NOT** true about the determinant of a given square matrix A?

A. The determinant of A is equal to the determinant of its transpose.

B. Interchanging two rows of A gives the same determinant.

C. If A contains a row of zeros, then its determinant is zero.

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D. If A has identical rows, then its determinant is zero.

Answer: A

25. Which of the following is the inverse of the matrix $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$?

A. $\begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix}$ B. $\begin{bmatrix} -4 & -3 \\ -3 & -2 \end{bmatrix}$ C. $\begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix}$ D. $\begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$

Solution:

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix} = -1\begin{pmatrix} 4 & -3 \\ -3 & 2 \end{pmatrix}$$

$$\therefore A^{-1} = \begin{pmatrix} -4 & 3 \\ 3 & -2 \end{pmatrix}$$

Answer: C

26. Consider the system of linear equation $\begin{cases} 3x - ay = 1 \\ 6x + 4y = 5 \end{cases}$. Then which of the following is true?

A. If $a = 2$, then the system has no solution.

B. If $a > 2$, then the system has infinitely many solution.

C. If $a > -2$, then the system has a unique solution.

D. If $a = -2$, then the system has a unique solution.

Solution:

Let $D = \begin{vmatrix} 3 & -a \\ 6 & 4 \end{vmatrix}$

The system has

i) No solution if $D = 0$, i.e $12 + 6a = 0 \Rightarrow a = -2$

ii) Solution if $D \neq 0$ that is $12 + 6a \neq 0 \Rightarrow a \neq -2$

Answer: C

27. On the graph of the cumulative frequency distribution of a certain data, it is observed that, the mean, median, mode and standard deviation are 9.8, 10.0, 10.4, and 2.1, respectively. Which of the following is a correct conclusion about this distribution?

A. The distribution is skewed to the positive.

B. The data values are symmetrically distributed.

C. Most of the data values lie to the left of the modal value.

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D. Most of the data values lie to the right of the modal value.

Solution:

Mean = 9.8, Mode = 10.4

Median = 10, SD = 2.1

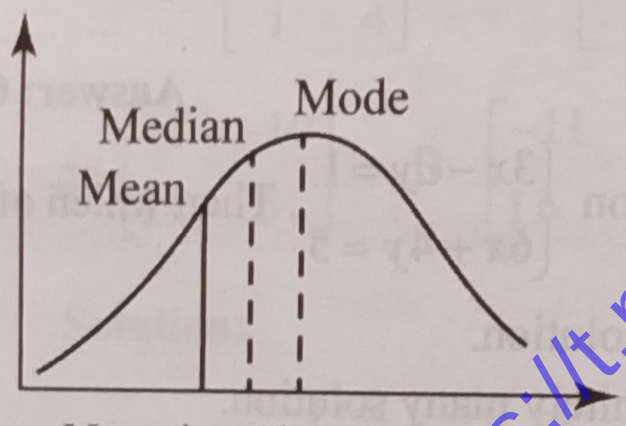
Mean, median, and mode of a normal distribution are equal. i.e the distribution is a symmetrical distribution, conversely, when values of mean, median and mode are not equal the distribution is skewed distribution

If the mean > the mode ← positively skewed.

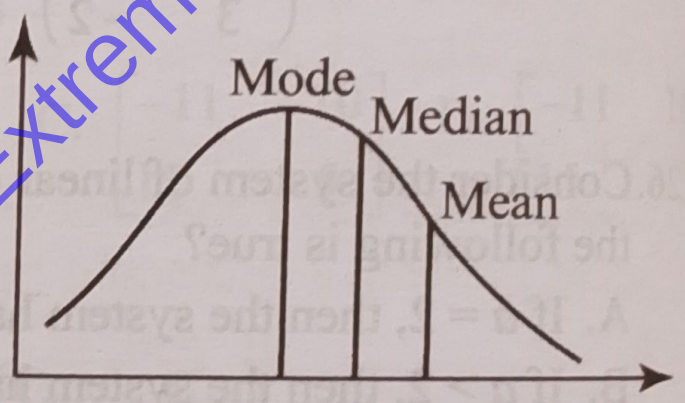
If the mean < the mode ← negatively skewed.

If the mean > the median ← positively skewed.

If the mean < the median ← negatively skewed.



Negative Skewness



Positive Skewness

Since, 9.8 < 10 < 10.4, the distribution is negatively skewed.

This means most of the data values lie to the left of the modal value.

Answer: C

28. The following table shows the mean and standard deviation of expenditures in Birr of three **F1, F2**, and **F3**, in a week.

Solution:

	F1	F2	F3
Mean	12,500	12,500	2,500
Standard deviation	3.62	3.56	3.6

Which factory of factories show more consistency in their expenditures than the other (s)?

A. Factory **2**

B. Factory **1**

C. Factories **F1** and **F2**

D. Factories **F1** and **F3**

Solution:- The standard deviation is a number which measured how far the data are spread from the mean. The S.D is small when the data

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are all concentrated close to the mean, and is larger when the data values show more variation from the mean. A smaller SD means greater consistency.

Answer: A

29. In leasing a building, tax is required from rental income. The following is the tax rate to a total amount of rent per year in Birr.

Amount taxable income	Up to 1,800	1,801 – 7,800	7,801–16,800	16,801–28,200
Tax rate	0%	10%	15%	20%
Deducted amount	0	180	570	1,410

If a shop is rented at 2,350 Birr per month for one year, what amount of tax is payable in Birr?

A. 4,230 B. 5,460 C. 2,420 D. 3,750

Solution:- Total amount of rent per year = $2350 \times 12 = 28{,}200$.

Tax = $28{,}200 \times 20\% = 5640$ Birr

Payable amount of tax = $5640 - 1410 = 4230$ Birr.

Answer: A

30. The amount of sales in a business enterprise is directly proportional to the number of items sold and inversely to the amount of tax paid, which is 5% of the amount sold.
If the sales amount to 15000 Birr and the number of items sold is 3000, what is the constant of proportionality in this situation?

A. 3750 B. 3650 C. 7500 D. 5700

Solution:

$$A = \text{sales of amount sold} = 15{,}000$$

$$\text{Let } x = \text{number of items sold} = 3000$$

$$y = \text{amount of tax paid} = 50\% \cdot A = \frac{5}{100} \times 15{,}000 = 750$$

$$A \; \alpha \frac{x}{y} \Rightarrow A = \frac{kx}{y}$$

$$15{,}000 = \frac{k \cdot 3000}{750} \Rightarrow k = \frac{15 \times 750}{3} = 3{,}750$$

Answer: A

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31. Which one of the following is the fourth term of the sequences $\left\{\cos\left(\frac{n\pi}{2}\right)\right\}$?

A. 1 B. 0.5 C. 0 D. −1

Solution:

$$a_n = \left\{\cos\left(\frac{n\pi}{2}\right)\right\}_{n=1}^{\infty} = \{0,-1,0,1,\cdots\}$$

Answer: A

32. Which one of the following is a geometric sequence?

A. 8, − 4, 2, −1 ...
C. 1, 1.5, 2.5, 3.5, ...
B. 0.1, 0.11, 0.12, 0.13, ...
D. 0.1, 0.01, 0.0011, 0.00011, ...

Solution:- A geometric sequences is identified by a common ratio (r) between successive term.

$\therefore$ $8, -4, 2, -1, \cdots$ is geometric sequence with common ratio $-\frac{1}{2}$

Answer: A

33. A factory that produces electric cable had sales Birr 1,000,000 the first day. During each successive days the sales increased by Birr 10,000 every day. What is the total sales of the factory in the first 10 days?

A. Birr 20,900,000
C. Birr 10,000,000
B. Birr 10,450,000
D. Birr 1,090,000

Solution:

Let A_1 = First day = 1,000,000.

A_2 = Second day = 1,010,000.

A_3 = Third day = 1,020,000.

Since it is an arithmetic sequence of d = 10,000 with A_1 = 1,000,000

$$A_n = A_1 + (n-1)d = 1{,}000{,}000 + (n-1)\cdot 10{,}000.$$
$$= 990{,}000 + 10{,}000n.$$

Hence, $S_n = \frac{n}{2}[A_1 + A_n]$

$$S_{10} = \frac{10}{2}[1{,}000{,}000 + 1{,}090{,}000] = 10{,}450{,}000.$$

Answer: B

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34. What is the least upper bound of the sequence $\left\{2+\frac{n}{n+1}\right\}_{n=1}^{\infty}$?

A. 3 B. 2 C. 1 D. 0

Solution:

Here $a_n=\left\{2+\frac{n}{n+1}\right\}_{n=1}^{\infty}=\left\{\frac{5}{2},\frac{8}{3},\frac{11}{4},\cdots 3\right\}$

$\Rightarrow \frac{5}{2}\le\left\{2+\frac{n}{n+1}\right\}_{n=1}^{\infty}<3.$

$\therefore$ The least upper bound is 3.

Answer: A

35. Which one of the following sequences is a null sequence?

A. $\left\{\frac{n}{n+1}\right\}$ B. $\left\{\frac{\cos n}{n}\right\}$ C. $\left\{n\sin\left(\frac{1}{n}\right)\right\}$ D. $\left\{1-\frac{1}{n}\right\}$

Solution:-A sequence $\{a_n\}$ is said to be null sequence if $\lim_{n\to\infty} a_n=0$

Hence, $\lim_{n\to\infty}\frac{\cos n}{n}=0$

Answer: B

36. Suppose $\{a_n\}$ and $\{b_n\}$ are sequences with $\lim_{n\to\infty} a_n=L$ and $\lim_{n\to\infty} b_n=M$. Then which of the following is **NOT** necessarily true?

A. $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)=\frac{L}{M}$

B. $\lim_{n\to\infty} a_n b_n=LM$

C. $\lim_{n\to\infty}(3a_n+2b_n)=3L+2M$

D. $\lim_{n\to\infty}(a_n-b_n)=L-M$

Solution:

Given: $\lim_{n\to\infty} a_n=L$ and $\lim_{n\to\infty} b_n=M$

Then

- $\lim_{n\to\infty}(a_n-b_n)=\lim_{n\to\infty} a_n-\lim_{n\to\infty} b_n=L-M$
- $\lim_{n\to\infty}(3a_n+2b_n)=\lim_{n\to\infty}3a_n+\lim_{n\to\infty}2b_n$
  $=3\lim_{n\to\infty} a_n+2\lim_{n\to\infty} b_n$
  $=3L+2M$

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- $\lim_{n\to\infty} a_n b_n = \lim_{x\to\infty} a_n \cdot \lim_{n\to\infty} b_n = L \cdot M$
- $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} = \frac{L}{M}$ Where $M \neq 0$.

Answer: A

37. $\lim_{x\to\infty} \frac{3x^2+2x}{5x-x^2}$ is equal to

A. 1 B. 0 C. –1 D. –3

Solution:

$$\lim_{x\to\infty} \frac{3x^2+2x}{5x-x^2} = \lim_{x\to\infty} \frac{\frac{3x^2}{x^2}+\frac{2x}{x^2}}{\frac{5x}{x^2}-\frac{x^2}{x^2}}$$

$$= \frac{\lim_{x\to\infty} 3+\frac{2}{x}}{\frac{5}{x}-1} = \frac{\lim_{x\to\infty} 3+\frac{2}{x}}{\lim_{x\to\infty} \frac{5}{x}-1} = \frac{3}{-1} = -3$$

Answer: D

38. Which of the following is the same as the derivative of a function f at a point P where P is on the graph of f?

A. The slope of the tangent line though P.
B. The graph of f has no jump at P.
C. The continuity of the function at P.
D. The slope of a secant line through P.

Solution:- Geometric interpretation of derivative of a function at a point is the slope of the tangent line at the given point.

Answer: A

39. What is the derivative of the function $f(x) = 3x^2 + 2\sqrt{x} - 4x$ at $x = 1$?

A. 3 B. 1 C. –1 D. –3

Solution:

$$f(x) = 3x^2 + 2\sqrt{x} - 4x$$

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$$f'(x)=\left(3x^2\right)'+\left(2\sqrt{x}\right)'-(4x)'=6x+2\cdot\frac{1}{2\sqrt{x}}-4$$

Then $f'(1)=6\cdot1+\frac{1}{\sqrt{1}}-4=3$

Answer: A

40. What are the 2nd and the nth derivative of $f(x)=e^{3x}$ respectively?

A. $9f(x)$ and $3^n f(x)$
B. $9f(x)$ and $n^2 f(x)$
C. $4f(x)$ and $2^n f(x)$
D. $2f(x)$ and $nf(x)$

Solution:

$f(x)=e^{3x}$

$f'(x)=\left(e^{3x}\right)'=e^{3x}\cdot(3x)'=3e^{3x}=3f(x)$

$f''(x)=\left(3e^{3x}\right)'=3\left(e^{3x}\right)'=3\left(3e^{3x}\right)'=9e^{3x}=3^2 f(x)$

$f'''(x)=\left(9^{3x}\right)'=9\left(e^{3x}\right)'=9\left(3e^{3x}\right)=27e^{3x}=3^3 f(x)$

$f^n(x)=3^n e^{3x}=3^n f(x)$

Answer: A

41. Which of the following is true about the zero of a function?

A. It is the concavity point of the function.
B. It is the critical point of the function.
C. It is the y - intercept of the function.
D. It is the x - intercept of the function.

Solution:- Let $f(x)$ be a given function.

If $f(c)=0$, then c is called the zero of a function. The point (c, 0) is the point where the graph meets the x - axis which is also called the x - intercept of the function.

Answer: D

42. Which of the following is true about the function defined by $f(x)=x^3-3x^2$ on $[-1, 3]$?

A. It has critical numbers at 0 and 2 and its maximum value is 0.
B. Its critical numbers are 0 and 2 and its maximum value is –4.
C. Its critical numbers are 0 and 2 and its maximum value is 15.
D. Its critical numbers are 0, 2, 3 and its maximum value is 0.

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Solution:

$$f(x)=x^3-3x^2,\ [-1,3]$$

- f is differentiable on $[-1, 3]$
- $f'(x)=3x^2-6x$

$$\Rightarrow f'(x)=0 \Leftrightarrow 3x^2-6x=0 \Leftrightarrow 3x(x-2)=0$$
$$\Rightarrow x=0 \text{ and } x=2$$

Since $0,2 \in [-1,3]$, $x=0,2$ are critical

- The maximum value of on $f[-1, 3]$ lies either at the end points or at the critical points.

$$\Rightarrow f(-1)=(-1)^3-3(-1)^2=-1-3=-4$$
$$f(3)=3^3-3\cdot 3^2=3^3-3^3=0$$
$$f(0)=0^3-3\cdot 0^2=0$$
$$f(2)=2^3-3\cdot 2^2=8-12=-4$$

$\therefore$ The maximum value is 0.

Answer: A

43. In a book shop, 15 copies of an algebra book costs 105 Birr and 20 copies of a geometry book costs 125 Birr. A customer needs to buy 6 algebra and 4 geometry books from the respective copies. What is the total amount that he has to pay?

A. 69 Birr B. 67 Birr C. 77 Birr D. 72 Birr

Solution:

	Amount of each book
Algebra	$\frac{105}{15}$ Birr $= 7$ Birr
Geometry	$\frac{125}{20}$ Birr $= 6.25$ Birr

Total amount $= 6A + 4G = 6 \times 7 + 4 \times 6.25 = 67$ Birr

Answer: B

44. What are the length L and width W of a rectangle with perimeter 10,000m that maximizes the area?

A. L = 5,000 m and W = 5,000 m

B. L = 3,000 m and W = 2,500 m

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C. L = 2,500 m and W = 3,000 m

D. L = 2,500 m and W = 2,500 m

Solution:

Perimeter = 2 (L + W) = 10,000

L + W = 5000 and L = 5000 – W

Area (A) = LW = (5000 – W) W

$A(W) = 5000W - W^2$

$A'(W) = 5000 - 2W \Leftrightarrow W = 2500m$

L = 5000 – 2500 = 2500m

Answer: D

45. If the radius r of a sphere is increasing at the rate of 2 cm/min, then the rate of change of the volume when r = 1 cm is

A. $12\pi\ cm^3/min$

B. $8\pi\ cm^3/min$

C. $6\pi cm^3/min$

D. $4\pi cm^3/min$

Solution:

Volume of sphere: $V = \frac{4}{3}\pi r^3$

$$\frac{dv}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Since $r = 1, \frac{dr}{dt} = 2cm/\min$

$$\Rightarrow \frac{dv}{dt} = 4\pi \cdot 1^2 \cdot 2 = 8\pi cm^3/\min$$

Answer: B

46. Which of the following is equal to $\int(\cos x - 2x)dx$ where c is a constant?

A. $\sin x - x^2 + c.$

B. $-\sin x + x^2 + c.$

C. $\sin x + x^2 + c.$

D. $-\sin x - x^2 + c.$

Solution:

$$\int(\cos x - 2x)dx = \int \cos x\,dx - \int 2x\,dx$$

$$= -\sin x + c_1 - \frac{2x^2}{2} + c_2$$

$$= -\sin x - x^2 + c, \text{ where } c_1 + c_2 = c$$

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Answer: D

47. What is the area of the region bounded by $y = 0$ and the graph of $f(x) = 4 - x^2$ on the interval $[-1, 1]$?

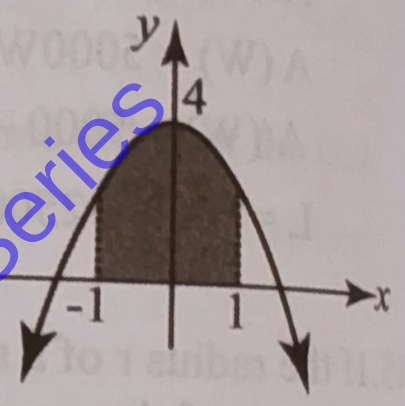
A. $\frac{26}{3}$ B. 8 C. $\frac{22}{3}$ D. $\frac{2}{3}$

Solution:

$$\text{Area} = \int_{-1}^{1} f(x)dx = \int_{-1}^{1} (4 - x^2)dx$$

$$= \int_{-1}^{1} 4dx - \int_{-1}^{1} x^2 dx = 4x\Big|_{-1}^{1} - x^2\Big|_{-1}^{1}$$

$$= [4 - (-4)] - [1 - 1] = 8$$



Answer: B

48. Suppose $\int_1^3 f(x)dx = 6$ and $\int_1^2 f(x)dx = 10$, what is the value of $\int_2^3 f(x)dx$?

A. 16 B. 4 C. 2 D. –4

Solution:

Given: $\int_1^3 f(x)dx = \int_1^2 f(x)dx + \int_2^3 f(x)dx$

Then, $\int_2^3 f(x)dx = \int_1^3 f(x)dx - \int_1^2 f(x)dx = 6 - 10 = -4$

Answer: D

49. Which of the following statements has a truth value T?

A. $(p \vee q) \Rightarrow (p \wedge q)$ C. $(p \wedge q) \Rightarrow (p \vee q)$

B. $(p \Rightarrow q) \Rightarrow (p \vee q)$ D. $(p \wedge q) \Rightarrow \neg(p \vee q)$

Solution:- $p \Rightarrow q$ is F is p if T and q is F. Otherwise $p \Rightarrow q$ is T.

p	q	$p \vee q$	$p \wedge q$	$p \Rightarrow q$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	T
F	F	F	F	T

From the table we see that $(p \wedge q) \Rightarrow (p \vee q)$ is always T.

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Answer: C

50. The inverse of the function $y = 3x - 5$ is equal to

A. $y = \frac{x-5}{3}$

B. $y = -5x + 3$

C. $y = \frac{x+5}{3}$

D. $y = \frac{x+3}{5}$

Solution:

Given: $y = 3x - 5$

i) Interchange x and y: $x = 3y - 5$

ii) Solve for y: $x = 3y - 5 \Leftrightarrow y = \frac{x+5}{3}$

iii) Make y as the inverse

$\therefore\ y = \frac{x+5}{3}$ is the inverse of $y = 3x - 5$

Answer: C

51. Which of the following is the universal set of the expression $\frac{1}{x^2} + \frac{x+1}{x^2+x}$?

A. $\mathbb{R} \setminus \{0, 1, -1\}$

B. $\mathbb{R} \setminus \{0, -1\}$

C. $\mathbb{R} \setminus \{0, 1\}$

D. $\mathbb{R} \setminus \{0\}$

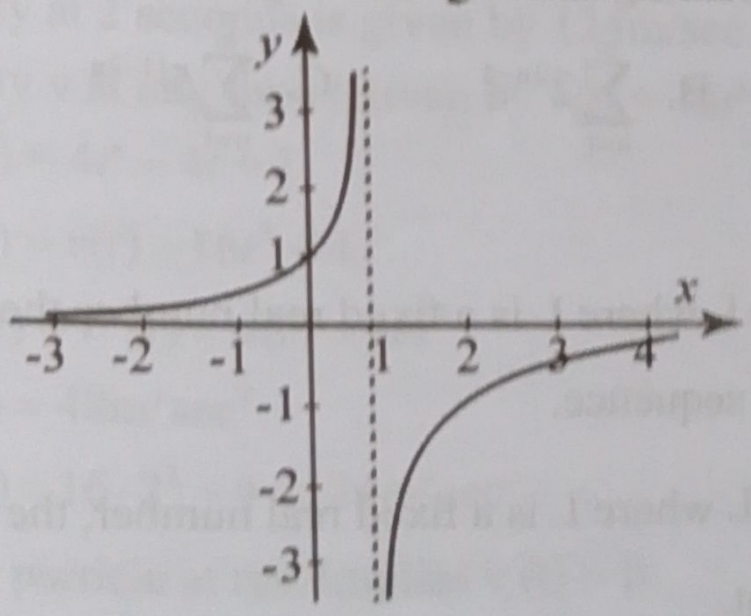
Solution:

Given: $\frac{1}{x^2} + \frac{x+1}{x^2+x} = \frac{1}{x^2} + \frac{x+1}{x(x+1)}$

$\therefore\ D = \mathbb{R} \setminus \{-1, 0\}$.

Answer: B

52. The graph of a rational function is given as in the Figure below.



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Which of the following is **NOT** true about the graph?

A. It is decreasing on $(1, \infty)$.

B. It is increasing on $(-\infty, 1)$.

C. $x = 1$ is its vertical asymptote.

D. $y = 0$ is its horizontal asymptote.

Solution:- By refering the graph:

- ✓ the line $x = 1$ is vertical asymptote.
- ✓ the line $y = 0$ is horizontal asymptote.
- ✓ the function is increasing on $(-\infty, 1)$ and $(1, \infty)$

Answer: A

53. The equation of a line is given by $3x - 2y + 6 = 0$. Then which of the following is true about the line?

A. $(2, 5)$ lies on the line.

B. Its x- intercept is 2.

C. Its y-intercept is -3.

D. Its slope is $\frac{3}{2}$

Solution:

- ✓ The point $(2, 5)$ does not lie on the line because $3(2) - 2(5) + 6 = 2 \neq 0$
- ✓ To find x-intercept, set $y = 0$
  $\Rightarrow 3x - 2(0) + 6 \Leftrightarrow x = -2.$
- ✓ To find y-intercept, set $x = 0$
  $\Rightarrow 0 - 2y + 6 = 0 \Leftrightarrow y = 3.$
- ✓ Its slope is $\frac{3}{2}$, since $3x - 2y + 6 = 0$ is $y = \frac{3}{2}x + 3.$

54. Which of the following series is convergent?

Answer: D

A. $\sum_{n=1}^{\infty}\left(\frac{3}{2}\right)^n$ B. $\sum_{n=1}^{\infty} 2^{2n-5}$ C. $\sum_{n=1}^{\infty} 5^{11-2n}$ D. $\sum_{n=1}^{\infty}\left(\frac{2}{5}\right)^{-n}$

Solution:

- If $\lim_{n\to\infty} a_n = L$ where L is a fixed real number then $\{a_n\}$ is a converged sequence.
- If $\sum_{n=1}^{\infty} a_n = L$ where L is a fixed real number, the series is convergent.

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• An infinite geometric series is convergent if $|r| < 1$.

$$\sum_{n=1}^{\infty} 5^{11-2n} = \sum_{n=1}^{\infty} 5^{11} \cdot 5^{-2n} = 5^{11} \sum_{n=1}^{\infty} \left(\frac{1}{25}\right)^n$$

$$r = \frac{1}{25} < 1. \text{ It is convergent.}$$

Answer: C

55. Let f be a function defined on a closed bounded interval $[a, b]$. Which of the following describes that f is continuous on $[a, b]$?

A. f is continuous on (a, b), f is continuous from the left at a and from the right at b.

B. f is continuous on (a, b), f is continuous from the left at a and from the left at b.

C. f is continuous on (a, b), f is continuous from the right a and from the left at b.

D. f is continuous at each point in (a, b) and at both a and b.

Solution:- f is continuous on $[a, b]$, then:

✓ f is continuous on (a, b)

✓ $\lim_{x \to a^+} f(x) = f(a)$ (f continuous at a from the right).

✓ $\lim_{x \to b^-} f(x) = f(b)$ (f is continuous at b from the left).

Answer: C

56. A position of a particle is given by the equation $S(t) = 4t^4 - 4t + 1$ where t is measured in seconds and S in meters. Which of the following is **NOT** true about the motion of the particle?

A. The acceleration of the particle at 1 sec is $48 m/sec^2$.

B. The particle is at $t = 0.5$ seconds.

C. The velocity at 2 seconds is given by 124m/sec.

D. The velocity v at any time t given by $v(t) = 16t^3 - 4$.

Solution:- $S(t) = 4t^4 - 4t + 1$

$$S'(t) = v(t) = 16t^3 - 4$$

$$S''(t) = V'(t) = a(t) = 48t^2$$

$$a(1) = 48 m/sec^2$$

$$v(2) = 16 \cdot 2^3 - 4 = 124 m/sec$$

The particle at rest implies v (t) = 0

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$$\Rightarrow 16t^3 - 4 = 0 \Leftrightarrow 4\left(4t^3 - 1\right) = 0$$

$$\Rightarrow t^3 = \frac{1}{4} \text{ and } t = \frac{1}{\sqrt[3]{4}} \text{ sec.}$$

Answer: B

57. If $f'(1) = 4$, $f(1) = 3$ and $g'(3) = 5$ then $(gof)'(1)$ is equal to

A. 20 B. 15 C. 5 D. 3

Solution:

Given: $f'(1) = 4$, $f(1) = 3$, $g'(3) = 5$

$$(gof)'(x) = g'(f(x)) \cdot f'(x)$$

$$(gof)'(1) = g'(f(1)) \cdot f'(1) = g'(3) \cdot 4 = 5 \cdot 4 = 20$$

Answer: A

58. Which of the following is true where c is a constant?

A. $\int \ell nx dx = \frac{1}{x} + c$

B. $\int xe^x dx = \frac{1}{2}x^2 e^x + c$

C. $\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx = -2\cos\sqrt{x} + c$

D. $\int (3x+5)^6 dx = \frac{(3x+5)^7}{7} + c$

Solution:

$$\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx \quad \text{Let } U = \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$2\sqrt{x}du = dx \Leftrightarrow 2udu = dx$$

$$\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx = \int \frac{\sin u}{u} \cdot 2udu = 2\int \sin u du$$

$$= -2\cos u + c = -2\cos\sqrt{x} + c$$

Answer: C

59. The acceleration of a body at any time t is given by $a(t) = 4 - 2t + 3t^2$ with initial velocity $v(0) = 5$. Then what is the velocity of v of the body as a function of time?

A. $v(t) = -2 + 6t$

B. $v(t) = 9 - t^2 + t^3$

C. $v(t) = 1 + 4t - t^2 + t^3$

D. $v(t) = 5 + 4t - t^2 + t^3$

Solution:

$$a(t) = 4 - 2t + 3t^2$$

$$v(t) = \int a(t) = \int \left(4 - 2t + 3t^2\right) dt$$

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$$4t-\frac{2t^2}{2}+\frac{3t^3}{3}+c=4t-t^2+t^3+c$$

$$v(0)=4\cdot 0-0^2+0^3+c=5\Rightarrow c=0$$

$$\therefore\ v(t)=5+4t-t^2+t^3$$

Answer: D

60. Which of the following argument is valid?

A. $p, q\vdash q\Rightarrow\neg q$

B. $p, \neg q\vdash p\wedge q$

C. $p, \neg q\vdash p\Rightarrow q$

D. $\neg p, q\vdash p\Rightarrow q$

Solution: -Assume that all the premises are true. If the conclusion is true, then the argument is valid. If the conclusion is false, then the argument is invalid.

$\neg p, q\vdash p\Leftrightarrow q$

$\neg p = T$ (premise)

$\therefore\ p = F$ (definition of $\neg$)

$q = T$ (premise)

$p \Rightarrow q$

$F \Rightarrow T = T$ (conclusion)

$\therefore$ Valid argument

Answer: D

61. A painter agreed to work 40 hours per week at an hourly rate of 60 Birr including Saturdays, and earns an overtime of time and a half. If he is committed to work for 9 hours each day from Monday to Friday and 5 hours on Saturday, how much (in Birr) will he earn just after 4 weeks?

A. 12,400 B. 12,200 C. 13,200 D. 12,800

Solution:

Total work time: $9 \times 5 + 5 = 50$hours.

Thus, 10 hours is the overtime.

Total Birr earned per week $= (40 \times 60) + (10 \times 90$ overtime$) = 3300$

Total Birr earned after 4 weeks: $3300 \times 4 = 13{,}200$Birr

Answer: C

62. Math test scores distribution of 20 students is given as in Table 2 below?

Scores	1 - 5	6 - 10	11 - 15	16 - 20
Frequency	4	6	8	2

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What is the standard deviation of the scores?

A. $\sqrt{21}$ B. $\sqrt{20}$ C. 5 D. 4

Solution:

$$S\cdot D=\sqrt{\frac{\sum f_i\left(x_i-\bar{x}\right)^2}{\sum f_i}}$$

Where x_i is the class mark of each mark

$$S\cdot D=\sqrt{\frac{4(3-10)^2+6(8-10)^2+8(13-10)^2+2(18-10)}{4+6+8+2}}$$

$$=\sqrt{\frac{4(49)+6(4)+8(9)+2(64)}{20}}$$

$$=\sqrt{\frac{196+24+72+128}{20}}=\sqrt{21}$$

Answer: A

63. What is the coefficient of x^2 in the binomial expansion of $\left(2x+\frac{3}{x^3}\right)^8$?

A. 576 B. 500 C. 6 D. 0

Solution:- $\left(2x+\frac{3}{x^3}\right)^8$?

The $(r+1)^{th}$ term,

$$T_{r+1}=\binom{n}{r}x^{n-r}y^r=\binom{8}{r}(2x)^{8-r}\left(\frac{3}{x^3}\right)^r$$

$$=\binom{8}{r}2^{8-r}x^{8-r}3r\cdot x^{-3r}$$

$$x^{8-r-3r}=x^2\Leftrightarrow 8-4r=2$$

$$6=4r\Rightarrow r=\frac{3}{2}\notin W$$

$\Rightarrow$ The expansion does not contain an x^2 term

$\Rightarrow$ Coefficient of $x^2=0$

64. A shop sales a pack of 20 articles for 5,500 Birr. It planned to clear

Answer: D

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the articles during a holiday. If the shop sells each for 220 Birr, what percent discount has it made?

A. 18% B. 15% C. 22% D. 20%

Solution:

$$\text{Price of each article}: \frac{5,500}{20}\text{ Birr} = 275\text{ Birr}$$

$$\text{Sale difference}: 275\text{ Birr} - 220\text{ Birr} = 55\text{ Birr}$$

$$\text{Percent discount} = \frac{55}{275} \times 100\% = 20\%$$

Answer: D

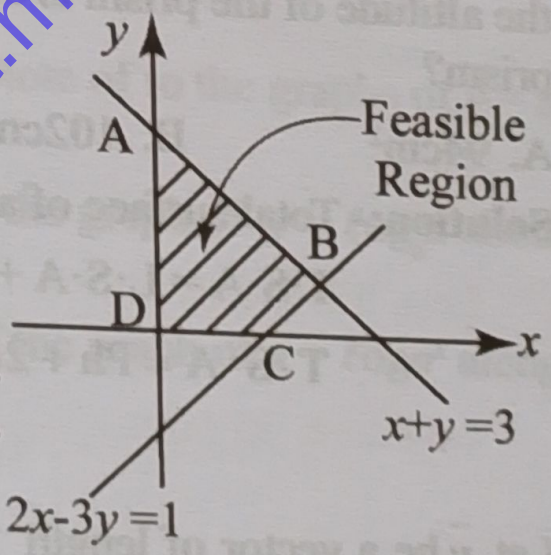
65. What is the maximum value of the function $Z = 3x + y$ subject to: $2x - 3y \geq 1, x + y \leq 3$, *for* $x \geq 0$ *and* $y \geq 0$?

A. $\frac{9}{2}$ B. $\frac{3}{2}$ C. 9 D. 7

Solution:

$Z = 3x + y$ ← Objective function

$2x - 3y \geq 1$
$x + y \leq 3$
$x \geq 0$
$y \geq 0$ ← subjective function



A ,B, C and D are called corner points.

The maximum value of Z lies in either at A or at B.

$$A: \begin{cases} x = 0 \\ x + y = 3 \end{cases}$$

$\Rightarrow y = 3$

$\therefore A(0,3)$

$$B: \begin{cases} x + y = 3 \\ 2x - 3y = 1 \end{cases}$$

$$- \begin{cases} 2x + 2y = 6 \\ 2x - 3y = 1 \end{cases}$$

$$5y = 5,\ y = 1, x = 2$$

$\therefore B(2,1)$

Then , $Z = 3x + y$

Z at A: $3(0) + 3 = 3$ and Z at B $= 3(2) + 1 = 7$

$\therefore$ The maximum value is 7

Answer: D

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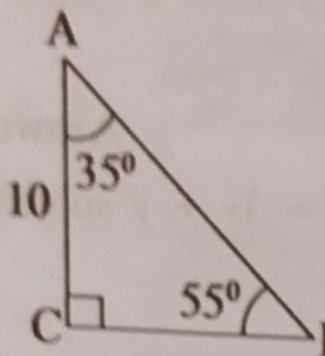
1. Consider ΔABC with right angle at C and $m(\angle ABC) = 55^0$. If the length of $\overline{AC}$ is 10 units, then what is the area of the triangle? (Given $\tan 35^0 = 0.7$ and $\tan 55^0 = 1.43$)

A. 35sq. units
B. 70sq. units
C. $5\sqrt{149}$ sq.units
D. $10\sqrt{149}$ sq.units

Solution:- We have given ΔABC right angle at C;



$\tan 35^\circ = \frac{BC}{AC}$ implies $BC = AC \tan 35^\circ$

$\text{Area}(\Delta ABC) = \frac{1}{2} AC \times CB = \frac{1}{2}(10)(AC \tan 35^\circ)$

$\therefore \text{Area}(\Delta ABC) = \frac{1}{2}(10)(10)(0 \cdot 7) = 35$ sq.unit

Answer: A

2. The base of a right prism is a rectangle with sides 3 cm and 4 cm and the altitude of the prism is 5cm. What is the total surface area of the prism?

A. $94cm^2$
B. $102cm^2$
C. $70cm^2$
D. $82cm^2$

Solution:- Total surface of a rectangular prism is given by

$T \cdot S \cdot A = L \cdot S \cdot A + 2$ (Base area)

$T \cdot S \cdot A = Ph + 2\ell W = (3+3+4+4)5 + 2(3 \times 4) = 94cm^2$

Answer: A

3. Let $\vec{u}$ be a vector of length $3\sqrt{3}$ cm directed to the East $\vec{v}$ be a vector of length 3cm directed to the north. Then $\vec{u} + \vec{v}$ is equals to a vector of length:

A. 3cm in the direction East 60^0 North
B. 3cm in the direction East 30^0 North
C. 6cm in the direction East 60^0 North
D. 6cm in the direction East 30^0 North

Solution:- We have $|\vec{u}| = 3\sqrt{3}$cm and $|\vec{v}| = 3$cm

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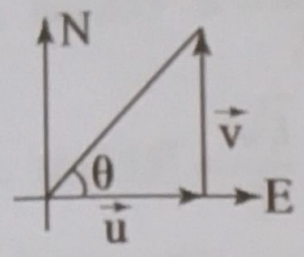

$$\text{Then}: \left|\vec{u}+\vec{v}\right|^2 = \left(\vec{u}+\vec{v}\right)\cdot\left(\vec{u}+\vec{v}\right)$$

$$= \left|\vec{u}\right|^2 + \left|\vec{v}\right|^2 + 2\left|\vec{u}\right|\left|\vec{v}\right|\cos 90^\circ$$

$$= \left(3\sqrt{3}\right)^2 + 3^2 + 2(0)$$

$$\therefore \left|\vec{u}+\vec{v}\right| = \sqrt{27+9} = \sqrt{36} = 6\text{cm}$$

$$\text{Also, } \tan\theta = \frac{\left|\vec{v}\right|}{\left|\vec{u}\right|} = \frac{3}{3\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ and } \theta = 30^\circ$$

The angle measured from north line is 60°.

Answer: C

4. Consider the graphs of the functions $y = a^x$ and $y = \log_a x$ for $a > 0$ and $a \neq 1$. Which of the following is **true** about the functions?

   A. For $a > 1$, the graphs of both $y = a^t$ and $y = \log_a x$ are increasing on $(0, \infty)$

   B. For $0 < a < 1$, the y-axis is an asymptote of to the graphs of $y = a^x$ and $y = \log_a x$.

   C. The graph of $y = a^t$ is a reflection of the graph of $y = \log_a x$ about the region.

   D. The graph of $y = a^x$ is a reflection of the graph of $y = \log a^x$ along $y = -x$.

Solution:

- $y = a^x$ is the inverse of $y = \log_a^x$
- one of them is obtained by reflecting the other on the line $y = x$.
- For $a > 1$, both $y = a^x$ and $y = \log_a^x$ are increasing in their respective domain.

Answer: A

5. Below is a histogram showing the time taken to complete a test.

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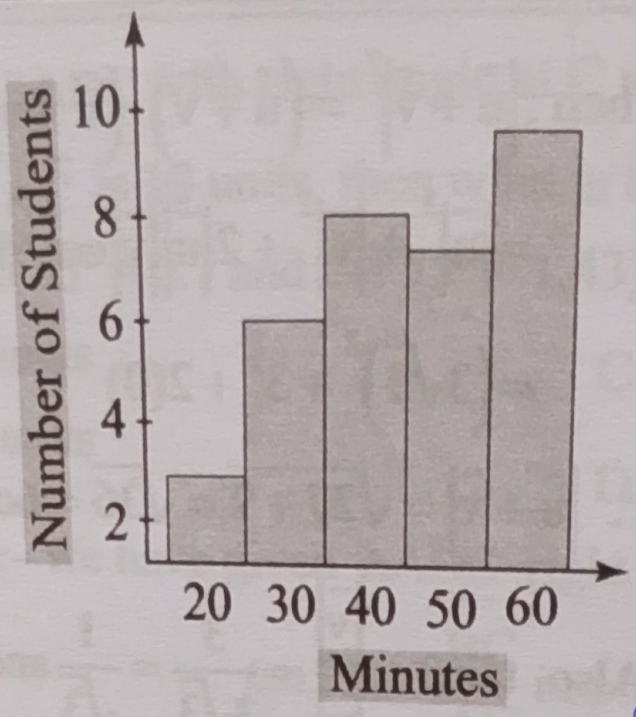

Based on the above histogram, which of the following is **true**?

A. 30% of the students completed the test with less than 40 minutes.

B. 25% of the students completed the test with more than 50 minutes.

C. 50% of the students completed the test with less than 50 minutes.

D. 40% students completed the test with more than 40 minutes.

Solution:

Minute	20	30	40	50	60
Number of students	3	6	9	8	10
Cum. frequency	3	9	18	26	36

Less than 40 minute $= \frac{9}{36} \times 100\% = 25\%$

Less than 50 minute $= \frac{18}{36} \times 100\% = 50\%$

Answer: C

6. What is solution set of equation $3^{x-x^2} = \frac{1}{9}$?

A. $\{1, -2\}$ B. $\{-1, 2\}$ C. $\{0, 1\}$ D. $\{-1, 0\}$

Solution:

$$3^{x-x^2} = \frac{1}{9} \Rightarrow 3^{x-x^2} = 3^{-2}$$

$\Rightarrow x - x^2 = -2$ implies $x^2 - x - 2 = 0$

$\Rightarrow (x-2)(x+1) = 0$

$\therefore \quad x = 2$ or $x = -1$

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Answer: B

7. Which of the following is **true** about the trigonometric values of 855^{o}?

A. $\cos 855^{0} = -\frac{\sqrt{2}}{2}$

B. $\sin 855^{0} = \frac{\sqrt{3}}{2}$

C. $\tan 855^{o} = 1$

D. $\cot 855^{o} = -\sqrt{2}$

Solution:-We use the relation of co terminal angles:

$$\sin\theta = \sin(\theta \pm 360^{o}K) \text{ and } \cos\theta = \cos(\theta \pm 360^{o}K)$$

$$\sin 855^{o} = \sin(855^{o} - 720^{o}) = \sin 135^{o} = \frac{\sqrt{2}}{2}$$

$$\cos 855 = \cos(855 - 720^{o}) = \cos 135^{o}$$

since $\cos 135^{o}$ is in 2nd quadrant and cosine negative value

$$\therefore \cos 855 = \cos 135^{o} - \cos 45^{o} = -\frac{\sqrt{2}}{2}$$

Answer: A

8. Which one of the following is true about the function $y = \sin x$?

A. Its range is $[-1, 1]$

B. Its domain is $(0, \infty)$

C. Its domain is $[-1, 1]$

D. Its period is π

Solution:- Properties of $y = \sin x$

- The domain is the set of real number
- The range = $\{y: -1 \leq y \leq 1\}$.
- The function is periodic, with period 2π.

Answer: A

9. The lower and upper bases of the frustum of a regular pyramid are equilateral triangles of side 3 cm long and 2 cm long respectively. If its slant height is 4cm, then what is the lateral surface area of the frustum?

A. $30cm^2$ B. $48cm^2$ C. $72cm^2$ D. $96cm^2$

Solution:- Lateral surface area of frustum of a regular pyramid is given by

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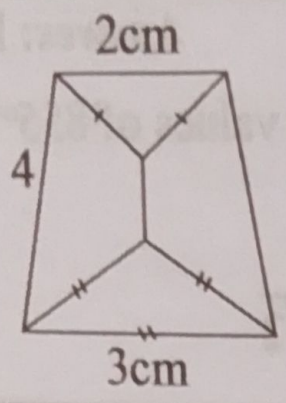

$$L\cdot S\cdot A=\frac{1}{2}(\text{slant height})\left[\begin{matrix}\text{perimeter of}\\ \text{lower base}\end{matrix}+\begin{matrix}\text{perimeter of}\\ \text{upper base}\end{matrix}\right]$$

$$=\frac{1}{2}\ell(P_1+P_2)=\frac{1}{2}(4)[(3+3+3)+(2+2+2)]=30cm^2$$

For brief case refer extreme maths Grade 9 -10 page 638

Answer: A

10. Which one of the following is **true** about the signum function, $f(x)=\operatorname{sgn} x$?

A. Its domain is $\{-1, 0, 1\}$

B. Its range is $[0,\infty]$

C. Its domain is the set of real numbers.

D. Its graph is the set of real numbers.

Solution:

$$f(x)=\operatorname{sgn}(x)=\begin{cases}\frac{|x|}{x}\\ 0\end{cases}=\begin{cases}1, \text{ if } x>0\\ 0, \text{ if } x=0\\ -1, x<0\end{cases}$$

$\therefore$ Domain $=\mathbb{R}$ and Range $=\{-1,0,1\}$

Answer: C

11. Which one of the following is the equation of a circle with center $(2,-1)$ and radius 3 units?

A. $x^2+y^2+2x+2y-4=0$

B. $x^2+y^2-4x+2y-4=0$

C. $x^2+y^2+4x-2y-5=0$

D. $x^2+y^2-4x+2y-5=0$

Solution:- Equation of a circle with center (h, k) and radius r is given by

$$(x-h)^2+(y-k)^2=r^2\Rightarrow(x-2)^2+(y+1)^2=3^2$$

$$\Rightarrow x^2-4x+4+y^2+2y+1=9$$

$$\Rightarrow x^2+y^2-4x+2y+5-9=0$$

$$x^2+y^2-4x+2y-4=0$$

Answer: B

12. Consider the ellipse given by the equation $4(x+1)^2+(y-1)^2=16$.

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Which one of the following is the standard form of equation of the ellipse?

A. $\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{4^2}=1$

B. $\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{1^2}=1$

C. $\frac{(x+1)^2}{4^2}+\frac{(y-1)^2}{1^2}=1$

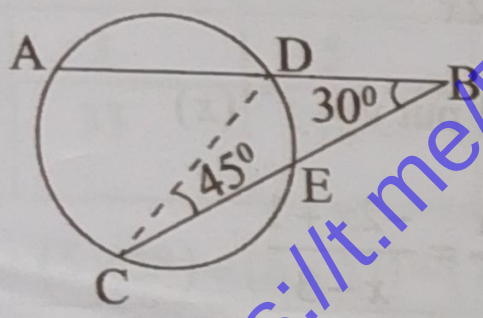
D. $\frac{(x+1)^2}{4^2}+\frac{(y-1)^2}{4^2}=1$

Solution:- Divide both side by 16

$$\Rightarrow \frac{4(x+1)^2}{16}+\frac{(y-1)^2}{16}=\frac{16}{16} \text{ implies } \frac{(x+1)^2}{4}+\frac{(y-1)^2}{16}=1$$

Answer: A

13. Suppose that the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$ intersect at B outside the circle as shown in the figure below.



If $m(\angle ABC)=30^0$ and $m(\angle BCD)=45^0$, then what is $m\left(\overset{\frown}{AC}\right)$?

A. 90^0 B. 105^0 C. 120^0 D. 150^0

Solution:

$$m(\angle DCE)=\frac{1}{2}m\left(\overset{\frown}{DE}\right)\Rightarrow m\left(\overset{\frown}{DE}\right)=2m(\angle DCE)$$

$$\therefore\ m\left(\overset{\frown}{DE}\right)=2(45^\circ)=90^\circ$$

$$\text{Also, } m(\angle B)=\frac{1}{2}\left(m\overset{\frown}{AC}-m\overset{\frown}{DE}\right)$$

$$30^\circ=\frac{1}{2}\left(m\overset{\frown}{AC}-90\right)$$

$$\therefore\ m\left(\overset{\frown}{AC}\right)=60+90=150$$

Answer: D

14. Let $f(x)=\frac{3x+1}{x+2}$. Which one of the following is the inverse of f?

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